A PRACTITIONER'S AND STUDENT'S GUIDE

Discounted Cash Flow Valuation

Theory, Technique and Worked Practice — from First Principles to Real-World Judgement

Thirty-three chapters, two complete case studies, a library of compact valuations, practice problems with solutions, formula sheets and a glossary

Contents

Preface 4
How to Use This Book 6

Part I — Foundations 7
Chapter 1 — What Is Valuation? 8
Chapter 2 — The Mathematics of Discounting 14
Chapter 3 — The Architecture of a DCF 20

Part II — Cash Flows 26
Chapter 4 — Reading Financial Statements for Valuation 27
Chapter 5 — Free Cash Flow to the Firm 34
Chapter 6 — Free Cash Flow to Equity and the Dividend Discount Model 41
Chapter 7 — Forecasting: Growth, Margins and Reinvestment 46
Chapter 8 — Normalising Earnings and Cleaning the Base Year 53

Part III — Discount Rates and Terminal Value 58
Chapter 9 — The Cost of Equity 59
Chapter 10 — Beta: Estimation, Un-levering and Re-levering 65
Chapter 11 — Cost of Debt, Capital Structure and WACC 70
Chapter 12 — Terminal Value 76

Part IV — Building and Testing the Valuation 83
Chapter 13 — From Enterprise Value to Equity Value per Share 84
Chapter 14 — Complete Case Study: Valuing Meridian Foods Ltd 90
Chapter 15 — Sensitivity, Scenarios and Simulation 96
Chapter 16 — Cross-Checks: Multiples, Reverse DCF and Triangulation 102

Part V — Special Situations 107
Chapter 17 — Adjusted Present Value and Leveraged Transactions 108
Chapter 18 — Valuing Banks, NBFCs and Insurers 113
Chapter 19 — Cyclical, Commodity, Infrastructure and Real-Estate Businesses 117
Chapter 20 — Start-ups, High-Growth Companies and Digital Platforms 122
Chapter 21 — Valuing Distressed Companies 129
Chapter 22 — Valuation in India and Other Emerging Markets 135
Chapter 23 — Real Options, Intangibles and ESG in DCF 142
Chapter 24 — Pitfalls, Governance and Professional Practice 147

Part VI — Applications and Extensions 153
Chapter 25 — Valuation in Mergers and Acquisitions 154
Chapter 26 — Private Companies, Minority Stakes and Disputes 160
Chapter 27 — Second Case Study: A Listed Pharmaceutical Company 166
Chapter 28 — A Library of Compact Valuations 173
Chapter 29 — The Integrated Model: Three Statements, FCFF, FCFE and Reconciliation 179
Chapter 30 — Taxes, Tax Losses and Deferred Taxes in DCF 186
Chapter 31 — Sector Primers: What Drives Value in Fourteen Industries 191
Chapter 32 — The Limits of DCF: Critiques, Behaviour and Judgement 199
Chapter 33 — Growth, Competitive Advantage and the Value of ROIC 204

Appendices 211
Appendix A — Formula Sheet 212
Appendix B — Practice Problems with Solutions 215
Appendix C — Glossary 219

Appendix D — Discount Factor Tables 223
Appendix E — Building the Model: A Spreadsheet Blueprint 225
Appendix F — Answers to Numerical Review Questions 228
Appendix G — Further Practice Problems with Solutions 230
Appendix H — Short Answers to Chapter Review Questions 233
Appendix I — Projects and Exercises for Self-Study 244
Index of Terms 248

Preface

Every investment decision, every acquisition, every capital-budgeting exercise and every listing rests on a single question: *what is this worth?* Discounted cash flow (DCF) valuation is the most complete answer finance has to that question. Its logic is a single sentence: **an asset is worth the cash it will generate for its owners, adjusted for the timing and the risk of that cash.** Everything else in this book — free cash flows, betas, terminal values, bridges from enterprise to equity — is a way of making that sentence precise and usable.

DCF has a curious reputation. To its admirers it is the only rigorous valuation method, the one that forces an analyst to state explicitly what she believes about growth, profitability, reinvestment and risk. To its critics it is "garbage in, garbage out", a spreadsheet that can be tuned to deliver any number its author desires. Both views contain truth. DCF is rigorous in structure and fragile in inputs. The craft of valuation lies in disciplining the inputs: tying growth to reinvestment, tying margins to competitive advantage, tying discount rates to the risk of the cash flows actually being valued, and tying the whole story to evidence.

This book is written for three kinds of reader:

- **Students** preparing for professional examinations (CA, CFA, CMA, MBA finance) who need a clear, structured treatment with worked examples and practice problems.
- **Analysts and finance professionals** who build DCF models and want a reference for the many judgement calls that textbooks gloss over.
- **Business owners, founders and managers** who must understand what a valuation report is saying, and what it is *not* saying.

How the book is organised

Part I – Foundations (Chapters 1–3) explains what valuation is, reviews the mathematics of discounting, and sets out the architecture of a DCF: enterprise versus equity, FCFF versus FCFE, and the consistency rules that keep a model honest.

Part II – Cash Flows (Chapters 4–8) shows how to turn financial statements into free cash flows, how to forecast, and how to normalise accounting data so that it reflects economic reality.

Part III – Discount Rates and Terminal Value (Chapters 9–12) covers the cost of equity, beta, the cost of debt, WACC and the terminal value — which, in most valuations, carries more than half of the answer.

Part IV – Building and Testing the Valuation (Chapters 13–16) takes the model from enterprise value to a value per share, works a complete case study end to end, and teaches sensitivity analysis, scenarios, simulation and cross-checks.

Part V – Special Situations (Chapters 17–24) adapts DCF to leveraged capital structures (APV), financial institutions, cyclical firms, start-ups, distressed companies, emerging-market and Indian contexts, real options, ESG, and finally to the question of professional practice and common errors.

Part VI – Applications and Extensions (Chapters 25–33) puts the tools to work: mergers and acquisitions and synergy valuation, private companies and disputes, a second full case study (a listed pharmaceutical company), a library of six compact valuations, a fully integrated three-statement model that reconciles FCFF with FCFE, taxes and tax losses, sector primers for fourteen industries, a chapter on growth, competitive advantage and the value of ROIC, and a closing chapter on the limits of DCF.

Appendices provide formula sheets, two sets of practice problems with full solutions, a glossary, discount-factor tables, a spreadsheet blueprint and answers to the numerical review questions.

The running case

Throughout the book we follow a fictional company, **Meridian Foods Ltd**, a packaged-foods manufacturer. Its numbers are introduced in Chapter 3 and developed chapter by chapter until the complete valuation in Chapter 14, and returns in Chapters 25 and 29. All figures are in ₹ crore unless stated. Meridian is entirely invented; any resemblance to a real company is coincidental, and the figures are for teaching, not investment advice.

A note on numbers

Worked examples are rounded for readability, so a spreadsheet rebuild may differ in the last decimal. Every market input (risk-free rate, equity risk premium, tax rate) is an *illustration*, not a forecast; source current inputs when you value a real business.

How to Use This Book

If you are new to valuation, read Part I carefully, then Chapters 4, 5, 9, 10, 11 and 12, then the case study in Chapter 14. Work the practice problems in Appendix B before reading further.

If you build models for a living, read the "Pitfalls" boxes in each chapter, Chapters 8, 12, 13 and 15, and Chapter 24. They are where most real-world valuation errors occur.

If you are revising for an examination, use the chapter summaries and key-formula boxes, the formula sheet in Appendix A, and the practice problems. Examination questions typically test (i) computing free cash flow from a set of data, (ii) computing WACC and un-levering/re-levering beta, (iii) terminal value, and (iv) going from enterprise value to equity value.

Conventions. Formulas are set in bold within the text. *Key idea* boxes summarise the central point of a section. *Pitfall* boxes flag common errors. *Worked example* boxes give fully-solved numerical illustrations. At the end of each chapter, a short list of *review questions* tests recall.

Part I — Foundations

Chapter 1 — What Is Valuation?

1.1 Value, price and cost

Three words are used loosely in everyday speech and must be kept apart in valuation.

- **Cost** is what was paid to acquire or build an asset. It is a historical fact.
- **Price** is what an asset trades for in a market today. It is an observation.
- **Value** is what an asset is *worth* given the cash flows it will produce and the risk of those flows. It is an estimate — a judgement.

A company's accounting book value is a cost-based number (adjusted for depreciation). Its market capitalisation is a price. Its DCF value is an estimate of worth. The three coincide only by accident. The whole discipline of investing is built on the possibility that price and value diverge; the whole discipline of valuation is the attempt to estimate the second so as to compare it with the first.

> **Key idea.** Price is what you pay; value is what you get. A valuation is a *judgement about value,* supported by evidence, expressed with an honest range.

1.2 Why value a business?

Valuations are performed for many reasons, and the purpose shapes the method, the standard of value and the level of precision required.

Purpose	Typical question	Notes
Mergers & acquisitions	What price should we offer / accept?	Often includes synergies; control premium; negotiated range
Public offerings (IPO/FPO)	What is a defensible issue price?	Cross-checked against peer multiples; investor appetite matters
Private equity / venture capital	What entry price delivers the target return?	Exit-driven; leverage assumptions
Corporate strategy	Which businesses create value? Where do we deploy surplus cash?	Value-based management; segment-level DCF
Capital budgeting	Does this project earn more than its cost of capital?	Project-level DCF = NPV
Financial reporting	Fair value, impairment testing, purchase price allocation	Standards-driven (Ind AS 113, Ind AS 36, Ind AS 103)
Tax and legal	Gift/estate, share transfer, shareholder disputes, oppression and mismanagement	Prescribed standards of value apply

Purpose	Typical question	Notes
Lending / restructuring	Is collateral and enterprise value sufficient?	Liquidation vs going-concern values
Employee equity	Option strike price, ESOP accounting	Fair value on grant date
Dispute resolution	Dissenting shareholder appraisal	Courts examine method and inputs

1.3 Standards of value

"Value" has no single meaning in practice; professional standards define specific *standards of value*. The common ones are:

- **Fair market value:** the price at which property would change hands between a willing buyer and a willing seller, neither being under compulsion and both having reasonable knowledge of relevant facts. Used widely in tax and legal contexts.
- **Fair value (financial reporting):** the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date (Ind AS 113 / IFRS 13). Market-participant assumptions, not entity-specific ones.
- **Fair value (shareholder-rights context):** in statutory appraisal and oppression cases, courts may define fair value differently — typically *without* minority or marketability discounts.
- **Investment value:** value to a particular investor given that investor's own synergies, cost of capital and plans. This can differ from fair market value.
- **Intrinsic value:** the value implied by a rational analysis of fundamentals — the present value of expected cash flows. DCF produces an estimate of intrinsic value.
- **Liquidation value:** net proceeds from selling assets piecemeal (orderly or forced) after settling liabilities.

Pitfall. Always state the standard of value and the premise of value (going concern or liquidation) at the beginning of any valuation. Two valuers can reach different numbers for the same company simply because they were answering different questions.

1.4 The three families of valuation approach

Almost every valuation technique falls into one of three families.

1.4.1 Asset (cost) approach

The value of the business is built from the value of its assets less its liabilities. Variants: book value, adjusted net asset value (revaluing assets to market), replacement cost, and liquidation value. This approach is natural for holding companies, real-estate firms, investment funds and companies facing liquidation. It is a poor fit for businesses whose value lies in intangibles and future earnings.

1.4.2 Market (relative) approach

The value is inferred from what the market pays for similar assets, using multiples such as price-to-earnings (P/E), enterprise-value-to-EBITDA (EV/EBITDA), price-to-book (P/B) or EV/Sales. Either *trading multiples* of listed peers or *transaction multiples* from acquisitions are used. It is fast, intuitive and reflects current market sentiment — a strength and a weakness. It tells you what similar assets are *priced* at, not necessarily what they are worth.

1.4.3 Income approach (including DCF)

Value is the present value of the future economic benefits the asset will produce. DCF is the central technique: forecast cash flows, estimate a risk-appropriate discount rate, and discount. Simpler variants include capitalisation of maintainable earnings (a one-period DCF with constant growth).

Approach	Looks at	Strength	Weakness
Asset	Balance sheet	Objective, floor value	Ignores earning power and intangibles
Market	Peer prices	Reflects market conditions; simple	Peers are never identical; inherits market mispricing
Income / DCF	Future cash flows	Fundamental, explicit assumptions	Sensitive to inputs; terminal value dominance

Good practice is to use DCF as the primary tool where cash flows can be forecast, and to cross-check with relative valuation and, where relevant, asset values.

1.5 Intrinsic versus relative valuation

Intrinsic valuation asks: what are the cash flows, how risky are they, and what are they worth? It is independent of what the market currently thinks. **Relative valuation** asks: how does the market price comparable assets, and what does that imply for this one? The two are complementary.

A subtle point: relative valuation is not "assumption-free". A multiple *embeds* assumptions about growth, risk and returns. A high P/E signals the market expects high growth, low risk, or both. When you apply a peer's multiple to your company you are assuming your company's growth/risk/returns match the peer's. DCF makes those assumptions explicit; multiples hide them.

1.6 The logic of DCF in one page

Suppose you could buy a business that will pay its owner a stream of cash C1, C2, C3, … at the ends of years 1, 2, 3, …. What is the most you should pay today? Three observations answer the question.

1. **Cash today is worth more than cash tomorrow.** You could invest cash today and earn a return. So future cash flows must be *discounted*.
2. **Riskier cash is worth less than safer cash.** Investors demand compensation for bearing risk. The discount rate must therefore rise with risk.
3. **A business is worth the cash it can pay out, not the accounting profit it reports.** Profit is an accrual concept; owners are paid in cash. What counts is cash left after the business has reinvested enough to sustain its growth.

Combining these:

Value = Σ [Expected cash flow in year t ÷ (1 + r)^{t}], where r reflects the risk of those cash flows.

Everything in the remaining chapters elaborates this expression: *which* cash flow (to the firm or to equity), *which* discount rate (WACC or cost of equity), *how long* to forecast, *what happens after* the forecast ends (terminal value), and *how to bridge* from the value of operations to what belongs to shareholders.

1.7 The five-step DCF process

1. **Understand the business and its industry.** Competitive position, economics, regulatory environment, management quality.
2. **Forecast free cash flows.** Revenue, margins, taxes, reinvestment — for an explicit period (typically 5–10 years).
3. **Estimate the discount rate.** Cost of equity, cost of debt, capital structure → WACC (or cost of equity for equity cash flows).

4. **Estimate the terminal value.** What the business is worth beyond the forecast horizon.
5. **Discount, sum, and bridge** to equity value (subtract debt and other claims, add non-operating assets), then **test** with sensitivities and cross-checks.

1.8 Valuation is both science and art

The *science* is the mathematics: discounting, algebra of WACC, un-levering betas. The *art* is judgement: how fast will this business grow, will its margins hold, how long can it earn excess returns, which peers are truly comparable? Two analysts with the same spreadsheet will produce different values because they hold different views of the future. A good valuation makes these views explicit, internally consistent and defensible. It also says plainly how much of the answer depends on them.

> **Pitfall.** False precision. A DCF output of ₹51.27 per share does not mean the value is known to the paisa. It means the value is about ₹50 given a specific set of assumptions; a different but equally reasonable set might give ₹42 or ₹63. Always present a range.

1.9 What can go wrong — a preview

The most common failures in DCF practice, each treated in depth later, are:

- Growth assumed without the reinvestment needed to fund it (Chapter 7).
- Terminal growth above the long-term growth of the economy, or terminal reinvestment inconsistent with terminal returns (Chapter 12).
- Mismatched cash flows and discount rates — for instance discounting FCFE at WACC (Chapter 3).
- Mismatched currencies — rupee cash flows discounted with a dollar rate (Chapter 22).
- Double-counting or omitting items in the equity bridge — cash, minorities, pensions, options (Chapter 13).
- Beta and capital-structure inconsistencies (Chapters 10–11).
- Confusing accounting profit with cash flow, ignoring working capital and capex (Chapters 4–5).

Chapter 1 summary

- Cost, price and value are different things; DCF estimates value.
- Three approaches exist — asset, market, income. DCF is the main income-approach tool.
- Always specify the standard and premise of value.
- DCF = present value of expected cash flows at a risk-appropriate rate.
- It is part science, part judgement; present a range, not a point.

Review questions

1. Distinguish between cost, price and value, with an example of each from a listed company.
2. List three purposes of valuation and say how each might affect the chosen method.
3. Why is relative valuation not free of assumptions?
4. What are the five steps in a DCF valuation?
5. Why might two competent valuers reach different DCF values for the same company?

Chapter 2 — The Mathematics of Discounting

DCF is built on a handful of formulas. They are simple, but the habits of using them correctly — consistent timing, consistent rates, awareness of compounding — separate clean models from broken ones. This chapter is a self-contained toolkit.

2.1 Compounding and discounting

If you invest ₹100 at 10% per year, after one year you have ₹110, after two ₹121, after three ₹133.10. In general:

Future value: $FV = PV \times (1 + r)^n$

Running the logic backwards, the present value of a sum to be received in n years is:

Present value: $PV = FV \div (1 + r)^n$

The quantity $1 \div (1 + r)^n$ is the **discount factor**. At r = 10%, the discount factors for years 1 to 5 are 0.9091, 0.8264, 0.7513, 0.6830 and 0.6209. The *rate* r is called the **discount rate**, the **required return** or the **opportunity cost of capital** — different names for the same thing seen from different angles.

> **Worked example 2.1.** A debenture will pay ₹1,000 in 4 years. At a required return of 9%, its present value is $1{,}000 \div 1.09^4 = 1{,}000 \div 1.4116 =$ **₹708.43**.

Frequency of compounding

If interest is compounded m times a year at nominal rate r, the effective annual rate is $\mathbf{(1 + r/m)^m - 1}$. For 12% nominal compounded monthly the effective rate is $(1.01)^{12} - 1 = 12.68\%$. In valuation, always convert to an *effective annual* rate before discounting annual cash flows. Continuous compounding uses e^{rt}, occasionally seen in option-pricing and some theoretical work.

2.2 Net present value (NPV)

For a set of cash flows CF_0 (today), CF_1, …, CF_n:

$NPV = CF_0 + CF_1/(1+r) + CF_2/(1+r)^2 + \ldots + CF_n/(1+r)^n$

An investment that costs ₹500 today and returns ₹200, ₹250 and ₹300 at the ends of years 1–3, at 12%, has:

NPV = −500 + 200/1.12 + 250/1.2544 + 300/1.4049 = −500 + 178.57 + 199.30 + 213.53 = **₹91.40**.

Decision rule: accept projects with positive NPV; reject negative. NPV measures the value created (in today's rupees) over and above the return required by investors. A DCF valuation of a company is an NPV calculation in which the "investment" is the price you would pay.

2.3 Internal rate of return (IRR)

The IRR is the discount rate at which NPV = 0. For the project above, IRR solves $-500 + 200/(1+i) + 250/(1+i)^2 + 300/(1+i)^3 = 0$. Trial and error (or a calculator) gives IRR ≈ **23.2%**. Since this exceeds the 12% required return, the project is acceptable.

Strengths: intuitive percentage figure; no need to know the discount rate to compute it. Weaknesses:

- **Multiple IRRs** when cash flows change sign more than once.
- **Reinvestment assumption:** IRR implicitly assumes interim cash is reinvested at the IRR itself, which is rarely realistic. (NPV assumes reinvestment at the discount rate.)
- **Scale blindness:** a small project can have a high IRR but low NPV.

For ranking mutually exclusive projects use NPV. In valuation, IRR is used for *investor return* analysis (e.g., "what IRR does a buyer earn at this price?"), not for estimating value.

2.4 Perpetuities and annuities

Many valuation shortcuts rely on closed-form sums.

Perpetuity (level cash flow forever)

PV = C ÷ r

A security paying ₹80 a year forever, at 10%: PV = ₹800.

Growing perpetuity (the Gordon growth formula)

If the first cash flow arrives one year from now and then grows at constant rate g forever:

PV = $C_1 \div (r - g)$, valid only if $r > g$.

Derivation. PV = $C_1/(1+r) + C_1(1+g)/(1+r)^2 + C_1(1+g)^2/(1+r)^3 + \ldots$ is a geometric series with first term $C_1/(1+r)$ and ratio $(1+g)/(1+r) < 1$. Its sum is $[C_1/(1+r)] \div [1 - (1+g)/(1+r)] = C_1/(r - g)$.

This formula, the backbone of terminal-value calculation, deserves two warnings. First, it is extremely sensitive to the gap $(r - g)$: when r = 11% and g = 5% the multiplier is 1/0.06 = 16.7; when g rises to 6% the multiplier is 1/0.05 = 20 — a 20% increase in value for a one-point change in growth. Second, it assumes the cash flow C_1 is *consistent* with growth g (growth needs reinvestment; see Chapter 12).

Ordinary annuity (level cash flow for n years)

PV = $C \times [1 - (1+r)^{-n}] \div r$

An annuity of ₹100 for 5 years at 10%: $100 \times (1 - 0.6209)/0.10$ = **₹379.08**.

Growing annuity

PV = $C_1 \times [1 - ((1+g)/(1+r))^n] \div (r - g)$

An annuity beginning at ₹100 growing 5% for 10 years at 12%: $100 \times [1 - (1.05/1.12)^{10}]/0.07 = 100 \times (1 - 0.5156)/0.07$ = **₹692.0** (approximately).

2.5 Timing conventions

2.5.1 End-year convention

The simplest convention treats each year's cash flow as arriving on the last day of the year. This understates value for a business whose cash arrives evenly through the year.

2.5.2 Mid-year convention

Cash flows accrue roughly evenly through the year, so on average half a year earlier. Discount factor for year t becomes $1/(1+r)^{t-0.5}$. The mid-year convention raises value by about $(1+r)^{0.5} - 1 \approx r/2$ — around 5% at an 11% discount rate. Most professional valuations use mid-year discounting for operating businesses.

Terminal value under mid-year discounting. If TV is computed by the Gordon formula (cash flows continuing mid-year), it is discounted with the *same* mid-year factor as the final-year cash flow $(n - 0.5)$. If TV is

computed by an exit multiple applied to year-n EBITDA (assumed sale at year-end), it is discounted with the end-year factor (n). This asymmetry is a frequent source of error.

2.5.3 Stub periods

If the valuation date is not a fiscal-year end — say you are valuing on 31 December with a March fiscal year — the first forecast period is a *stub*: the remaining three months. Take only the fraction of year-1 cash flow that falls after the valuation date (25% here), discount it for half the stub length (mid-point), and discount later years by stub + (t − 1) years, again adjusted for mid-year.

> **Worked example 2.2 (stub).** Valuation date 31 Dec 2026; FY ends 31 Mar. FY2027 FCFF = ₹400 cr, FY2028 = ₹460 cr, r = 11%. Stub fraction for FY2027 = 0.25 → cash flow = 100, mid-point at 0.125 years → factor $1/1.11^{0.125}$ = 0.9869 → PV = 98.69. FY2028 cash flows at mid-point 0.25 + 0.5 = 0.75 years → factor $1/1.11^{0.75}$ = 0.9247 → PV = 425.4. The pattern continues for later years.

2.6 Real versus nominal; currency consistency

Nominal cash flows include expected inflation; **real** cash flows are stated at constant prices. A rule that must never be broken: *nominal cash flows are discounted at nominal rates; real cash flows at real rates.* The relationship is the Fisher equation:

(1 + nominal) = (1 + real) × (1 + inflation)

At nominal 11% and inflation 5%, real = 1.11/1.05 − 1 = 5.71%.

The same principle applies to **currency**: cash flows in rupees need a rupee discount rate; cash flows in dollars, a dollar rate. The two rates differ by approximately the expected inflation differential (interest-rate parity). Chapter 22 discusses this for Indian valuations.

2.7 Present value of a growing stream with changing growth

Real companies do not grow at one rate forever. A common structure is a *two-stage* model: explicit growth for n years, then stable growth. Value = PV of explicit-period cash flows + PV of terminal value.

For instance, a stream beginning at ₹100 (year 1), growing at 15% for 5 years, then at 5% forever, with r = 11%:

- Year 1–5 cash flows: 100, 115, 132.25, 152.09, 174.90.
- Discount factors at 11%: 0.9009, 0.8116, 0.7312, 0.6587, 0.5935.
- PV of explicit flows: 90.09 + 93.34 + 96.70 + 100.18 + 103.81 = **484.12**.
- Year 6 cash flow = 174.90 × 1.05 = 183.65; TV at year 5 = 183.65/(0.11 − 0.05) = 3,060.8; PV = 3,060.8 × 0.5935 = **1,816.6**.
- Total = **₹2,300.7**.

Notice that the terminal component is 79% of the total. This dominance of terminal value is typical and returns as a theme in Chapter 12.

2.8 Discount factors and the yield curve

A single discount rate for all maturities is a simplification. Strictly, a cash flow at year t should be discounted at the spot rate for maturity t (taken from the government zero-coupon curve) plus a risk premium. In practice, valuers use one rate because (i) the yield curve is relatively flat over the relevant range, (ii) risk premiums matter more than term structure, and (iii) the spread of estimation error in the premium dwarfs curve effects. In unusual conditions — a steeply inverted curve, or when valuing a bond-like asset — consider term-structure-consistent discounting.

2.9 Common spreadsheet implementations

- **PV(rate, nper, pmt, fv)** values a stream of equal payments.
- **NPV(rate, values)** discounts a range assuming the first value is at the end of period 1 (a frequent trap: the initial outlay at time 0 must be added *outside* the function).
- **XNPV(rate, values, dates)** discounts irregular dates using actual days/365 — the best tool for stub periods and mid-year timing.
- **IRR / XIRR** compute internal rates; supply a guess if there are sign changes.

> **Pitfall.** Excel's `NPV` function treats the first cell as period 1, not period 0. If you pass it a range starting with today's investment, every cash flow is discounted one period too many.

2.10 Practice drill

Consider cash flows of ₹150, ₹180, ₹210, ₹240 and ₹260 crore at the ends of years 1–5, followed by growth of 4% forever, at a discount rate of 10%.

1. Discount factors: 0.9091, 0.8264, 0.7513, 0.6830, 0.6209.
2. PV of explicit flows: 136.36 + 148.76 + 157.77 + 163.92 + 161.43 = **768.24**.
3. TV = 260 × 1.04 ÷ (0.10 − 0.04) = 4,506.7; PV = 4,506.7 × 0.6209 = **2,798.3**.
4. Total value = **₹3,566.5 crore**. Terminal value share ≈ 78%.

Chapter 2 summary

- PV = FV/(1+r)^n; NPV sums discounted flows; IRR is the rate that sets NPV to zero.
- Growing perpetuity: $C_1/(r - g)$ with $r > g$ — powerful and dangerous.
- Use mid-year discounting for operating businesses; mind the stub period.
- Keep nominal/real and currency consistent between cash flows and discount rates.
- In a typical two-stage model, the terminal value is the majority of total value.

Review questions

1. Derive the growing-perpetuity formula from the geometric series.
2. Why does Excel's NPV function often produce a wrong DCF value if used naively?
3. A TV is computed by the Gordon formula in a mid-year model. At which period is it discounted? What if it is computed by an exit multiple?
4. Convert a nominal rate of 12% with 6% inflation to a real rate.
5. Explain why IRR can mislead when ranking projects.

Chapter 3 — The Architecture of a DCF

Before computing anything, a DCF requires a decision about *whose* cash flow is being valued. This choice determines the discount rate, the treatment of debt, and the final bridge to per-share value. Get it wrong and the numbers, however carefully calculated, are meaningless.

3.1 Claims on a business

A business's operating cash flows are claimed by several parties.

Claimant	Receives	Typical form
Lenders	Interest and principal	Contractual; senior
Preference shareholders	Fixed dividend	Contractual-ish; senior to equity
Equity shareholders	Residual cash flow	Dividends, buybacks; unlimited upside
Minority shareholders in subsidiaries	Share of subsidiary profits	Claim on part of consolidated cash flows
Government	Taxes	Paid before distributions to equity

Valuation can focus on *all* claimants together (the "firm" or "enterprise") or only on equity holders. Both are legitimate; they must not be mixed.

3.2 Free cash flow to the firm (FCFF)

FCFF is the cash flow generated by operations available to *all* providers of capital — debt and equity — after paying taxes on operating profit and after reinvestment in the business. It is independent of capital structure (apart from the effect of taxes, handled in the discount rate).

FCFF = EBIT × (1 − t) + Depreciation & amortisation − Capital expenditure − Increase in net working capital

Equivalently: **FCFF = NOPAT − Net reinvestment**, where net reinvestment = capex − depreciation + ΔNWC.

FCFF is discounted at the **weighted average cost of capital (WACC)**. The result is the **enterprise value (EV)** — the value of operating assets. To reach equity value, subtract debt and other non-equity claims and add non-operating assets (Chapter 13).

3.3 Free cash flow to equity (FCFE)

FCFE is the cash flow left for equity holders after the firm has paid taxes, met reinvestment needs and serviced debt (interest and net principal).

FCFE = Net income + D&A − Capex − ΔNWC + Net borrowing

(where net borrowing = new debt issued − debt repaid).

Equivalently, **FCFE = FCFF − Interest × (1 − t) + Net borrowing**.

FCFE is discounted at the **cost of equity**, and the result is directly the **equity value**. No bridge for debt is needed, because debt service has already been deducted from the cash flows.

3.4 Dividend discount model (DDM)

The DDM discounts *dividends* actually paid at the cost of equity:

$$\text{Equity value} = \Sigma\ \text{Dividend}_t \div (1 + Ke)^t$$

The DDM is a special case of FCFE valuation: when the firm pays out all FCFE as dividends, they are identical. When dividends are less than FCFE (cash accumulates) the DDM understates value unless the retained cash is treated carefully; when dividends exceed FCFE (the firm borrows or issues shares to pay) it overstates. The DDM works well for stable, high-payout firms and for financial institutions where FCFE is hard to define.

3.5 Which to use? FCFF versus FCFE

Situation	Preferred model	Reason
Stable leverage; clear debt ratio	Either	Should give consistent answers
Leverage changing materially	FCFF (with changing WACC) or APV	FCFE requires continuous modelling of debt flows
Highly leveraged or distressed	FCFF/APV	Equity is a residual of a residual; noisy
Financial institutions	FCFE / DDM	Debt is operating raw material; FCFF undefined
Firm with negative FCFE but positive FCFF	FCFF	FCFE negative due to debt paydown misleads
Multiple business units	FCFF by unit	Different risks, different WACCs

FCFF is the workhorse of professional practice for non-financial firms because it separates the *operating* valuation from the *financing* decision, and because it does not require forecasting the debt schedule in detail.

3.6 The consistency rules

A DCF has internal consistency requirements that are easy to state, easy to violate and often decisive for the result.

1. **Cash flow ↔ discount rate.** FCFF → WACC. FCFE → Ke. Dividends → Ke.
2. **Cash flow ↔ claim.** FCFF → EV (then subtract debt etc.). FCFE → equity value (do *not* subtract debt).
3. **Cash flow ↔ currency and inflation.** Nominal cash flows with nominal rate in the same currency.
4. **Growth ↔ reinvestment.** Growth in operating income requires reinvestment (Chapter 7). A terminal growth rate must carry matching terminal reinvestment.
5. **Discount rate ↔ risk of cash flows.** Use a rate that reflects the *business risk* of the cash flows, with leverage consistent with the capital structure assumed.
6. **Tax treatment ↔ tax rate.** Use operating tax (on EBIT) for FCFF; the tax shield on interest enters via after-tax cost of debt in WACC. Don't double-count by also reducing FCFF for interest.
7. **Cash and non-operating assets ↔ income.** If interest income on cash is excluded from FCFF (as it should be), add the *value* of cash and investments separately.
8. **Timing ↔ convention.** Mid-year cash flows with mid-year discounting; the valuation date matches the balance-sheet date used for the bridge.

> **Pitfall.** The classic violation: discounting FCFE at WACC "to be conservative". The WACC is lower than the cost of equity, so this *overstates* value. FCFE at WACC is wrong in both directions of intuition.

3.7 The structure of a DCF model

A professional DCF workbook typically has these modules:

1. **Inputs and assumptions sheet** — growth, margins, tax rate, working-capital days, capex ratio, discount-rate inputs, scenario switches.
2. **Historical analysis** — three to five years of financial statements reformulated into operating and financing components; ratios (ROIC, margins, turnover).

3. **Forecast** — revenue, operating costs, EBIT, tax, NOPAT; reinvestment (capex, depreciation, working capital).
4. **Free cash flow** — FCFF for each year.
5. **Discount-rate module** — cost of equity (CAPM), cost of debt, weights, WACC (by year if leverage changes).
6. **Terminal value** — Gordon and (as cross-check) exit multiple.
7. **Valuation and bridge** — EV, adjustments, equity value, per-share value.
8. **Sensitivity and scenarios** — data tables on WACC × g and key operating drivers.
9. **Checks** — balance sheet balances, cash reconciles, implied multiples and returns are reasonable.

3.8 Introducing Meridian Foods Ltd

Our running case is a packaged-foods manufacturer with the following history and attributes. All figures are ₹ crore.

Business. Meridian makes branded biscuits, ready-to-cook mixes and snacks, sold through distributors and modern trade across India. It operates five plants and has strong brands in several regional categories.

Latest financials (base year, FY2026).

Item	₹ crore
Revenue	2,000
EBITDA	340
Depreciation & amortisation	100
EBIT	240 (margin 12.0%)
Interest expense	54
Profit before tax	186
Tax @ 25%	46.5
Net income	139.5

Balance sheet (31 March 2026).

Assets	₹ cr	Liabilities & equity	₹ cr
Net fixed assets	900	Equity (share capital + reserves)	1,100
Net working capital (operating)	200	Borrowings (all debt)	600
Goodwill & intangibles	100		
Long-term investments (non-operating)	100		
Cash & equivalents	150		

Assets	₹ cr	Liabilities & equity	₹ cr
Other operating assets	250		
Total	**1,700**	**Total**	**1,700**

Operating invested capital = net fixed assets 900 + operating NWC 200 + intangibles 100 + other operating assets 250 = **₹1,450 crore**. As a cross-check, equity 1,100 + debt 600 − cash 150 − non-operating investments 100 = 1,450. Base-year NOPAT is 240 × 0.75 = 180, so the base-year return on invested capital is about 12.4%.

Shares outstanding: **50 crore**. Face value ₹10.

Strategy. Management plans to grow revenue by 14%, 12%, 10%, 9% and 8% over the next five years (premiumisation, distribution expansion, new categories), with EBIT margin rising from 12.0% to 14.5% on operating leverage and mix. Every ₹1 of additional revenue requires ₹0.50 of additional invested capital (a sales-to-capital ratio of 2.0). The corporate tax rate is 25%.

Market inputs. Risk-free rate (10-year government bond) 7.0%; equity risk premium 6.0%; industry unlevered beta 0.75; target debt-to-equity 25% (20% debt-to-capital); pre-tax cost of debt 9%.

These inputs will be justified, challenged and refined in Chapters 4–13; the final valuation appears in Chapter 14.

3.9 Looking ahead

The next five chapters concentrate on cash flows: how to read statements through a valuation lens, how to compute FCFF and FCFE, how to forecast them, and how to clean accounting data so that the forecast starts from truth rather than from reported noise.

Chapter 3 summary

- FCFF belongs to all capital providers → discount at WACC → enterprise value → bridge to equity.
- FCFE belongs to equity → discount at Ke → equity value directly.
- DDM is a special case of FCFE when payout equals FCFE.
- Consistency rules (cash flow ↔ discount rate ↔ claim ↔ currency ↔ growth ↔ reinvestment) keep a model honest.
- Meridian Foods Ltd is the running example for Parts II–IV.

Review questions

1. State the formula for FCFF and explain what it represents.
2. Why does discounting FCFE at WACC overstate value?
3. For which kinds of firms is FCFE or DDM preferred and why?
4. List four consistency rules a DCF must obey.
5. Compute FCFE for a firm with net income 120, depreciation 40, capex 90, ΔNWC 15, new debt 30, repayments 10.

Part II — Cash Flows

Chapter 4 — Reading Financial Statements for Valuation

A statutory balance sheet and income statement are prepared for compliance, creditors and tax. Valuation needs them reorganised around a different question: *what does the business do operationally, and how is it financed?* This chapter teaches that reorganisation and the ratios that follow from it.

4.1 Why reformulate?

A published income statement mixes operating items (sales, cost of goods, depreciation) with financing items (interest expense, interest income) and non-recurring items. A published balance sheet mixes operating assets (plant, inventory, receivables) with non-operating ones (surplus cash, investments in other companies). A DCF values the *operating business*, so these must be separated:

- Operating profit → NOPAT → FCFF → enterprise value.
- Financing items → capital structure → discount rate.
- Non-operating assets → valued separately and added in the bridge.

4.2 Operating versus financial items

Operating assets are used to generate revenue: property, plant and equipment; intangible assets used in operations; inventories; trade receivables; prepaid expenses; operating cash needed to run the business.

Operating liabilities arise from operations and carry no explicit interest: trade payables, accrued expenses, advances from customers, provisions for warranties, deferred revenue.

Financial assets and liabilities: cash beyond operating need, marketable securities, loans receivable (financial assets); bank loans, debentures, finance leases, preference shares (financial liabilities).

Non-operating assets are held for reasons other than the core business: investments in unconsolidated associates, idle land, surplus properties, loans to related parties.

How much cash is operating? Some cash is needed to run the business (cash in tills, minimum balances). A common treatment is to treat all cash above, say, 1–2% of revenue as excess. Practitioners vary; the key is consistency between the cash treated as operating in forecasts and the cash added in the bridge.

4.3 The invested-capital identity

From the operating side:

Invested capital = Net fixed assets + Net working capital + Goodwill and intangibles + Other operating assets − Other operating liabilities

From the financing side:

Invested capital = Equity + Debt (incl. leases) + Preference + Minorities − Cash and non-operating investments

The two sides must agree. For Meridian: operating side = 900 + 200 + 100 + 250 = 1,450; financing side = 1,100 + 600 − 150 − 100 = 1,450. ✔

Invested capital is the base on which returns are measured and on which reinvestment builds.

4.4 NOPAT

NOPAT = EBIT × (1 − operating tax rate)

NOPAT is what operating profit would be if the company had no debt: the profit available to *all* capital providers. For Meridian: EBIT 240 × (1 − 0.25) = **180**.

Which tax rate? Use the rate applicable to operating profit. Candidates: the statutory marginal rate (25.17% for most Indian domestic companies under the concessional section 115BAA regime; around 34.94% under the older regime with surcharge and cess), or the effective rate (cash taxes ÷ PBT). In the near term, forecasts should reflect actual expected cash taxes, including any carried-forward losses and incentives; in the terminal period, drift toward the marginal rate because deferrals and incentives cannot last forever.

4.5 Return on invested capital (ROIC)

ROIC = NOPAT ÷ Invested capital (usually opening or average).

ROIC measures how efficiently the business turns capital into operating profit. It decomposes into two components:

ROIC = NOPAT margin × Capital turnover = (NOPAT ÷ Revenue) × (Revenue ÷ Invested capital).

Meridian: NOPAT margin = 180/2,000 = 9.0%; capital turnover = 2,000/1,450 = 1.38×; ROIC = 9.0% × 1.38 = 12.4%.

Comparing ROIC with WACC tells whether the business creates value: if ROIC > WACC, each rupee of growth *adds* value; if ROIC < WACC, growth *destroys* it. Meridian's ROIC 12.4% versus WACC ≈ 11.2% implies a small positive spread today; the plan to raise margins is what lifts ROIC toward 15%.

> **Key idea.** Growth creates value only when new capital earns more than its cost. A DCF forecast that has revenue growing rapidly while ROIC stays below WACC is a recipe for value destruction, however good the revenue story sounds.

4.6 Working capital

Operating net working capital (NWC) = (Inventory + Trade receivables + Other operating current assets) − (Trade payables + Other operating current liabilities).

Exclude cash, short-term debt and current portions of long-term debt (financing items). Include only what the business needs for operations.

Metrics:

- **Receivable days** = Receivables ÷ Revenue × 365
- **Inventory days** = Inventory ÷ Cost of goods sold × 365
- **Payable days** = Payables ÷ Cost of goods sold (or purchases) × 365
- **Cash conversion cycle** = Receivable days + Inventory days − Payable days.

A growing company ties up more cash in working capital; a shrinking one releases it. For forecasting, model NWC as days or as a percentage of revenue, and compute the *change* each year (Chapter 7).

4.7 Fixed assets, capex and depreciation

Capex = Closing net fixed assets − Opening net fixed assets + Depreciation (for tangible assets, ignoring revaluations and disposals). Compare capex to depreciation:

- **Capex / Depreciation > 1:** the business is investing for growth (or inflation of asset costs).
- **Capex / Depreciation ≈ 1:** maintenance mode.
- **Capex / Depreciation < 1:** harvesting; shrinking the asset base (sustainable only temporarily).

Meridian's depreciation is 100 on opening net fixed assets of roughly 860 (about 11.6%). A steady-state reinvestment will keep capex above depreciation in line with growth and inflation.

4.8 Leases and Ind AS 116

Under Ind AS 116 (and IFRS 16) most leases are on the balance sheet as a right-of-use asset and lease liability. The income statement shows depreciation on the right-of-use asset and interest on the lease liability instead of rent. Consequences for valuation:

- EBITDA is *higher* than under the old standard (rent no longer deducted). EBIT is only slightly affected.
- Lease liabilities are **debt** (a financing claim).
- FCFF is *before* lease payments (lease principal and interest are financing flows), and **EV includes the lease claim**; so lease liability must be subtracted in the bridge.
- WACC should include lease debt in capital structure and cost of debt.

The alternative is to treat leases as operating: subtract lease payments as an operating cost, exclude lease liability from debt, and use EBITDA *after* rent. Either is valid *if applied consistently*, including in any multiples used for cross-checks. Mixing (EBITDA pre-rent with EV excluding leases) is a classic error.

4.9 Intangibles, R&D and brand spending

Accounting treats most internally generated intangibles (R&D, marketing building a brand, training) as expenses, not assets. This understates both invested capital and NOPAT, since the spending is really an investment.

For firms where such spending is material (pharma, software, FMCG advertising), analysts sometimes *capitalise* R&D or brand spending, amortise it over an estimated useful life, and add the unamortised balance to invested capital and the net addition to operating income. Doing so makes ROIC more meaningful and brings reinvestment closer to economic reality. It must be done for both history and forecast, and consistently with the terminal value.

4.10 Non-recurring and non-operating items

Strip from operating profit anything that will not recur or does not belong to operations:

- Gains/losses on sale of assets or investments.
- Restructuring and impairment charges (but consider: a company with *recurring* "one-off" charges has recurring costs).
- Government grants and tax refunds, if one-time.
- Fair-value gains on investments, exchange gains on borrowings.
- Share of profit of associates (value the associate separately).
- Exceptional items disclosed under Ind AS.

Judgement point. Do not remove costs merely because they are lumpy. Management's habit of "adjusting" repeated expenses (restructuring every year) inflates apparent profitability. A useful test: if the item has occurred in three of the last five years, treat it as recurring.

4.11 Common-size analysis and trend analysis

Express each line as a percentage of revenue across years: gross margin, employee cost %, selling and distribution %, EBITDA margin, depreciation %, EBIT margin. Changes reveal operating leverage, mix shifts, input-cost pressure. Cross-sectional comparison against peers highlights whether margins are above or below the industry and why.

Meridian	FY2024	FY2025	FY2026
Revenue growth	11.0%	12.5%	13.1%
Gross margin	41.0%	41.8%	42.5%
EBITDA margin	15.0%	16.1%	17.0%
D&A / revenue	5.0%	5.0%	5.0%
EBIT margin	10.0%	11.1%	12.0%
Capex / revenue	8.0%	7.5%	7.2%

Meridian	FY2024	FY2025	FY2026
NWC / revenue	10.5%	10.2%	10.0%
ROIC	10.8%	11.6%	12.4%

The trend: improving gross margin and operating leverage, stable working-capital intensity, capex intensity trending down as the capacity build-out matures. These patterns are the starting point for forecasts.

4.12 Quality-of-earnings checks

Before relying on historical numbers, test them.

1. **Cash conversion.** Compare operating cash flow with EBITDA/net income over several years. Persistent shortfall suggests aggressive revenue recognition or working-capital build-up.
2. **Accruals.** Large, growing accruals (profit ahead of cash) are a warning sign.
3. **Receivables versus sales.** Receivables growing faster than revenue may signal channel stuffing or collection problems.
4. **Capitalised costs.** Rising capitalised development or software costs may flatter profit.
5. **Related-party transactions.** Pricing at non-arm's-length can distort margins.
6. **Auditor's qualifications and emphasis of matter.** Read the notes.
7. **Tax rate.** Unusually low effective tax may reflect temporary benefits.

4.13 Segment information

When a company has several businesses with different growth, margins and risk, value them separately (sum-of-the-parts), each with its own WACC and cash flows, and subtract unallocated corporate costs. Segment disclosures (Ind AS 108) give revenue, profit and assets by segment — enough for a first cut. Where segments are not separable, value the company as a whole but blend the risk profile.

4.14 Putting it together: Meridian's reformulated statements

Income statement (₹ crore, FY2026).

Revenue	2,000.0

Operating costs excluding D&A	(1,660.0)
EBITDA	340.0
D&A	(100.0)
EBIT	**240.0**
Tax on EBIT @ 25%	(60.0)
NOPAT	**180.0**
Interest expense	(54.0)
Tax shield on interest	13.5
Net income	**139.5**

NOPAT 180 − after-tax interest 40.5 = 139.5 ✔. Reformulation does not change net income, it only reveals the operating and financing layers.

Chapter 4 summary

- Separate operating from financing and non-operating items.
- Invested capital can be built from either side of the balance sheet; the two must tie.
- NOPAT = EBIT × (1 − t); ROIC = NOPAT ÷ invested capital = margin × turnover.
- Working capital, capex and depreciation are the bridge from profit to cash.
- Leases, intangibles and non-recurring items require deliberate, consistent treatment.
- Test earnings quality before extrapolating history.

Review questions

1. Why is interest income on surplus cash excluded from FCFF?
2. Compute ROIC given EBIT 300, tax rate 25%, invested capital 1,800.
3. How do Ind AS 116 leases change EBITDA, debt and EV?
4. When is it legitimate to adjust earnings for "one-off" items?
5. What does capex/depreciation of 0.7 for several years imply?

Chapter 5 — Free Cash Flow to the Firm

FCFF is the central input of enterprise DCF. This chapter derives it from several starting points, shows that they agree, and works through the practical complications: tax, working capital, capex, and non-cash items.

5.1 The definition

FCFF = NOPAT − Net investment

where

Net investment = (Capex − Depreciation) + Increase in operating NWC + Increase in other operating assets (e.g., capitalised intangibles).

Equivalently, with all items shown:

FCFF = EBIT(1 − t) + D&A − Capex − ΔNWC

The ingredients correspond to a simple story. The business earns operating profit; taxes are paid on it; some profit is "non-cash" (depreciation) and is added back; the business must then reinvest in fixed assets and working capital to sustain and grow. What is left is *free* — available to pay lenders and shareholders without hurting the business's operating capacity.

5.2 Four ways to reach FCFF

Because accounting identities link the statements, FCFF can be computed from several starting points. A professional model computes at least two as a check.

5.2.1 From EBIT

FCFF = EBIT(1 − t) + D&A − Capex − ΔNWC

5.2.2 From EBITDA

FCFF = EBITDA(1 − t) + t × D&A − Capex − ΔNWC

(The depreciation tax shield t × D&A is the only cash effect of depreciation.)

5.2.3 From net income

FCFF = Net income + Interest × (1 − t) + D&A − Capex − ΔNWC

(Add back after-tax interest because interest is a payment to a capital provider, so it belongs in FCFF. If there are non-cash charges beyond D&A, add them back; if there are non-operating gains, remove them.)

5.2.4 From cash flow from operations (CFO)

FCFF = CFO + Interest × (1 − t) − Capex

(Under Ind AS, interest paid may be classified in financing; adjust accordingly. CFO already reflects ΔNWC and non-cash items.)

Meridian FY2026 illustration. Suppose last year's capex was 144 (7.2% of revenue) and NWC rose by 20 (NWC grew from 180 to 200).

- From EBIT: 180 + 100 − 144 − 20 = **116**.
- From EBITDA: 340 × 0.75 + 0.25 × 100 − 144 − 20 = 255 + 25 − 144 − 20 = **116** ✔.
- From net income: 139.5 + 54 × 0.75 + 100 − 144 − 20 = 139.5 + 40.5 + 100 − 144 − 20 = **116** ✔.

All three agree. The business generated 116 of free cash for lenders and shareholders; it paid interest of 54 (40.5 after tax shield), leaving 75.5 for equity before any net borrowing.

5.3 Taxes in FCFF

FCFF uses **taxes on operating profit** — what the firm would pay if it were unlevered:

Operating taxes = EBIT × t

Not the tax actually paid. Actual taxes are lower because interest is deductible. The interest tax shield is real but it is captured in the *after-tax cost of debt* in WACC, not in FCFF. Counting it in both places double-counts it.

Which rate, again? Three candidates:

1. **Statutory marginal rate** — appropriate for the terminal period and for the tax shield of debt.
2. **Effective rate** — total tax expense ÷ PBT; reflects permanent differences and incentives. Good for history.

3. **Cash tax rate** — taxes actually paid ÷ PBT; reflects timing differences (deferred taxes). Appropriate for the near term where deferrals are expected to reverse slowly.

A sensible approach: near-term forecasts use cash tax guidance (with any MAT credit or loss carry-forward modelled explicitly); the rate converges linearly to the marginal rate by the terminal year.

Tax losses and credits

If a firm has accumulated tax losses, the present value of the tax savings is an asset. Either (a) model the reduction in cash taxes year by year in FCFF, or (b) value the tax shield separately and add it in the bridge. Do not do both.

Deferred taxes

Deferred tax liabilities that will reverse (timing differences) are not a debt-like claim if they will persist as the business grows (a perpetual deferral). Where reversal is expected, treat them as a cash outflow in the forecast or deduct their present value. The key is to *decide once* how deferred tax is handled and apply it consistently in FCFF and in the bridge.

5.4 Depreciation and its tax shield

Depreciation is a non-cash expense; adding it back is correct because the cash left the business in the year of the capex. But depreciation matters in two ways:

1. It reduces taxable profit (tax shield).
2. It provides a benchmark for maintenance capex.

If depreciation (accounting) differs materially from tax depreciation (written-down-value method under the Income-tax Act), the cash tax differs from book tax. A rigorous model separates the two.

Sustainable relationship. In steady state with growth g, capex ≈ depreciation × (1 + something). If assets last L years and grow at rate g, the steady-state ratio of capex to depreciation exceeds 1 and approximates $(1 + g)^{L/2}$ for straight-line depreciation. For a 10-year-life, 5% growth firm this gives about 1.28. A terminal capex equal to depreciation is therefore not consistent with growth.

5.5 Working capital in detail

ΔNWC is the *year-on-year increase* in operating net working capital. An increase is a cash outflow.

ΔNWC = NWC_t − NWC_(t−1)

Using Meridian's NWC of 10% of revenue: with revenue growing from 2,000 to 2,280 (Year 1), NWC rises from 200 to 228, ΔNWC = 28 — a cash outflow.

Which items belong? Keep the definition consistent between history and forecast. Exclude cash, short-term borrowings, current maturities of long-term debt, dividends payable and current tax payable (although many treat tax payable as operating). Include inventory, receivables, prepaid expenses, payables, accrued expenses, customer advances.

Working-capital seasonality. If the balance sheet date falls at a seasonal low or high, the closing NWC misrepresents the average; adjust or use a year-average.

5.6 Capex: maintenance versus growth

Total capex = maintenance capex (to replace worn assets and sustain current output) + growth capex (to expand capacity).

- **Maintenance capex** is roughly depreciation adjusted for inflation in asset replacement costs.
- **Growth capex** relates to capacity additions needed for revenue growth.

For forecasting, either (a) model capex directly from capacity plans, or (b) tie reinvestment to revenue growth via a sales-to-capital ratio (Chapter 7). Method (b) automatically links growth to investment, helping avoid unrealistic growth-without-capex forecasts.

5.7 Other adjustments to FCFF

- **Stock-based compensation (ESOPs).** An expense that reduces EBIT; adding it back as non-cash overstates FCFF unless dilution is also deducted from value. The sound approach: treat SBC as a real operating cost (leave it deducted) and do not add it back. Dilution from *existing* outstanding options is handled in the bridge (Chapter 13).

- **Provisions** (bad debts, warranty): operating items. Changes are part of working capital or of other operating liabilities. Non-cash provisions reduce profit without cash outflow; if the provision is a true accrual of a future cash outflow, it is correct to treat it as a cost.
- **Operating lease payments** — see Chapter 4.
- **Pension and post-retirement costs** — operating service cost belongs in EBIT; the unfunded deficit is debt-like (bridge).
- **Foreign-exchange gains/losses on operating items** — retain if recurring; remove if one-off or financing-related.

5.8 From FCFF to claims: reconciliation

FCFF equals what is paid to or retained for capital providers:

FCFF = Interest × (1 − t) + Net debt repayment + Dividends + Buybacks − New equity issued + Increase in cash

(roughly; with proper treatment of the tax shield). This identity is a useful check: the sources and uses of cash must tie.

5.9 The first-year FCFF for Meridian

Revenue grows 14% to 2,280. EBIT margin 12.5% → EBIT = 285.0. NOPAT = 285.0 × 0.75 = 213.75. Reinvestment = (Revenue increase 280) ÷ sales-to-capital 2.0 = 140. FCFF = 213.75 − 140 = **73.75**.

Cross-check by components: capex − depreciation + ΔNWC must equal 140. With NWC at 10% of revenue, ΔNWC = 28; so net capex (capex − depreciation) = 112. If depreciation is 5% of revenue = 114, capex = 226. That is 9.9% of revenue — significant, reflecting the growth phase.

The model has not forecast capex separately; it has only imposed a *total* reinvestment consistent with growth. Later we will break it down.

₹ crore	FY2026A	FY2027E
Revenue	2,000.0	2,280.0
EBIT	240.0	285.0
NOPAT	180.0	213.75
Net reinvestment	—	140.0
of which ΔNWC	—	28.0
of which net capex	—	112.0
FCFF	—	73.75

5.10 Typical errors in FCFF

1. **Using book tax instead of tax on EBIT.** Understates tax and double-counts the interest shield.
2. **Adding back all non-cash items blindly.** Some "non-cash" items (ESOP expense, provisions) represent real economic costs.
3. **Omitting working-capital build** in a high-growth forecast.
4. **Capex below depreciation** in the terminal year with positive growth.
5. **Including interest income** on surplus cash in EBIT and also adding the cash in the bridge (double counting).
6. **Including cash flows from non-operating assets** (associates) in FCFF *and* valuing them separately.
7. **Inconsistent treatment of leases** across EBITDA, FCFF and debt.

5.11 Exercises

Exercise 5.1. EBIT = 500; tax rate 30%; depreciation 120; capex 200; NWC rose from 300 to 345. Compute FCFF.

Solution. NOPAT = 350; ΔNWC = 45; FCFF = 350 + 120 − 200 − 45 = **225**.

Exercise 5.2. From the same company: net income = 260; interest = 80. Check that FCFF from net income gives 225.

Solution. 260 + 80 × 0.7 + 120 − 200 − 45 = 260 + 56 + 120 − 200 − 45 = 191. This does **not** equal 225 — so the data are inconsistent: NOPAT of 350 implies EBIT − tax = 350, so net income should be 350 − 56 = 294 (not 260). Always reconcile: (EBIT − interest) × (1 − t) = (500 − 80) × 0.7 = 294. This is how a cross-check exposes an input error.

Chapter 5 summary

- FCFF = NOPAT − net investment; NOPAT uses tax on EBIT; the interest shield is in WACC.
- FCFF can be derived from EBIT, EBITDA, net income or CFO; use two methods to check.
- Net investment = net capex + ΔNWC (+ other operating assets).
- Be explicit about ESOPs, leases, pensions, deferred taxes and tax losses.

Review questions

1. Why is the interest tax shield excluded from FCFF?
2. Derive the EBITDA-based FCFF formula from the EBIT-based one.
3. What is the steady-state relation between capex and depreciation under growth?
4. How should ESOP cost be treated in FCFF?
5. A company's FCFF is positive but its net income is negative. Is this possible? Give an example.

Chapter 6 — Free Cash Flow to Equity and the Dividend Discount Model

Chapter 5 valued the whole firm. This chapter values the *equity directly*. FCFE and DDM matter for banks and financial companies, for firms with stable leverage, for dividend-paying blue chips, and as a check on FCFF valuations.

6.1 Defining FCFE

FCFE = Net income + D&A − Capex − ΔNWC + Net borrowing

This is the cash that could be paid to shareholders *after* everything else — operations, reinvestment, debt service — has been taken care of. It is a *potential* dividend. Actual dividends may be lower or higher.

Net borrowing = new debt raised − principal repaid. If the firm maintains a constant debt-to-capital ratio, then net borrowing is a fixed fraction of net reinvestment, and the formula simplifies:

FCFE = Net income − (1 − δ) × Net reinvestment

where δ is the debt ratio applied to *new* investment (the share of reinvestment funded by debt) and net reinvestment = capex − D&A + ΔNWC. The equity funds only the (1 − δ) fraction.

Example

Net income 150, capex 200, depreciation 80, ΔNWC 30. Net reinvestment = 200 − 80 + 30 = 150. If 40% of net reinvestment is debt-funded: FCFE = 150 − 0.6 × 150 = **60**. Check using the long formula: 150 + 80 − 200 − 30 + 60 (net borrowing = 0.4 × 150) = 60 ✔.

6.2 The FCFE valuation

Equity value = Σ $FCFE_t$ ÷ $(1 + Ke)^t$ + Terminal value ÷ $(1 + Ke)^n$

where the terminal value is **$FCFE_{n+1}$ ÷ (Ke − g)**. Ke is the cost of equity, *consistent with the leverage assumed in the forecast*. If leverage changes during the forecast, Ke changes too (through re-levered beta).

The valuation gives equity value directly; subtract nothing for debt. Add the value of non-operating assets *only if* their cash flows are not in net income (i.e., if investment income is excluded).

6.3 Terminal FCFE and the sustainable growth rate

Terminal-year FCFE must be consistent with stable growth. In stable growth, expected growth in net income is:

g = Retention ratio × Return on equity — the fundamental growth equation.

If a firm retains 40% of earnings and earns 15% ROE, g = 6%. FCFE in the terminal year, with payout ratio = 1 − g/ROE:

Terminal FCFE = Net income × (1 − g/ROE)

(for a firm that maintains constant leverage, so that the equity is financing exactly the retained fraction of net income). This is the FCFE counterpart of the FCFF formula FCFF = NOPAT × (1 − g/ROIC) of Chapter 12.

6.4 The dividend discount model

The DDM discounts *actual or expected dividends*:

$P_0 = D_1/(1+Ke) + D_2/(1+Ke)^2 + \ldots$

6.4.1 Gordon growth form

For a stable firm: **$P_0 = D_1 \div (Ke - g)$**

Example. D_1 = ₹40; Ke = 12%; g = 6%. P_0 = 40/0.06 = ₹666.7. If the market price is ₹800, the market either expects a higher g, lower Ke, or both. Solving for the implied Ke at P = 800 and g = 6%: Ke = 40/800 + 0.06 = 11%.

6.4.2 Two-stage DDM

Dividends grow at g_1 for n years, then at the stable rate g_2:

$P_0 = \Sigma_{t=1}^{n} D_0(1+g_1)^t/(1+Ke)^t + [D_(n+1)/(Ke - g_2)] \div (1+Ke)^n$

Example. D_0 = 10; g_1 = 15% for 5 years; g_2 = 5%; Ke = 12%. Dividends: 11.5, 13.23, 15.21, 17.49, 20.11. Discount factors at 12%: 0.8929, 0.7972, 0.7118, 0.6355, 0.5674. PVs: 10.27, 10.55, 10.83, 11.12, 11.41 → sum **54.18**. D_6 = 20.11 × 1.05 = 21.12; TV = 21.12/0.07 = 301.7; PV = 301.7 × 0.5674 = 171.2. **$P_0 \approx$ ₹225.4.**

6.4.3 When DDM fails

- Firms that don't pay dividends or pay far less than FCFE: DDM undervalues unless retained cash is valued separately. The remedy is FCFE.
- Firms that pay out more than they earn (borrowing to pay): DDM overstates.
- Firms with highly irregular dividends.
- Firms in which control or buyback policy changes the payout.

6.5 Financial institutions

For banks, NBFCs and insurers, "debt" is the raw material of the business: deposits and borrowings fund loans, and interest expense is an operating cost. FCFF and enterprise value are therefore ill-defined. Valuation focuses on equity: either DDM or an *excess return model* in which **Value of equity = Book equity + PV of expected excess returns (ROE − Ke) × book equity**. The regulatory capital requirement means cash flow to equity is limited: FCFE = Net income − retained earnings needed to keep regulatory capital ratios as assets grow. Chapter 18 develops this fully.

6.6 FCFE versus FCFF: the reconciliation

A well-built model should produce the same equity value both ways, within sampling noise, if the assumptions are consistent. The reasons they often don't:

1. **Different debt assumptions.** FCFF/WACC implicitly assumes a target capital structure, maintained through time. FCFE models the *actual* debt schedule. If debt is paid down while WACC keeps assuming a constant ratio, the two methods diverge.
2. **Inconsistent cost of equity.** Ke used in the FCFE model must correspond to the equity beta at the modelled leverage; the WACC's embedded Ke uses target leverage.
3. **Inconsistent terminal growth or reinvestment.**
4. **Different treatment of the cash balance.**

Differences of a few percent are normal; large gaps indicate an error. Reconciling them is one of the best audits of a DCF.

6.7 Worked example: Meridian as an FCFE model

Take Meridian with year-1 net income. Interest on 600 debt at 9% = 54 (assume debt grows with the business at 20% of capital).

Year 1: EBIT 285.0. Debt rises from 600 to ... (we hold D/V at 20% of *invested capital*, so net borrowing = 20% × net reinvestment = 0.2 × 140 = 28). Interest at 9% on average debt ≈ 0.09 × 614 ≈ 55.3; round to 55.0. PBT = 230.0; tax 25% = 57.5; NI = 172.5.

FCFE = NI − (1 − δ) × net reinvestment = 172.5 − 0.8 × 140 = **60.5**. Cross-check with FCFF: FCFF 73.75 − interest after tax (55 × 0.75 = 41.25) + net borrowing 28 = **60.5** ✔.

The full FCFE valuation, discounting at Ke = 12.34%, gives a result close to the FCFF value; the small difference is due to the simplified debt schedule. We leave the detail to the exercises.

6.8 Dividends, buybacks and payout

Total *cash returned* to shareholders = dividends + buybacks. Using dividends alone understates what shareholders receive when buybacks are significant. The "augmented" DDM uses total payout. For Indian companies, buyback tax and dividend tax regimes affect both firm and shareholder; a valuation at the company level typically ignores shareholder-level taxes unless the standard of value requires them.

6.9 Relating the models

Model	Cash flow	Discount rate	Output
FCFF	To all capital providers	WACC	Enterprise value
FCFE	To shareholders (potential)	Ke	Equity value
DDM	Dividends paid	Ke	Equity value
Excess return	Book value + PV of excess returns	Ke	Equity value

Under consistent assumptions, all four give the same equity value. They are different windows onto the same house.

Chapter 6 summary

- FCFE = net income − (1 − δ) × net reinvestment, where δ is the debt share of new investment.
- Discount FCFE at Ke; the result is equity value directly.

- Stable-growth FCFE = NI × (1 − g/ROE).
- DDM is the special case of FCFE paid out; it fails when payout ≠ FCFE.
- Banks and NBFCs are valued with equity-based models.
- FCFE and FCFF should reconcile; gaps expose inconsistencies.

Review questions

1. Compute FCFE: NI 200; depreciation 60; capex 140; ΔNWC 20; borrowing 50; repayment 30.
2. Why must the cost of equity be consistent with leverage in an FCFE model?
3. If ROE is 18% and g is 6%, what payout ratio is sustainable?
4. When is DDM inappropriate?
5. Why are FCFF models ill-suited to banks?

Chapter 7 — Forecasting: Growth, Margins and Reinvestment

A DCF is only as good as its forecast. Yet forecasting is where valuations most often go wrong — not through arithmetic, but through stories that are internally inconsistent. This chapter builds a disciplined forecasting framework in which growth, profitability and reinvestment are tied together.

7.1 The forecasting philosophy

Three principles guide good forecasts.

1. **Start from the outside in.** Begin with the market: size, growth, competitors, regulation. Then position the company within it. A forecast that has a company growing faster than its market indefinitely must explain whose share it is taking.
2. **Tie the three pillars together.** Growth in operating income comes from revenue growth, margin change, or both. Revenue growth requires reinvestment. Margin and capital efficiency determine how much value the growth creates. Never forecast these independently.
3. **Let assumptions fade toward the sustainable.** Competitive forces pull growth toward the economy's rate, margins toward industry norms and returns toward the cost of capital — except where durable advantages justify otherwise. The forecast should show *how and when* the convergence occurs.

7.2 The explicit forecast period

How many years should the explicit forecast cover? The answer depends on how long the company's current growth and margins will differ from steady state.

Company type	Typical explicit period
Mature, stable (utilities, staples)	5 years
Moderate growth	5–7 years
High-growth or margin-expansion story	10 years or more
Start-ups, turnarounds	10–15 years, with a fade

A short explicit period forces the terminal value to carry heavy weight and to embed non-steady-state assumptions; a very long period invites spurious precision. Many practitioners use five years of detailed forecast plus a *fade* period (years 6–10) in which growth and margins interpolate to terminal values.

7.3 Forecasting revenue

7.3.1 Top-down

Revenue = Market size × Market share. For Meridian: the Indian packaged-foods categories in which it operates are expected to grow roughly 9–10% in nominal terms; Meridian's growth above that (14% in year 1) reflects share gains and category expansion, which should *decline* toward the market rate. That is why the plan has growth fading from 14% to 8%.

7.3.2 Bottom-up

Revenue = Volume × Price, by segment, channel or geography. E.g., number of outlets × average throughput × price per unit. Bottom-up forecasts expose capacity constraints and price/volume trade-offs.

7.3.3 Cross-checks

- Compare forecast revenue growth with historical growth, industry growth and peer growth.
- Compute implied market share and check plausibility.
- Check absolute levels: ₹3,300 crore in five years — does the distribution reach and plant capacity support it?
- Check that revenue growth in the fade years converges toward nominal GDP.

7.4 Forecasting margins

Operating margin = Gross margin − (Selling, general & administrative ÷ Revenue) etc. Forecast margin by cost line, or directly as EBIT margin with justification.

Sources of margin change:

- **Operating leverage.** Fixed costs spread over a larger revenue base raise margin as sales grow. Estimate the fixed/variable split.

- **Mix.** Shift toward higher-margin products or channels.
- **Pricing power and input costs.** For FMCG: commodity costs (wheat, palm oil, packaging), pass-through ability and lag.
- **Scale and efficiency programmes.** Procurement, automation.
- **Competitive intensity.** Margin gains are typically competed away over time unless there is a moat.

For Meridian the plan has EBIT margin rising from 12.0% (FY2026) to 14.5% (FY2031), a gain of 2.5 percentage points over five years. A sanity check: leading packaged-food peers earn EBIT margins of 15–20%; Meridian at 14.5% remains below them, so the target is plausible, though not guaranteed.

> **Pitfall.** *Hockey-stick margins.* Forecasts often show margins flat for two years and then jumping. Unless a specific event justifies a step change, margin paths should be smooth and attributable to identified drivers.

7.5 Forecasting reinvestment

Reinvestment is the cash the business must put back to create the forecast growth. There are three ways to forecast it, each of which should lead to similar answers.

7.5.1 Line-by-line

Forecast capex, depreciation and working capital separately. Capex comes from plant capacity needs; depreciation from the asset schedule; working capital from days or percentage of revenue. Rigorous but requires detailed data.

7.5.2 Sales-to-capital ratio

Reinvestment = ΔRevenue ÷ (Sales-to-capital ratio)

The sales-to-capital ratio measures incremental revenue per rupee of incremental invested capital. It is the marginal counterpart of capital turnover. If Meridian generates ₹2.00 of additional revenue for each ₹1.00 of net new investment (net capex plus ΔNWC), the sales-to-capital ratio is 2.0.

Historical check: average capital turnover is 1.38× (2,000 ÷ 1,450), lower than the marginal 2.0, because the *marginal* investment is expected to be more efficient than the existing stock of capital (which includes goodwill

and older, costlier assets). Is 2.0 plausible? Compare history: over the last three years revenue rose from about 1,416 to 2,000 (+584), while invested capital rose from 1,100 to 1,450 (+350), a marginal ratio of 1.67. Meridian's plan assumes a *better* marginal ratio (2.0) than it has achieved recently. This is an aggressive assumption we will test in sensitivity analysis. For now, we adopt it as management's plan; a more cautious analyst would use 1.6–1.8.

7.5.3 Reinvestment rate and the growth equation

Growth in NOPAT = Reinvestment rate × ROIC

where the reinvestment rate = Net investment ÷ NOPAT. If a firm reinvests 40% of NOPAT at an ROIC of 15%, NOPAT grows 6% a year. Conversely, to grow at 6% with ROIC 15% the firm needs a 40% reinvestment rate. This identity is the single most useful consistency check in forecasting.

> **Key idea.** *Growth is not free.* For given ROIC, more growth requires more reinvestment, leaving less FCFF. A valuation that raises growth without raising reinvestment is simply creating value from nothing.

7.6 The expected growth identity applied to Meridian

Year 1: NOPAT growth = (213.75 ÷ 180) − 1 = 18.75%. This exceeds revenue growth (14%) because the margin also expands. The reinvestment rate is 140 ÷ 213.75 = 65.5%. Growth fed by margin expansion is not the same as growth fed by reinvestment, and the growth identity (g = reinvestment rate × ROIC) applies to growth that comes from *new capital*; margin gains are a one-off lift that must eventually stop. In the terminal state, a reinvestment rate of 33.3% at an ROIC of 15% gives g = 5% (Chapter 12). ✔

7.7 Scenario-based forecasting

Rather than one forecast, build several scenarios, each internally coherent, and attach probabilities.

Scenario	Revenue CAGR (5 yrs)	FY2031 EBIT margin	Sales-to-capital	Probability
Bear	7%	12.0%	1.5	25%
Base	10.6%	14.5%	2.0	50%
Bull	13%	16.0%	2.2	25%

A probability-weighted value is the expected value; the spread shows the uncertainty. Chapter 15 returns to this.

7.8 Building the Meridian forecast

Using the plan's assumptions:

₹ crore	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E
Revenue growth	13.1%	14.0%	12.0%	10.0%	9.0%	8.0%
Revenue	2,000.0	2,280.0	2,553.6	2,809.0	3,061.8	3,306.7
EBIT margin	12.0%	12.5%	13.0%	13.5%	14.0%	14.5%
EBIT	240.0	285.0	332.0	379.2	428.6	479.5
NOPAT (25% tax)	180.0	213.8	249.0	284.4	321.5	359.6
Net reinvestment	—	140.0	136.8	127.7	126.4	122.5
FCFF	—	**73.7**	**112.2**	**156.7**	**195.1**	**237.1**
Reinvestment rate	—	65.5%	54.9%	44.9%	39.3%	34.1%

Reinvestment is large early (66% of NOPAT) and falls as growth slows — typical for a company growing out of a heavy investment phase. FCFF rises more than threefold over five years even though NOPAT rises 1.7×, because the *reinvestment burden falls as growth fades*.

Cross-check of ROIC. Invested capital grows: 1,450 + 140 = 1,590 (FY27), 1,727 (FY28), 1,855 (FY29), 1,981 (FY30), 2,104 (FY31). ROIC on opening capital: FY27: 213.8/1,450 = 14.7%; FY28: 249.0/1,590 = 15.7%; FY29: 284.4/1,727 = 16.5%; FY30: 321.5/1,855 = 17.3%; FY31: 359.6/1,981 = 18.2%.

ROIC climbs to about 18%, above the 15% assumed in the terminal period, and that is a warning worth heeding. ROIC rises because margin expansion and a high sales-to-capital ratio coexist. In the terminal period we assume ROIC *fades* to 15% (still well above WACC 11.2%), a conservative convergence. If we believed Meridian could sustain 18%, terminal value would be higher. Document the decision either way.

7.9 The fade period

For many companies, a sharp drop from 8% growth to 5% growth at year 6 is unrealistic; a fade period smooths it. A simple 5-year fade would take growth from 8% to 5% in equal steps (7.4%, 6.8%, 6.2%, 5.6%, 5.0%) and margins from 14.5% toward a terminal level, and ROIC from 18% to 15%. The valuation effect is usually modest but the forecast is more defensible.

In the Meridian base case we keep the five-year explicit period for clarity and encode the fade in the terminal assumptions; Chapter 14 shows the fade variant as a sensitivity.

7.10 Forecasting with competitive-advantage logic

The length of time a company can earn ROIC above its cost of capital is its *competitive-advantage period*. Sources: brands, switching costs, network effects, scale, regulation, patents. Evidence: persistent high ROIC across a business cycle, pricing power, stable share. A DCF should explicitly link the assumed advantage period to these sources. For commoditised businesses with no moat, ROIC should converge to WACC after a short period; for strong brands it may stay elevated for decades.

7.11 Checklist for a defensible forecast

1. Revenue growth consistent with market growth and share.
2. Margin changes attributed to identified drivers.
3. Reinvestment consistent with growth (via sales-to-capital or line items).
4. ROIC trajectory stated and plausible.
5. Working capital days and capex intensity compared with history and peers.
6. Tax rate path stated.
7. Growth and ROIC fade toward steady state.
8. Scenarios and sensitivities identified.

Chapter 7 summary

- Forecast from the outside in; tie growth, margins and reinvestment together.
- Reinvestment = ΔRevenue ÷ sales-to-capital; growth = reinvestment rate × ROIC.
- Use fade periods and scenarios; avoid hockey-sticks.
- Check ROIC path against competitive-advantage logic.

Review questions

1. State the growth identity and use it to find the reinvestment needed for 7% growth at 14% ROIC.

2. Why must marginal sales-to-capital be compared with history?
3. How long should an explicit forecast period be for a stable consumer company? Why?
4. List three sources of margin expansion and a check on each.
5. What is a fade period and why is it used?

Chapter 8 — Normalising Earnings and Cleaning the Base Year

The base year is the platform on which every forecast stands. If it is distorted — by one-offs, acquisitions, accounting changes or cyclical peaks — every number above it inherits the distortion. This chapter covers the normalisation adjustments that professional valuers routinely make.

8.1 Why normalise?

Reported operating profit is a good starting point but rarely a *maintainable* level. Normalisation seeks the level of earnings that the business can sustain under normal conditions, free of unusual influences. The result is "maintainable EBIT" (or EBITDA), a central concept in income-approach valuation under Indian valuation practice as well as globally.

8.2 Categories of adjustments

8.2.1 Non-recurring items

Gains and losses from sale of property, one-time legal settlements, restructuring, impairments, insurance recoveries. Remove them from the base year; consider whether similar items occur regularly.

8.2.2 Non-operating items

Rental income from surplus property, dividend or interest income from investments, profits of associates. Remove from EBIT; value the underlying asset separately.

8.2.3 Accounting policy differences and changes

If the target uses a different depreciation policy or revenue-recognition approach from peers, adjust to a common basis. Where a standard changed during the historical window (e.g., adoption of Ind AS 116 in FY2020), restate history to be comparable.

8.2.4 Related-party and owner-related adjustments

In private and closely held companies, owner-managers may take salary above or below market, charge personal expenses to the company, or transact with related parties on non-arm's-length terms. Replace with

market-rate equivalents. In a valuation of an owner-managed business, a "replacement cost of management" is usually deducted.

8.2.5 Acquisitions and disposals

Pro-forma the base year to include a full year of acquired businesses (and exclude divested ones), so that growth is organic and margin is representative.

8.2.6 Inflation, currency and commodity effects

If the base year benefited from unusually favourable commodity prices or exchange rates, adjust to mid-cycle levels (see Section 8.5).

8.3 A worked example: normalising Meridian

Suppose Meridian's reported FY2026 profit statement included the following, in ₹ crore:

Item	Amount	Treatment
Reported EBIT	252.0	starting point
Gain on sale of a surplus warehouse	(14.0)	non-recurring, non-operating — remove
Restructuring cost: closure of an old plant	+6.0	one-off — add back
Insurance recovery for a fire	(3.0)	one-off — remove
Rental income on idle land (included in revenue)	(2.0)	non-operating — remove; value the land separately
Promoter-family salary above market	+4.0	related-party — normalise
Maintainable EBIT	**243.0**	

The result, 243, is close to but not identical with the 240 used in earlier chapters; our teaching case uses 240 for simplicity (as if the adjustments had already been made). In a real engagement the normalisation schedule is itself an exhibit in the valuation report, with evidence for each line.

8.4 Working-capital normalisation

If year-end working capital is unusually low (because of aggressive collection at year-end) or high (inventory build before a festival season), the base-year NWC percentage is unrepresentative. Use the average of monthly balances if available, or the three-year average ratio. The *change* in NWC matters in the first forecast year; a distorted base distorts year 1 cash flow.

8.5 Cyclical businesses: mid-cycle margins

For cyclical industries (metals, cement, shipping, capital goods, commodities) the base year may be at the peak or trough. Valuation based on a peak year overstates; a trough year understates. Normalise by:

1. **Averaging margins** over a full cycle (typically 7–10 years).
2. **Using mid-cycle commodity prices** (a price deck from forward curves or long-run cost curves).
3. **Modelling the cycle explicitly** — rising and falling for some years — and reverting to the mid-cycle in the terminal year.

Example. A steel producer earned EBIT margins of 6%, 14%, 22%, 17%, 4%, −3%, 8%, 18% over eight years. The average is 10.75%; use roughly 10–11% for the terminal year and normalise the base year accordingly. The terminal value should *never* be based on peak margins.

8.6 Capex normalisation

If a company underspent on maintenance capex in recent years, reported FCFF is flattering. Compare capex to depreciation and to peer capex intensity over the cycle; adjust to a sustainable level. Likewise, a one-off expansion year inflates capex; its growth benefit appears only in later revenue, so treat it as growth capex tied to specific revenue additions.

8.7 Normalising the tax rate

Reported effective tax rates include one-offs: deferred tax reversals, prior-period items, MAT credits, tax holidays. Normalise to a marginal rate in the terminal period; in the forecast, model known incentives with their expiry dates.

8.8 Stock-based compensation and employee costs

Cash salary is clearly an operating cost; stock compensation is also a cost, even though it involves no cash outlay. Do not add it back to EBIT. Failing to do so inflates profit and FCFF, particularly in technology firms where SBC may be 10–20% of revenue.

8.9 Adjustments for debt-like items

Normalisation also affects the bridge. Items that behave like debt and should be deducted from enterprise value include:

- Unfunded pension and gratuity obligations (post-tax).
- Lease liabilities (when treated as debt).
- Contingent liabilities that are probable and measurable.
- Deferred consideration for acquisitions.
- Overdue statutory dues, legal settlements payable.
- Redeemable preference shares and convertible instruments (see Chapter 13).

The principle: if the business will have to pay it with cash that is *not* in FCFF, it belongs in the bridge.

8.10 Documentation and sign-off

Each adjustment needs: (i) a source (annual report note, management certificate, ledger), (ii) a rationale, (iii) a quantification, (iv) a tax effect. Maintain a normalisation schedule in the workpapers. Under Indian valuation standards (ICAI Valuation Standards, Registered Valuers rules), the valuer must document the basis of each adjustment and its source.

8.11 Common mistakes

1. **Over-adjusting.** Adding back every unfavourable item and ignoring favourable ones.
2. **Under-adjusting.** Extrapolating a peak-year profit.
3. **Forgetting the tax effect** of each adjustment.
4. **Adjusting EBIT but not working capital or capex.**
5. **Not applying the same adjustments to peer data** used in multiples.

Chapter 8 summary

- Maintainable earnings, not reported earnings, form the base.
- Remove non-recurring and non-operating items; normalise owner-related and related-party items.
- For cyclical firms use mid-cycle margins.
- Normalise capex, working capital and tax.

- Capture debt-like items in the bridge; document each adjustment.

Review questions

1. List four adjustments to reported EBIT and say how each is treated in the bridge.
2. How would you normalise the margins of a cement company for a DCF?
3. Why must SBC not be added back?
4. Which items in a balance sheet are debt-like?
5. Why must tax effects be applied to each adjustment?

Part III — Discount Rates and Terminal Value

Chapter 9 — The Cost of Equity

The discount rate is the price of risk. Any error here compounds over every future cash flow and, through the terminal value, carries into infinity. This chapter develops the cost of equity — the return shareholders require — using the Capital Asset Pricing Model and its alternatives.

9.1 What the cost of equity is

The **cost of equity (Ke)** is the return that equity investors require for bearing the risk of the company's shares. It is an *opportunity cost*: what the investor could earn on other investments of equal risk. It is not an accounting number (dividend yield, ROE) and not a cost the company "pays" in cash; it is a required return, and the company must earn at least this on equity-financed investments to create value.

9.2 Risk and diversification

Total risk of a stock has two components. **Specific (idiosyncratic) risk** — the risk of a factory fire, a lost contract, a product failure — can be diversified away by holding many stocks. **Market (systematic) risk** — recessions, interest rate changes, commodity swings — cannot be diversified away. Finance theory says the market rewards only systematic risk, because investors can eliminate the rest for free. The cost of equity should therefore reflect only the risk that remains in a diversified portfolio.

9.3 The Capital Asset Pricing Model

Ke = Rf + β × (Rm − Rf)

- **Rf** — the risk-free rate.
- **β (beta)** — the sensitivity of the stock's returns to market returns; the measure of systematic risk.
- **(Rm − Rf)** — the equity risk premium (ERP): the extra return investors require for holding the market portfolio rather than risk-free securities.

For Meridian: Rf = 7.0%, ERP = 6.0%, levered beta 0.89 → Ke = 7.0% + 0.89 × 6.0% = **12.34%**.

Assumptions behind CAPM include investors holding diversified portfolios, one-period horizons, frictionless markets. These are imperfect, yet CAPM remains the dominant model in practice because it is simple, transparent

and gives a reasonable first approximation.

9.4 The risk-free rate

The risk-free rate is the return on an asset with no default risk and no reinvestment risk, in the same currency and with the same duration as the cash flows being valued.

Choosing it. For long-duration cash flows (a going concern), use the yield on a long-dated government bond, typically 10-year, and for India often a 10-year or 30-year G-Sec yield on the valuation date. The yield should be (i) in rupees if cash flows are in rupees, (ii) on the valuation date, and (iii) not distorted by temporary technical factors.

Default spread issue. A government that can default (or whose bond yields include a default spread) does not offer a truly risk-free rate in its own currency. Some practitioners subtract the sovereign default spread from the government bond yield to obtain a risk-free rate, then add a country risk premium to the ERP. For India, whose local-currency borrowing is rarely distressed, many valuers use the G-Sec yield as is. The approach must be consistent with the ERP choice (Section 9.5).

Normalised versus spot. In periods of extreme central-bank intervention, spot yields may be artificially low or high. Some valuers use a "normalised" rate (e.g., the long-run expected growth of nominal GDP). The rule: whatever is chosen, make sure *Rf + ERP* gives a sensible total expected return on the market.

9.5 The equity risk premium

The ERP is the most influential single parameter in the discount rate. A one-point change in ERP changes Ke by about β × 1 point. Estimation methods:

1. **Historical premium.** Average of (equity returns − government bond returns) over a long period. Problems: high standard error (even 50 years of data give an error of ±2%); survivorship bias; choice of bond maturity and averaging method (arithmetic vs geometric). Indian data suggest historical premia of roughly 5–8% depending on period.
2. **Implied premium.** Solve for the ERP that equates the present value of expected market cash flows with the current index level. This forward-looking premium responds to market conditions: lower when markets are expensive, higher after crashes.

3. **Survey-based.** Practitioner and academic surveys give a range (e.g., 5–7% for India).
4. **Mature market premium plus country risk premium.** For emerging markets: ERP = Mature market ERP (e.g., 4.5–5.5%) + Country risk premium (derived from sovereign default spread scaled up by relative equity-to-bond volatility). For India, the total ERP has recently been estimated in the neighbourhood of 7–8%, though professional bodies and valuers use differing figures.

For this book we use **6.0%** as an illustrative total ERP over a 7.0% risk-free rate, giving an expected market return of 13.0%. The Indian equity market's long-run nominal return has been broadly in this range.

> **Pitfall.** Mixing sources. Using a *dollar* ERP with a *rupee* risk-free rate, or a very high ERP with a low risk-free rate that was estimated in a different currency, yields an incoherent Ke.

9.6 Beta (preview)

Beta is estimated by regression of the stock's returns on market returns, or built bottom-up from comparable firms; Chapter 10 treats it fully. Beta values: market = 1.0; defensive sectors such as FMCG and pharma often 0.5–0.8; cyclicals, banks, metals, real estate often 1.1–1.5.

9.7 Adding other premia

9.7.1 Size premium

Historical evidence suggests small-cap stocks earned more than CAPM predicts. Many valuers add a size premium (e.g., 1–3% for small caps). Its theoretical justification is debated; its empirical stability has weakened. If used, state the data source and apply only where the company is truly small relative to its benchmark universe.

9.7.2 Company-specific risk premium (CSRP)

A premium for risks not captured by beta: key-person dependence, customer concentration, thin management, unproven technology. Theoretically unjustified for a diversified investor, but practically common in valuing private companies and start-ups (where investors are *not* diversified). Better practice is to reflect specific risks in **cash flow**

scenarios (probability-weighted forecasts) rather than as an arbitrary add-on to the discount rate, because an add-on conflates the risk of the *mean* cash flow with the *volatility* around it.

9.7.3 Illiquidity and marketability

Not a component of the discount rate in strict theory; handled as a discount to value in specific contexts (Chapter 13).

9.8 Alternatives to CAPM

9.8.1 Arbitrage Pricing Theory (APT) and multi-factor models

$$Ke = Rf + \beta_1 \lambda_1 + \beta_2 \lambda_2 + \ldots + \beta_k \lambda_k$$

where each factor (inflation, industrial production, interest-rate changes, credit spreads) has its own risk premium λ and the stock its own sensitivity β. APT is theoretically richer but needs the factors and premia to be specified and estimated, which is hard.

9.8.2 Fama–French three- and five-factor models

Add size, value (book-to-market), profitability and investment factors. Useful in academic work and in some valuation contexts; less common in standard practice because of data requirements and parameter instability.

9.8.3 Build-up method

Ke = Risk-free rate + ERP + Size premium + Industry premium + Company-specific premium. Common in private-company appraisals in the US and increasingly elsewhere; lacks an explicit beta and thus conflates risks, but is simple.

9.8.4 Dividend-yield-plus-growth / implied cost of equity

Back out Ke from the market price: $Ke = D_1/P_0 + g$. Useful as a sanity check for stable dividend payers.

9.9 Sanity-checking the cost of equity

A computed Ke should be tested:

1. **Relative to Rf.** Ke should exceed the risk-free rate by a margin consistent with business risk.
2. **Relative to the cost of debt.** Ke should exceed the pre-tax cost of debt (equity is riskier than debt). Meridian: Ke 12.34% vs Kd 9%.

3. **Relative to peers.** Compare with Ke used in broker reports and fairness opinions for similar companies.
4. **Relative to the return the market actually earns** (13% in our example): defensive companies below, cyclical above.

9.10 Worked example: building a cost of equity step by step

Value an unlisted regional dairy company on 31 March 2026.

1. Rf: 10-year G-Sec yield 7.0%.
2. ERP: 6.0% (total, for India).
3. Industry unlevered beta (food processors): 0.70. Dairy's target D/E: 30%. Tax 25%. Re-levered beta = 0.70 × [1 + 0.75 × 0.30] = 0.70 × 1.225 = 0.8575.
4. Ke (CAPM) = 7.0% + 0.8575 × 6.0% = 12.15%.
5. Size premium: the company has revenue of ₹300 crore; small-cap premium 1.5% (as per a chosen database). Ke = 13.65%.
6. CSRP: management concentration and a key-supplier dependency judged by the valuer to merit 1.0%. Ke = **14.65%**.

A review: is 14.65% reasonable? It is 5 percentage points above the 9.5% pre-tax cost of debt this small firm could achieve. That seems defensible. But note the 2.5% of add-ons are judgemental, and the report should show Ke without them (12.15%) as a reference and explain why they are used.

9.11 Cost of equity for different standards of value

For *fair value* in financial reporting, the discount rate is a *market participant* rate; a size premium and specific risks are included only if market participants would. For *investment value* to a specific buyer, the buyer's own required return applies. For a *minority stake*, some cost-of-equity add-ons may reflect the lack of control; for *control* value, the cash flows reflect control benefits. Always align discount rate and cash-flow assumptions with the standard of value.

Chapter 9 summary

- Ke is the return equity investors require; CAPM = Rf + β(ERP).
- Rf in the cash-flow currency, long maturity, on the valuation date.

- ERP: historical, implied or survey-based; add country risk for emerging markets.
- Size and specific premia are judgemental; prefer scenario-based treatment of specific risk.
- Sanity-check Ke against Rf, Kd, peers and market return.

Review questions

1. Why does CAPM reward only systematic risk?
2. Compute Ke with Rf 7%, ERP 6%, β 1.2.
3. What is the main argument against adding a company-specific risk premium?
4. List two ways to estimate the ERP and one weakness of each.
5. What checks would you apply to a computed Ke?

Chapter 10 — Beta: Estimation, Unlevering and Re-levering

Beta is the single input in CAPM that is specific to the company. It is also the most commonly mis-estimated. This chapter explains what beta measures, how it is estimated by regression, why regression betas are usually too noisy to trust on their own, and how the *bottom-up beta* method gives more reliable and more flexible answers.

10.1 What beta measures

Beta is the covariance of a stock's returns with the market's returns divided by the variance of the market's returns:

$$\beta = \text{Cov}(R_i, R_m) \div \text{Var}(R_m)$$

Beta says: when the market moves 1%, this stock tends to move β%. A beta of 1.0 is average market risk; below 1.0 is defensive; above 1.0 is aggressive. Beta does *not* measure total volatility, and it says nothing about whether a company is good.

10.2 Three determinants of beta

1. **The nature of the business — operating leverage and cyclicality.** Firms whose revenues swing strongly with the economy (autos, metals, construction) have higher business risk. Firms with high fixed costs magnify revenue swings into profit swings. Staples and healthcare have lower business risk.
2. **Financial leverage.** Debt raises equity risk because debt service is fixed: equity holders bear the volatility of operating profit *plus* the fixed claim.
3. **Discretionary versus necessary products.** Demand for essentials is stable; demand for luxuries is volatile.

Business risk and financial risk are separable, which is why the "unlevering" calculation below is possible.

10.3 Regression beta

The traditional estimate regresses periodic stock returns on index returns:

$$R_{stock,t} = \alpha + \beta \times R_{market,t} + \varepsilon_t$$

Choices: index (NSE Nifty 50 or Nifty 500, BSE Sensex or a broader index), return interval (daily, weekly, monthly), window (2, 3 or 5 years), adjustment (raw or Blume-adjusted). Typical practice for India is weekly returns over 2–5 years against the Nifty 50 or Nifty 500.

Problems with regression betas

- **High standard error.** A regression beta of 0.90 may have a standard error of 0.20, so a 95% range of roughly 0.5–1.3. This is too wide to settle between "defensive" and "aggressive".
- **Sensitivity to choices.** Switching from weekly to monthly returns or from 2 to 5 years can move beta by 0.2 or more.
- **Thin trading.** Illiquid stocks have betas biased downward because their prices do not respond to market moves immediately. Corrections exist (Dimson, Scholes–Williams).
- **Backward-looking.** It captures the firm's *past* business mix and leverage. If the firm has since changed, the beta is stale.
- **Index composition.** A concentrated index (a few heavy stocks) may be a poor proxy for the market portfolio.

Blume adjustment

Betas tend to revert toward 1 over time. The simplest adjustment: **Adjusted β = 0.67 × raw β + 0.33 × 1.0**. A raw beta of 0.60 becomes 0.73; a raw 1.50 becomes 1.34.

10.4 The bottom-up beta

A more reliable approach averages the betas of many comparable firms (reducing the noise, since standard error falls with the square root of the number of firms) after removing the effect of each firm's financial leverage, then re-applies the *target* firm's leverage.

Steps

1. **Identify comparable firms** in the same business(es). Use as many as are genuinely comparable.
2. **Collect regression betas** and each firm's capital structure (debt-to-equity at market values) and tax rate.
3. **Un-lever each beta** to obtain the *unlevered (asset) beta*, which reflects business risk alone.
4. **Average** (or take the median of) the unlevered betas.

5. **Adjust for cash** (optional): if peers hold large cash, un-lever to remove it, since cash has a beta of about zero.
6. **Re-lever** at the target firm's own capital structure to get its levered (equity) beta.

The un-levering formulas

Hamada (debt beta = 0, tax shield recognised):

$\beta_u = \beta_l \div [1 + (1 - t) \times D/E]$

$\beta_l = \beta_u \times [1 + (1 - t) \times D/E]$

With debt beta (βd):

$\beta_a = \beta_e \times E \div [E + D(1 - t)] + \beta_d \times D(1 - t) \div [E + D(1 - t)]$

For investment-grade firms, debt beta is small (0.0–0.2), and many valuers assume zero. For highly leveraged or distressed firms, ignoring debt beta overstates the unlevered beta. Some analysts use the *no-tax-shield* version ($\beta_u = \beta_l \div (1 + D/E)$) when they believe leverage does not create a permanent tax benefit.

10.5 Worked example: Meridian's bottom-up beta

Five listed packaged-foods peers (fictional data):

Peer	Observed (levered) β	D/E (market)	Tax	Unlevered β
A	0.98	40%	25%	0.98 ÷ 1.30 = 0.754
B	0.80	10%	25%	0.80 ÷ 1.075 = 0.744
C	1.10	60%	25%	1.10 ÷ 1.45 = 0.759
D	0.72	0%	25%	0.720
E	0.92	30%	25%	0.92 ÷ 1.225 = 0.751
Average				**0.746**
Median				**0.751**

The unlevered betas cluster tightly around **0.75** — the equity betas differ because the firms use different amounts of debt. Meridian's target D/E is 25%:

$\beta_l = 0.75 \times [1 + 0.75 \times 0.25] = 0.75 \times 1.1875 = 0.891$

and **Ke = 7.0% + 0.891 × 6.0% = 12.34%**, exactly as used in earlier chapters.

Why this is better: The standard error of the average of five unlevered betas is far smaller than that of Meridian's own regression beta; and the method lets us compute Ke at *any* leverage the firm might adopt.

10.6 Multi-business firms

If the company has several businesses (e.g., packaged foods, dairy and a small retail chain), compute the beta of each segment from pure-play comparables and combine with value weights:

$$\beta_p = \Sigma\, w_i \times \beta_i$$

where w_i is the segment's share of *firm value* (estimated from sales, EBITDA, or an iterative DCF). The combined unlevered beta is then re-levered at the firm's capital structure.

10.7 Private companies and divisions

A private company has no traded stock. The bottom-up approach is the only practical approach: take sector unlevered beta, re-lever at the company's *target* capital structure (not its current accidental one). If the owner is not diversified, total beta (market beta ÷ correlation with market) is sometimes used to reflect undiversifiable risk for that owner — an *investment value* concept, not a *market* concept (see Chapter 20).

10.8 Beta and cash

If a peer holds cash equal to 20% of equity, the beta of its operating assets is *higher* than its equity beta suggests. Correct with:

$$\beta_{operating} = \beta_{unlevered} \div (1 - \text{Cash/Firm value})$$

This matters for cash-rich sectors (technology, pharma) and is ignored in many simple analyses.

10.9 Beta over time and across the forecast

If leverage will change over the forecast, the levered beta and cost of equity change, and so does WACC (if computed bottom-up). In stable growth the firm should be assumed to have beta closer to 1 (as a firm converges toward the average firm in the economy) — an adjustment often made for the terminal period. For Meridian, with unlevered beta 0.75 well below 1, some analysts nudge the terminal beta toward 0.8–0.9. We keep

0.75 for the base case and test this in sensitivity (an unlevered terminal beta of 0.85 would raise the terminal WACC by about 0.6 percentage points (to 11.8%) and reduce enterprise value by roughly 7%).

10.10 Common errors in beta

1. **Using the target's own raw regression beta without checking the standard error.**
2. **Mixing book and market values of debt/equity** in un-levering. Use market values (or at least consistent estimates).
3. **Un-levering with a tax rate that differs from the one used elsewhere.**
4. **Not re-levering** to the target's leverage after averaging peer unlevered betas.
5. **Ignoring cash** in cash-rich peers.
6. **Using a beta computed against a foreign index** while using a local market premium.

Chapter 10 summary

- Beta measures systematic risk: covariance with the market divided by market variance.
- Regression betas are noisy; Blume-adjust or use bottom-up.
- Bottom-up: un-lever peers, average, re-lever at target leverage.
- Hamada: $\beta_l = \beta_u[1 + (1 - t)D/E]$; include debt beta where leverage is high.
- Meridian: unlevered 0.75 → levered 0.89 at D/E 25% → Ke 12.34%.

Review questions

1. A peer has levered beta 1.2, D/E 50%, tax 25%. What is its unlevered beta?
2. Re-lever that beta for a company with D/E 20%.
3. Why does averaging many peers reduce the error in beta?
4. How does cash affect the beta of a business?
5. What happens to beta in the terminal period and why might you adjust for it?

Chapter 11 — Cost of Debt, Capital Structure and WACC

FCFF is discounted at the weighted average cost of capital — the blended return required by all capital providers. This chapter shows how to build each component, how to choose the capital structure, and how to avoid the circularity that arises when value and WACC depend on each other.

11.1 The WACC formula

WACC = [E ÷ (D + E)] × Ke + [D ÷ (D + E)] × Kd × (1 − t) (+ preference share component, if any)

where E and D are *market values* of equity and debt, Ke is the cost of equity, Kd the pre-tax cost of debt, and t the marginal tax rate on interest. The factor (1 − t) captures the **interest tax shield**: interest is deductible, so each rupee of interest costs only (1 − t) after tax.

For Meridian: E/(D+E) = 80%, D/(D+E) = 20%, Ke = 12.34%, Kd = 9.0%, t = 25%.

WACC = 0.80 × 12.34% + 0.20 × 9.0% × 0.75 = 9.875% + 1.35% = **11.225%**.

11.2 The cost of debt

The cost of debt is the return lenders currently require on *new* borrowing at the firm's risk, not the historical coupon on old debt.

11.2.1 From a bond yield

If the firm has traded long-term bonds, use their yield to maturity.

11.2.2 From a credit rating

Obtain the firm's rating (or estimate a synthetic rating from an interest-coverage ratio), then use **Kd = Rf + default spread for that rating**. Example: Meridian's EBIT/interest = 240/54 = 4.4×, consistent with an A to AA rating in the Indian market. Typical spreads over G-Sec: AAA 0.6–1.0%; AA 1.2–1.8%; A 2.0–3.0%; BBB 3.0–4.5%. With Rf = 7.0% and a spread of 2.0%: **Kd = 9.0%**.

11.2.3 From the firm's bank borrowing cost

Use the incremental borrowing rate on term loans; this is the practical approach for unlisted firms. Check consistency with market rates for similar credits.

11.2.4 Adjust for default risk

The *promised* yield overstates the *expected* return to lenders when default is probable: Expected cost of debt = Promised yield − Expected loss rate. For investment-grade firms the difference is small; for distressed firms it is substantial and affects the choice of discount rate (Chapter 21).

11.2.5 Tax rate on interest

Use the marginal tax rate (25.17% for most domestic firms) if the firm is profitable and can use the full deduction. If the firm is loss-making or tax-exhausted, the tax shield is deferred or absent, and the after-tax cost of debt rises accordingly. Interest-deductibility limits (such as thin-capitalisation rules under section 94B for related-party debt) should be checked.

11.3 Other sources of capital

- **Preference shares:** treat as a separate component with cost = dividend yield (adjusted for expected redemption) — no tax shield on preference dividends.
- **Convertible debentures:** split into debt and equity components (option value).
- **Leases (Ind AS 116):** lease liability is debt; include at market value with its own cost (the incremental borrowing rate).
- **Minority interests and non-controlling stakes:** if the cash flows in FCFF belong to the parent and minorities, treat minorities as a capital provider and deduct their value in the bridge, rather than adding to WACC weights.

11.4 Choosing the capital structure weights

Three approaches:

1. **Current actual structure.** The market-value weights today. Suitable if the firm's financing is stable.

2. **Target/optimal structure.** The structure the firm is expected to maintain over time; typically the industry average or the firm's stated policy. Most appropriate for long-horizon DCF.
3. **Changing structure.** Use year-by-year weights when the firm is expected to de-lever or re-lever materially (e.g., LBO, post-IPO paydown). Then WACC varies by year and discount factors are cumulative products: $DF_t = \Pi\ 1/(1 + WACC_s)$. In this case APV (Chapter 17) is often cleaner.

Use market values, not book values. Equity at book understates the weight of equity for most profitable firms and overstates the weight of debt, lowering WACC spuriously. Debt at book is usually a fair approximation of market value unless rates have moved sharply or default risk is high.

11.4.1 The circularity problem

The equity weight needs the market value of equity, which is what the DCF is computing. Resolve by iteration:

1. Start with a guess for capital structure (e.g., industry average D/V).
2. Compute WACC and EV; derive equity value = EV − net debt.
3. Recompute D/V = Debt ÷ (Debt + Equity); recompute WACC.
4. Repeat until stable (typically 3–4 iterations; spreadsheets can do this with iterative calculation or a macro).

For Meridian: debt 600; DCF-implied equity ≈ 2,563 (Chapter 14), so actual D/V = 600 ÷ (600 + 2,563) ≈ 19.0% — close to the 20% target used; the iteration converges quickly and WACC moves by under 5 basis points.

11.5 Is there an "optimal" capital structure?

Debt provides a tax shield (benefit) but raises the probability and cost of financial distress (cost) and the costs of agency conflicts. The **trade-off theory** says there is an optimal debt ratio where marginal benefit equals marginal cost. In practice, the WACC curve is U-shaped but *flat* near the optimum: a firm at 15–30% debt often has virtually the same WACC. Hence precision in the target ratio is less important than getting Ke and Kd consistent with whatever ratio is chosen.

Modigliani–Miller (with taxes): Value of levered firm = Value of unlevered firm + PV(tax shield). It shows why more debt *appears* to lower WACC — because Kd(1 − t) < Ke — while *Ke itself rises* with leverage. Done consistently (Ke re-levered as D/E rises), the benefit is only the tax shield net of distress costs.

> **Pitfall.** Lowering WACC by raising the weight of cheap debt *without* raising Ke. This gives value increases that do not exist. Always re-lever beta when changing D/E.

11.6 Building WACC in sequence — Meridian

Step	Input	Result
1. Risk-free	10-year G-Sec	7.00%
2. ERP	Total ERP for India	6.00%
3. Unlevered beta	Bottom-up, 5 peers	0.75
4. Target D/E	25% (D/V = 20%)	
5. Levered beta	0.75 × [1 + 0.75 × 0.25]	0.891
6. Cost of equity	7.00% + 0.891 × 6.00%	12.34%
7. Pre-tax cost of debt	7.00% + 2.00% spread	9.00%
8. After-tax cost of debt	9.00% × (1 − 0.25)	6.75%
9. Weights	E 80%, D 20%	
10. **WACC**	0.8 × 12.34% + 0.2 × 6.75%	**11.225%**

11.7 WACC for different cash flows and periods

- **Different businesses** within a company have different risk: use segment-specific WACCs (sum-of-the-parts).
- **Stable-growth period:** Beta typically drifts toward 1, debt ratio toward industry norm, and Kd toward a stable spread. WACC in the terminal period may differ from the explicit period.
- **Foreign operations:** add a country risk premium to the portion of cash flows from riskier countries (revenue-weighted), or discount in the relevant currency (Chapter 22).
- **Project-level WACC:** use the hurdle rate for the project's risk, not the company-wide WACC (Chapter 3's consistency rule).

11.8 Hurdle rates vs. WACC: a practical caution

Many companies set a hurdle rate above the computed WACC (e.g., 14% vs 11.2%) as a buffer for estimation error. In valuation, the buffer *double-counts* risk if the cash flows are already prudent or scenario-weighted. Use the best-estimate WACC with best-estimate (expected) cash flows. Layer conservatism through scenarios and ranges, not through an inflated discount rate.

11.9 Sanity checks on WACC

- **Relative to peers' WACC** (as disclosed in brokers' notes and fairness opinions).
- **Relative to historical ROIC and growth:** a WACC far above ROIC for a stable firm implies value destruction.
- **Reasonableness of total return:** WACC for a low-beta consumer staples firm should typically lie between 9% and 12% in the Indian rupee market at recent interest rates; for cyclicals 11–14%.
- **Spread over Rf:** WACC − Rf should be modest for defensive businesses (e.g., 3–5 points).

11.10 Worked example: levered recapitalisation

Suppose Meridian re-levers to D/E = 50% (D/V = 33.3%), keeping unlevered beta at 0.75. New levered beta = 0.75 × (1 + 0.75 × 0.50) = 1.031. Ke = 7% + 1.031 × 6% = 13.19%. Suppose the extra debt raises the cost of debt to 9.75% (the firm's rating falls). WACC = 0.667 × 13.19% + 0.333 × 9.75% × 0.75 = 8.79% + 2.44% = **11.23%**.

Observe: *WACC is essentially unchanged*. The tax shield on additional debt is offset by the higher Ke and higher Kd. This is why many valuations are not very sensitive to the exact debt ratio — and why arguments that "more debt creates value" must be handled with care.

Chapter 11 summary

- WACC = E/V·Ke + D/V·Kd(1 − t); use market values and the marginal tax rate.
- Cost of debt = current borrowing rate at the firm's risk (Rf + default spread).

- Use target capital structure for long-horizon valuations; resolve circularity by iteration.
- Re-lever Ke whenever D/E changes; do not lower WACC by adding cheap debt alone.
- Meridian WACC = 11.225%.

Review questions

1. Compute WACC: E = 700, D = 300, Ke = 14%, Kd = 10%, t = 30%.
2. Why must Ke rise when leverage rises?
3. How do you resolve the circularity between EV and WACC?
4. What is wrong with adding a "buffer" to WACC for caution?
5. How do Ind AS 116 leases affect WACC inputs?

Chapter 12 — Terminal Value

In most DCF valuations, more than half the total value — often 70–85% — comes from the terminal value (TV): the value of all cash flows beyond the explicit forecast. It is therefore the most consequential and most abused part of the model. This chapter covers the principal methods, the consistency conditions they must meet, and ways to test them.

12.1 What terminal value represents

At the end of the explicit period (year N), the model assumes the business enters a *steady state*: it grows at a constant rate forever, earns a constant return on new capital, and maintains a constant capital structure. The terminal value at year N is the present value, at that date, of all subsequent cash flows. It is then discounted back to today.

$$\mathbf{EV = \Sigma_{t=1}^{N} FCFF_t \div (1+WACC)^t + TV_n \div (1+WACC)^N}$$

12.2 The Gordon growth (perpetuity growth) method

$$\mathbf{TV_n = FCFF_{n+1} \div (WACC - g)}$$

where $FCFF_{n+1}$ is the first cash flow *after* the explicit period — not simply $FCFF_n \times (1 + g)$ unless the year-N cash flow was already at steady state. The method assumes growth g forever.

12.2.1 Choosing g

The terminal growth rate must be no greater than the long-run growth rate of the economy in the currency of the cash flows. Rationale: a company cannot outgrow the economy forever, or it would eventually be larger than the economy. Practical anchors:

- **Risk-free rate** as a ceiling (a rough proxy for the nominal growth rate of the economy): $g \le Rf = 7.0\%$ for Meridian.
- **Nominal GDP growth** long-run forecasts for India (real growth 5–6% plus inflation 4–5%) suggest 9–11%, higher than Rf; but most valuers use a *much lower* g (4–6%) because a *single company* should converge below the economy as a whole and because the share of the economy that any firm can capture is finite.
- **Inflation plus modest real growth**: 4–5% for staple businesses; for declining industries, 0–3%.

Meridian uses **g = 5.0%**, which is below Rf (7.0%) and approximately inflation plus 1 percentage point of real growth.

12.2.2 Sensitivity

Terminal value is *highly* sensitive to g. With WACC = 11.225%: g = 4% gives TV multiple = 1/(0.07225) = 13.8; g = 5% gives 1/0.06225 = 16.1; g = 6% gives 1/0.05225 = 19.1. A 1-point change in g changes TV by 15–19%.

12.3 Consistency: growth requires reinvestment

The most important rule of terminal value: **FCFF in the terminal year must be consistent with the growth rate and the return on new investment.**

Reinvestment rate = g ÷ ROIC → $FCFF_{n+1}$ = $NOPAT_{n+1}$ × (1 − g/ROIC)

Substituting gives the **value-driver formula**:

TV_n = $NOPAT_{n+1}$ × (1 − g/ROIC) ÷ (WACC − g)

This formula makes the economics clear. Two special cases:

- **ROIC = WACC (no excess returns).** Then TV = $NOPAT_{n+1}$ × (1 − g/WACC) ÷ (WACC − g) = $NOPAT_{n+1}$ ÷ WACC. *Growth adds no value*, because new investment earns exactly its cost. The TV is independent of g.
- **ROIC > WACC.** Growth creates value; higher g raises TV.
- **ROIC < WACC.** Growth destroys value; TV *falls* as g rises.

> **Key idea.** Terminal growth is not a free lunch. Assuming 5% growth *and* low reinvestment implies infinite ROIC — which is neither plausible nor sustainable. Always compute implied ROIC from your terminal assumptions.

The usual error

Many models compute $FCFF_{n+1}$ = $FCFF_n$ × (1 + g) with year-N FCFF that still reflects high-growth reinvestment (too low a cash flow) or, conversely, with depreciation exceeding capex (too high a cash flow). Rebuild terminal FCFF from NOPAT and the reinvestment rate g/ROIC.

12.4 Terminal ROIC

What ROIC should be used? Options:

1. **ROIC = WACC.** Competitive advantages disappear; the most conservative and a common choice for commodity or intensely competitive industries.
2. **ROIC = a modest premium over WACC** (e.g., 3–5 points) for firms with durable brands.
3. **Current ROIC held constant.** Appropriate only if there is evidence of a lasting moat; otherwise aggressive.
4. **Fade** from current to a long-run level over the final years.

Meridian: ROIC reaches roughly 18% by the end of the forecast; we assume terminal ROIC of **15%**, a moderate fade that still leaves a substantial spread over WACC (3.8 points).

12.5 Meridian's terminal value

- Year-5 revenue: 3,306.7; terminal-year (year 6) revenue = 3,306.7 × 1.05 = **3,472.0**.
- EBIT margin (held at year-5 level): 14.5% → EBIT = 503.4; NOPAT = 503.4 × 0.75 = **377.6**.
- Reinvestment rate = g ÷ ROIC = 5% ÷ 15% = **33.3%**; reinvestment = 125.9.
- **$FCFF_6$ = 377.6 × (1 − 0.3333) = 251.7.**
- **TV_5 = 251.7 ÷ (0.11225 − 0.05) = 251.7 ÷ 0.06225 = ₹4,043.7 crore.**
- Discount factor, year 5 = $1 \div 1.11225^5 = 0.5875$.
- **PV of TV = 4,043.7 × 0.5875 = ₹2,375.6 crore.**

Share of enterprise value: 2,375.6 ÷ 2,913.3 = **81.5%**.

What if ROIC = WACC in the terminal period? $FCFF_6$ = 377.6 × (1 − 0.05/0.11225) = 377.6 × 0.5546 = 209.4; TV = 209.4 ÷ 0.06225 = 3,364; PV = 1,976. EV falls from 2,913 to about 2,514 (−14%). This 14% is the *value of Meridian's assumed terminal competitive advantage*. That quantification is useful in negotiation: the buyer should not pay for a moat the business may not have.

12.6 The exit-multiple method

TV_n = Multiple × $Metric_n$ (e.g., EV/EBITDA × $EBITDA_n$).

The multiple is taken from comparable companies or transactions. It is quick, tied to market pricing, and avoids the g-and-WACC debate. Limitations:

- It imports *current* market pricing (which may be inflated or depressed) into the terminal value, so the DCF is no longer purely intrinsic.
- The multiple embeds assumptions about growth, risk, and returns that may not match the terminal state of the company.
- It must be applied to a *normalised* EBITDA (mid-cycle) at the terminal date.

Use the two methods against each other. Compute implied exit multiple from the Gordon TV and compare with market multiples; compute implied g from an exit-multiple TV.

Meridian	Value
TV_5 (Gordon)	4,043.7
$EBITDA_5$ (EBIT 479.5 + D&A ≈ 153.3)	632.8
Implied EV/EBITDA at exit	**6.4×**
Implied EV/NOPAT (year 5)	11.2×
Implied EV/EBIT (year 5)	8.4×

For illustration, suppose listed leaders in the sector trade at EV/EBITDA multiples of 15–30× (check current data; such levels have been seen at times); an implied exit multiple of 6.4× is *very* conservative against that, which tells us either that the Gordon assumptions are conservative or that the peer multiples are very high because the market expects high long-term growth. Both views should be written into the report. The prudent response is to avoid setting the terminal value from peak peer multiples, but to *show* the sensitivity.

12.7 Implied growth from an exit multiple

If $TV_n = M \times EBITDA_n$, the implied perpetual growth is obtained by solving $TV_n = FCFF_{n+1} \div (WACC - g)$. A rough shortcut: **$g = (TV \times WACC - FCFF_{n+1}) \div (TV + FCFF_{n+1})$** (when $FCFF_{n+1} \approx FCFF_n(1 + g)$). If an exit multiple implies g of 8% for a company in an economy growing at 5–6% real, the multiple is too high.

12.8 Other terminal-value methods

12.8.1 Liquidation value

For finite-life or declining businesses: terminal value equals the estimated liquidation value of assets (net of costs) at the horizon. Relevant for mining, projects with concession ends, a business facing structural decline.

12.8.2 Zero-growth perpetuity / earnings power value

$TV = NOPAT_{n+1} \div WACC$: assumes no growth (equivalent to ROIC = WACC growth). The "earnings power value" is a conservative floor for a stable business; the difference between the DCF and EPV is the value of growth.

12.8.3 H-model and fading growth

Growth declines linearly from g_high to g_stable over H years: **$TV = FCFF_n \times [(1 + g_{stable}) + H \times (g_{high} - g_{stable})/2] \div (WACC - g_{stable})$**. A neat approximation for fade periods when full year-by-year modelling is unnecessary.

12.8.4 Finite-life terminal value

If competitive advantage ends at year N + K, grow at g for K years with ROIC > WACC, then at g with ROIC = WACC. This is the "competitive-advantage-period" model.

12.9 Mid-year discounting and TV

With mid-year convention, as noted in Chapter 2, a Gordon TV is discounted at N − 0.5 years (because flows continue mid-year), while an exit-multiple TV is discounted at N years. In Meridian's case with end-year discounting, the Gordon TV factor is 0.5875. Under mid-year discounting every PV, including the PV of TV, would be higher by $(1.11225)^{0.5} - 1 \approx 5.5\%$; on the terminal component alone that adds about ₹130 crore, and on total EV about ₹160 crore. Choose the convention consistent with when cash flows really arrive, and apply it to *all* flows.

12.10 Sanity checks on terminal value

1. **TV share of EV.** Over 75% is common; above 90% signals an under-specified explicit period.

2. **Implied multiples.** EV/EBITDA, EV/NOPAT at exit versus mature-industry multiples.
3. **Implied ROIC and reinvestment.** Terminal reinvestment rate positive; ROIC plausible.
4. **Terminal growth** ≤ Rf and ≤ long-run nominal GDP growth.
5. **Terminal margin** within the range peers sustain over cycles.
6. **Terminal capex vs depreciation** — capex > depreciation when g > 0.
7. **Terminal tax rate** at marginal.
8. **Terminal WACC** reflecting stable-state beta and leverage.

12.11 The Gordon formula and perpetual-growth pitfalls

- **WACC − g too small** (below ~4 points) explodes the multiple. With WACC 8% and g 6%, TV = 50 × FCFF. Be wary of low-WACC environments where g approaches WACC.
- **Using the nominal growth rate in a real model** or vice versa.
- **Applying g to non-recurring items** in the terminal-year cash flow.
- **Using terminal growth > inflation for a mature company** without justifying the real growth source.

12.12 Reporting terminal value

A good report shows the TV method, the key parameters (g, ROIC, WACC), TV as a % of EV, and a sensitivity table of EV versus (WACC, g) and (ROIC, g). Readers must be able to see at a glance how much of the value depends on the terminal assumptions.

Chapter 12 summary

- TV usually represents 70–85% of value; treat it with care.
- Gordon: $TV = FCFF_{n+1}/(WACC - g)$, with $FCFF_{n+1} = NOPAT_{n+1}(1 - g/ROIC)$.
- g ≤ risk-free / nominal economy growth; ROIC fades toward WACC unless there is a durable moat.
- Cross-check against exit multiples and implied ROIC.
- Meridian: TV 4,043.7; PV 2,375.6; 81.5% of EV; ROIC = WACC alternative cuts EV by 14%.

Review questions

1. Why must terminal reinvestment be tied to terminal growth and ROIC?
2. If ROIC = WACC in perpetuity, what is the TV and why is it independent of g?
3. Compute TV: $NOPAT_6$ = 200, g = 4%, ROIC = 12%, WACC = 10%.
4. How does mid-year discounting treat Gordon versus exit-multiple TV?
5. List four sanity checks on a TV.

Part IV — Building and Testing the Valuation

Chapter 13 — From Enterprise Value to Equity Value per Share

A DCF of FCFF produces *enterprise value* — the value of the operating business. But shareholders own the *equity*, and what matters to a buyer or a seller is the price per share. The step between the two — the **equity bridge** — is where many otherwise sound valuations lose money through omission or double-counting. This chapter lists every item that belongs in the bridge and shows how to treat it.

13.1 The bridge in one line

Equity value = Enterprise value − Debt and debt-like claims − Preference capital − Non-controlling interests + Cash and non-operating assets

and

Value per share = Equity value attributable to common shareholders ÷ Fully diluted shares

13.2 The golden rule of the bridge

> **An item belongs in the bridge if its value is not already in the FCFF being discounted.**

If an asset's income is excluded from FCFF (surplus cash, associate stakes, idle land), add its value. If a liability's cash outflows are excluded from FCFF (debt service, unfunded pension, lease payments treated as financing), deduct its value. If a cash flow is *inside* FCFF (operating lease rent when leases are treated as operating), do not deduct the liability. Every item must be treated exactly once.

13.3 Debt and debt-like claims

Item	Treatment	Notes
Bank loans, debentures, commercial paper	Deduct at **market value** (usually ≈ book)	Use the balance at the valuation date including accrued interest
Lease liabilities (Ind AS 116)	Deduct if EBITDA/FCFF are pre-lease-payments	Be consistent with WACC and multiples
Finance leases / hire purchase	Deduct	

Item	Treatment	Notes
Preference shares	Deduct at market or redemption value	
Convertible bonds	Deduct the debt component; treat the conversion option as dilution (see 13.7)	
Unfunded pension and gratuity deficit	Deduct post-tax deficit	Actuarial valuation; discount rate consistency
Deferred consideration, earn-outs payable	Deduct PV	Probability-weight contingent amounts
Provisions that will require cash (litigation, decommissioning)	Deduct PV if not in FCFF	Avoid double-counting if forecast cash flows include the payments
Dividends declared but unpaid	Deduct	
Overdue taxes and statutory dues	Deduct	
Guarantees given for others	Deduct the probability-weighted expected loss	Disclose the contingent exposure
Trade payables and operating accruals	**Do not deduct**	They are in working capital (FCFF)
Deferred tax liability	Usually not deducted if perpetual; deduct PV if reversing	See Chapter 5

13.4 Cash and non-operating assets

13.4.1 Cash

Add **surplus** cash and marketable securities. Operating cash needed to run the business is inside working capital and should not be added again. Two refinements:

- **Trapped cash.** Cash held in jurisdictions where repatriation is taxed or restricted is worth less than face value.
- **Minimum operating balance.** Treat as operating.

13.4.2 Non-operating investments

- **Listed investments:** at market value (post-tax if gains would attract tax on sale and a sale is realistic).
- **Unlisted investments, associates and joint ventures:** value separately by DCF or multiples; the share of profit must be excluded from EBIT. Value is share % × equity value of the associate (with any discount for lack of control or liquidity).
- **Surplus land and property:** market value less transaction cost and tax; exclude associated rental income from EBIT.

- **Loans receivable:** at fair value (credit-adjusted).
- **Tax assets** (carried-forward losses, MAT credit): PV of expected savings, if not already in FCFF.

13.4.3 Meridian's bridge items

- Debt: ₹600 crore (deduct).
- Surplus cash & equivalents: ₹150 crore (add).
- Long-term investments (non-operating, assumed fair value ≈ book): ₹100 crore (add).
- No preference capital or minority interest.

13.5 Non-controlling interests (minorities)

If the FCFF of the consolidated group includes 100% of a subsidiary that is only 70% owned, the equity bridge must remove the 30% claim:

NCI value = Minority % × Equity value of the subsidiary — ideally from a separate DCF or market multiple, not book value. Using book value of NCI from the balance sheet undervalues the claim when the subsidiary is highly profitable (and overstates it when it is not).

13.6 Associates, joint ventures and cross-holdings

For a 30% stake in an associate, *Ind AS* shows equity-accounted share of profit below operating profit. FCFF excludes it, so add: **% stake × equity value of the associate** (DCF/multiple), possibly discounted. Do not capitalise share of profit at the *parent's* WACC; the associate's own risk is what matters. Cross-holdings between group companies require care to avoid double counting.

13.7 Employee stock options and dilution

Existing options, warrants and convertibles are claims on equity.

13.7.1 Treasury-stock method (simple, approximate)

Diluted shares = Basic shares + Options × (1 − Strike ÷ Share price) for in-the-money options. This requires the share price — itself the output — so iterate.

13.7.2 Option-pricing method (better)

Value each option tranche with Black–Scholes or a binomial model (value > intrinsic value because of time value and volatility), then deduct the **total option value** from equity value before dividing by *basic* shares. Treats vested and unvested options consistently (unvested options may be probability-weighted for forfeiture).

13.7.3 Illustration (hypothetical extension to Meridian)

Assume Meridian has 2 crore outstanding options with an average strike of ₹30, and that equity value to all holders before options is ₹2,563.3 crore. Intrinsic value per option at the base-case price of ₹51.27 ≈ ₹21.3; total intrinsic ≈ ₹42.5 crore (an option-pricing model would give somewhat more because of time value, say ₹50 crore). Equity to common ≈ 2,563.3 − 50 = ₹2,513.3 crore, or **₹50.27 per share** over 50 crore basic shares, about 2% lower than ₹51.27. (The book's headline case has no outstanding options.)

> **Pitfall.** Ignoring *future* option grants. If ESOP expense is deducted in EBIT (as it should), future grants are already paid for in the cash flows; only *existing* options belong in the bridge.

13.8 Convertible instruments

A convertible debenture is part debt, part equity option. If conversion is far out of the money, treat as debt at its yield-adjusted value. If deep in the money, treat as equity (add shares, do not deduct the debt). In between, split into a straight-debt component and an option component and value the latter.

13.9 Control premiums and minority discounts

Standards of value (Chapter 1) determine whether further adjustment is needed.

- **DCF of the company's cash flows** typically gives a *controlling, marketable* value on a going-concern basis, if the cash flows are those a controller could achieve. If the forecast is the *status quo* management plan without any restructuring upside, it gives a *minority, marketable* value of the company's cash flows as they are.
- **Control premium:** the extra price a buyer will pay to obtain control, reflecting the ability to change strategy, cash-flow distribution, capital

structure and management. Observed in M&A transactions (often 20–40% over unaffected market price) but varies enormously; it is not a fixed number.

- **Discount for lack of control (DLOC):** applied to a minority stake when the value was computed on a controlling basis. The reciprocal of a control premium: DLOC = 1 − 1/(1 + control premium). A 25% premium corresponds to a 20% DLOC.
- **Discount for lack of marketability (DLOM):** private-company shares cannot be sold quickly. Empirical studies of restricted stock, pre-IPO transactions and put-option models suggest 10–35%, depending on holding period and volatility. Applied to a minority, unlisted stake; *not* normally applied when valuing the company as a whole for an acquisition by a strategic buyer.
- **Do not apply discounts mechanically.** In shareholder-dispute and statutory-appraisal contexts, courts may prohibit them; in reporting contexts, fair value of a controlling block may be based on market participant assumptions that include them.

13.10 The complete Meridian bridge

₹ crore	Amount
Enterprise value (Chapter 14)	2,913.3
Less: borrowings	(600.0)
Add: surplus cash & equivalents	150.0
Add: non-operating long-term investments	100.0
Equity value	**2,563.3**
Basic shares (crore)	50.0
Value per share (₹)	**51.27**

13.11 Reconciling to market price

Compare the DCF value per share with the share price (or with the latest transaction price, for a private company). A gap is not itself an error: it is a *question* — which of the model's assumptions differs from the market's? Chapter 16 turns the question into a reverse DCF.

13.12 A checklist for the bridge

1. Every debt-like item identified from the notes to accounts.
2. Lease treatment consistent across EBITDA, FCFF, WACC and bridge.

3. Cash split into operating vs surplus.
4. Non-operating assets valued separately; their income excluded from EBIT.
5. Minorities at market-based value, associates at proportional equity value.
6. ESOPs and convertibles handled with an option approach.
7. Valuation date balance-sheet items; subsequent events (dividends, share issues) accounted for.
8. Control and marketability adjustments only where the standard of value requires them.

Chapter 13 summary

- Bridge: EV − debt-like claims − preference − minorities + cash and non-operating assets = equity.
- Each item belongs *once*: in FCFF or in the bridge, never both and never neither.
- Treat leases, pensions, ESOPs, convertibles, associates carefully.
- DLOC/DLOM depend on the standard of value; apply with justification.
- Meridian: EV 2,913.3 → equity 2,563.3 → ₹51.27 per share.

Review questions

1. State the "golden rule" of the equity bridge.
2. Why is book value of NCI a poor proxy?
3. How are existing and future ESOP grants treated differently?
4. Why is a lease liability deducted when FCFF is before lease payments?
5. A 30% DLOC corresponds to what control premium?

Chapter 14 — Complete Case Study: Valuing Meridian Foods Ltd

This chapter assembles everything from Parts I–IV into one complete valuation. It is a template: the sequence, exhibits and checks are those that a professional valuation report would contain.

14.1 Engagement summary

- **Subject:** Meridian Foods Ltd, a packaged-foods manufacturer (fictional).
- **Valuation date:** 31 March 2026.
- **Purpose:** Estimate the intrinsic value of 100% of the equity for a strategic investor evaluating a majority purchase.
- **Standard of value:** Fair market value, going-concern premise, control basis (no minority discount applied).
- **Method:** Primary: DCF (FCFF, WACC). Cross-checks: trading multiples, reverse DCF, sensitivity and scenarios.
- **Currency and units:** ₹ crore, nominal, end-year discounting (mid-year variant shown).

14.2 Business and industry review

Meridian operates in the Indian packaged-foods market, which has grown at roughly 9–10% a year in nominal terms on rising incomes, urbanisation and formalisation of the grocery value chain. Meridian has regional leadership in biscuits and growing franchises in ready-to-cook mixes. Strengths: brand equity, distribution, plant efficiency. Risks: input-cost volatility (wheat, edible oil, packaging), competition from national and private-label brands, and distributor-credit exposure.

14.3 Historical analysis (summary)

Revenue growth 11–13% a year; EBIT margin up from 10.0% to 12.0% over three years; ROIC up from 10.8% to 12.4%; capex intensity falling from 8.0% to 7.2% of revenue; NWC stable at about 10% of revenue. Cash conversion has been healthy (CFO ≈ 95–100% of EBITDA after tax). No unusual items beyond those normalised in Chapter 8.

14.4 Forecast assumptions

Driver	FY27	FY28	FY29	FY30	FY31	Terminal
Revenue growth	14.0%	12.0%	10.0%	9.0%	8.0%	5.0%
EBIT margin	12.5%	13.0%	13.5%	14.0%	14.5%	14.5%
Tax rate (operating)	25%	25%	25%	25%	25%	25%
Sales-to-capital (marginal)	2.0	2.0	2.0	2.0	2.0	via g/ROIC
Terminal ROIC						15%

Rationale: growth fades from 14% toward the market rate and then to a perpetual 5%; margin gains reflect operating leverage and premiumisation but stay below leading peers' 15–20%; the marginal capital efficiency is slightly above recent experience and is flagged for sensitivity.

14.5 Free cash flow forecast

₹ crore	FY27E	FY28E	FY29E	FY30E	FY31E
Revenue	2,280.0	2,553.6	2,809.0	3,061.8	3,306.7
EBIT	285.0	332.0	379.2	428.6	479.5
Less: tax on EBIT @25%	(71.3)	(83.0)	(94.8)	(107.1)	(119.9)
NOPAT	**213.8**	**249.0**	**284.4**	**321.5**	**359.6**
Less: net reinvestment	(140.0)	(136.8)	(127.7)	(126.4)	(122.5)
FCFF	**73.7**	**112.2**	**156.7**	**195.1**	**237.1**

14.6 Discount rate

WACC = **11.225%** (Chapter 11: Rf 7.0%, ERP 6.0%, unlevered β 0.75 → levered 0.891, Ke 12.34%; Kd 9.0% pre-tax, 6.75% after tax; D/V 20%).

14.7 Present value of explicit cash flows

	FY27	FY28	FY29	FY30	FY31
FCFF	73.7	112.2	156.7	195.1	237.1
Discount factor @ 11.225%	0.8991	0.8083	0.7268	0.6534	0.5875
PV of FCFF	**66.3**	**90.7**	**113.9**	**127.5**	**139.3**

Sum of PVs = **₹537.7 crore**.

14.8 Terminal value

Item	Value
Terminal-year revenue (FY32 = FY31 × 1.05)	3,472.0

Item	Value
Terminal EBIT @ 14.5%	503.4
Terminal NOPAT	377.6
Reinvestment rate = g/ROIC = 5%/15%	33.3%
Terminal FCFF	251.7
TV at FY31 = 251.7 ÷ (11.225% − 5%)	**4,043.7**
Discount factor (5 years)	0.5875
PV of TV	**2,375.6**

14.9 Enterprise value and equity bridge

₹ crore	Amount	% of EV
PV of explicit FCFF	537.7	18.5%
PV of terminal value	2,375.6	81.5%
Enterprise value	**2,913.3**	100%
Less: borrowings	(600.0)	
Add: surplus cash	150.0	
Add: non-operating investments	100.0	
Equity value	**2,563.3**	
Shares (crore)	50.0	
Value per share	**₹51.27**	

14.10 Implied multiples (reasonableness checks)

Metric	Meridian at DCF value	Comment
EV / FY26 EBITDA (340)	8.6×	Low versus assumed listed-leader multiples (illustratively 15–25×)
EV / FY26 EBIT (240)	12.1×	
EV / FY26 Sales	1.46×	
P/E on FY26 EPS (₹2.79)	18.4×	Within a plausible 16–22× range
Terminal EV/EBITDA	6.4×	Conservative

The P/E of 18.4× is sensible. The EV/EBITDA of 8.6× is well below the multiples at which the largest listed packaged-food companies have traded; that gap is examined in the cross-checks in Section 14.13 and, as a reverse DCF, in Chapter 16. Part of it is explained by Meridian's smaller scale and lower margin (EBITDA margin 17% versus 20–25% for leaders), and part by a deliberately conservative terminal assumption.

14.11 Sensitivity analysis

Value per share (₹): WACC versus terminal growth

WACC \ g	4.0%	4.5%	5.0%	5.5%	6.0%
10.25%	58.0	60.2	62.9	66.2	70.1
10.75%	52.7	54.4	56.5	58.8	61.7
11.225%	48.4	49.7	**51.3**	53.1	55.2
11.75%	44.2	45.2	46.4	47.7	49.2
12.25%	40.7	41.5	42.4	43.4	44.5

Value per share (₹): terminal ROIC versus g

ROIC \ g	4.0%	5.0%	6.0%
11.225% (= WACC)	42.9	43.3	43.7
13%	45.9	47.6	49.9
15%	48.4	**51.3**	55.2
17%	50.3	54.1	59.2
19%	51.8	56.3	62.4

Observe that when ROIC = WACC, value barely responds to g (as theory predicts); with ROIC above WACC, higher g adds value.

One-at-a-time operating sensitivities (₹ per share)

Change	Value	vs. base
Base case	51.27	—
Sales-to-capital 1.67 (recent history)	49.35	−3.7%
Sales-to-capital 1.60	48.85	−4.7%
EBIT margin stays at 12.5% (no expansion)	43.22	−15.7%
Terminal EBIT margin 13.0%	46.35	−9.6%
Terminal ROIC = WACC	43.28	−15.6%
Mid-year discounting	54.45	+6.2%
Extended 10-year fade (growth 8% → 5%)	55.05	+7.4%

Margin expansion and terminal ROIC are the two assumptions that matter most; sales-to-capital matters less, because the marginal capital intensity affects only a few years of reinvestment.

14.12 Scenario analysis

Scenario	Probability	Revenue growth FY27–31	FY31 margin	Sales-to-capital	Terminal g / ROIC	EV (₹ cr)	Per share (₹)
Bear	25%	9%, 8%, 7%, 7%, 6%	12.0%	1.5	4% / 12%	1,849.9	30.0
Base	50%	14%, 12%, 10%, 9%, 8%	14.5%	2.0	5% / 15%	2,913.3	51.3
Bull	25%	15%, 14%, 13%, 12%, 11%	16.0%	2.2	5% / 17%	3,713.3	67.3
Probability-weighted							**49.95**

The probability-weighted value (₹49.95) is slightly below the base-case value because the downside is larger than the upside in this configuration. The range ₹30–₹67 is wide — the honest statement of uncertainty.

14.13 Cross-check with multiples

Method	Range of value per share (₹)
P/E 16×–22× on FY26 EPS of ₹2.79	44.6 – 61.4
EV/EBITDA 8×–10× on FY26 EBITDA of 340, then the same bridge (− debt 600 + cash and investments 250)	47.4 – 61.0
DCF, WACC ±0.5 pt, g 4.5–5.5% (central block above)	45.2 – 58.8
DCF scenarios (bear–bull)	30.0 – 67.3

The DCF base value of ₹51.27 sits comfortably in the overlap of the market-based ranges.

14.14 Conclusion

Concluded value of 100% of the equity of Meridian Foods Ltd as at 31 March 2026: ₹2,300–₹2,900 crore (₹46–₹58 per share), with a point estimate of ₹2,563 crore (₹51 per share), on a controlling, marketable, going-concern basis. The range reflects the base-case DCF with reasonable sensitivity to WACC (±0.5 point) and terminal growth (4.5–5.5%) and is supported by cross-checks against peer multiples.

The valuer should record: (i) the key assumptions that drive the answer (margin expansion, terminal ROIC); (ii) the portion of value in the terminal period (81.5%); (iii) limitations (reliance on management's plan, peer comparability); and (iv) subsequent-event considerations.

14.15 What the case study teaches

1. **Most of the value is in the terminal value**, so terminal assumptions deserve most of the attention.
2. **Margins and returns, not revenue growth, drive value.** Growth at ROIC close to WACC adds little.
3. **Cross-checks matter.** The implied P/E and EV/EBITDA tell whether the DCF is in the market's neighbourhood and, if not, where to look.
4. **Sensitivity tables are part of the answer, not an appendix.**
5. **Every number has a source** — a schedule, a note, a benchmark — and every judgement is stated.

Chapter 14 summary

- A professional DCF report: engagement terms, business review, historical analysis, forecast, WACC, TV, EV, bridge, sensitivities, scenarios, cross-checks, conclusion.
- Meridian: EV ₹2,913 crore; equity ₹2,563 crore; ₹51.27 per share; TV 81.5% of EV.
- Margin expansion and terminal ROIC are the key sensitivities.

Review questions

1. Why is the probability-weighted value lower than the base case here?
2. Which two assumptions would you most want evidence on, and why?
3. What does the sensitivity to ROIC at ROIC = WACC reveal about the value of growth?
4. Rebuild the FY27 FCFF from the stated assumptions.
5. Write a two-paragraph conclusion for a client who needs a single price per share.

Chapter 15 — Sensitivity, Scenarios and Simulation

A DCF gives a number; the real question is how much that number can be trusted. This chapter covers three tools — sensitivity analysis, scenario analysis and Monte Carlo simulation — that turn a point estimate into a statement about uncertainty.

15.1 Why uncertainty analysis is not optional

Every input to a DCF is an estimate: growth, margins, reinvestment, WACC, terminal growth. Small errors compound — particularly in the terminal value. A valuation that reports ₹51.27 per share without saying "and the 90% range is ₹36–₹69" hides most of what a decision-maker needs to know. Uncertainty analysis also helps the valuer *prioritise research*: if the value is insensitive to an input, do not spend a week refining it.

15.2 One-way sensitivity (tornado analysis)

Vary one input at a time through a plausible range, holding others at base, and record the change in value. Plot the results as a *tornado chart* with the most influential inputs at the top.

Meridian — tornado ranges (₹ per share):

Input	Low	High	Value at low	Value at high	Swing
WACC	12.25%	10.25%	42.4	62.9	20.5
Terminal ROIC	11.2% (= WACC)	19%	43.3	56.3	13.0
Terminal-year EBIT margin	13.0%	16.0%	46.4	56.2	9.8
Terminal growth	4.0%	6.0%	48.4	55.2	6.8
Revenue growth FY27–31 (±2 pts every year)	−2 pts	+2 pts	48.1	54.7	6.6
Sales-to-capital (FY27–31)	1.6	2.4	48.8	52.9	4.1

WACC is the largest single driver, followed by terminal ROIC and the terminal margin. Sales-to-capital and the near-term growth rate matter least — because the explicit period carries only 18% of value.

Lesson for effort allocation: spend time on WACC and terminal economics, not on shaving 0.5 percentage points off year-2 growth.

15.3 Two-way sensitivity tables

Show the interplay of two inputs — most commonly WACC × terminal growth, WACC × exit multiple, or margin × growth. Chapter 14 has WACC × g and ROIC × g tables. Present them as heat maps if possible, and *mark the base case*. A table with extreme combinations (WACC − g < 3 points) should be flagged as unreliable.

15.4 Break-even and goal-seek analysis

Ask: *what would have to be true for value to equal the market price (or the offer price)?* Solve for one variable at a time:

- the WACC that equates DCF value to price (implied discount rate);
- the terminal margin required;
- the growth uplift required;
- the terminal growth rate implied.

Chapter 16 develops this as the reverse DCF.

15.5 Scenario analysis

Scenarios change several inputs together, reflecting how the world could unfold. A good scenario has a *narrative* (what happens in the market, to costs, to competition) and *consistent numbers*.

15.5.1 Design principles

1. **Few and distinct.** Three scenarios (bear/base/bull) is usually enough. Five are better only if each has a distinct story.
2. **Coherent.** Margins, growth, capital intensity and terminal ROIC should tell one story. A "boom" scenario with 20% growth but unchanged capex is incoherent.
3. **Anchored.** Link scenarios to external drivers (commodity price decks, demand outcomes, regulatory events).
4. **Probabilities, if used, are judgements.** State the basis; avoid false precision (25/50/25 is a convention, not a measurement).

15.5.2 Meridian's scenarios

Bear: intense competition and commodity inflation squeeze margin; growth slows to 6–9%; capital intensity rises (sales-to-capital 1.5); terminal ROIC fades to 12%, g = 4%. Value ₹30.0 per share.

Base: management plan. ₹51.3.

Bull: successful premiumisation, export growth and new categories; growth 11–15%; margin 16%; sales-to-capital 2.2; terminal ROIC 17%. ₹67.3.

Probability weights 25/50/25 give an expected value of ₹49.95. The value per share is *not* symmetric around the base: downside ₹21 below base, upside ₹16 above. That asymmetry (downside bigger) is typical of businesses where the plan is already optimistic.

15.5.3 Scenario analysis vs. a higher discount rate

Some practitioners add a premium to WACC to reflect risk. Scenario analysis is superior: it separates the *expected* cash flow (probability-weighted) from the *discount rate* for systematic risk. An arbitrary add-on penalises long-dated cash flows and the terminal value disproportionately, and cannot distinguish risks that are diversifiable (should not be priced) from systematic ones.

15.6 Monte Carlo simulation

Monte Carlo assigns **probability distributions** to uncertain inputs, draws thousands of combinations, computes value for each, and builds a distribution of outcomes.

15.6.1 Steps

1. Identify the key uncertain inputs (3–8 is typical).
2. Specify a distribution for each: normal (symmetric), triangular (min/most-likely/max), lognormal (positive skew), or empirical.
3. Specify *correlations* between inputs (e.g., higher commodity prices → lower margin; higher inflation → higher WACC and higher nominal growth).
4. Run 5,000–50,000 trials.
5. Summarise: mean, median, percentiles, probability of falling below a threshold.

15.6.2 Meridian simulation

Inputs and distributions (illustrative):

Input	Distribution
Revenue-growth shift (all years)	Normal, mean 0, s.d. 2 pts

Input	Distribution
EBIT-margin shift (all years)	Normal, mean 0, s.d. 1 pt
Sales-to-capital	Triangular, 1.4 / 1.9 / 2.3
WACC	Normal, mean 11.225%, s.d. 0.75 pts (floored at g + 3 pts)
Terminal growth	Triangular, 3.5% / 5.0% / 6.0%
Terminal ROIC	Triangular, 11% / 15% / 19%

Results from 10,000 trials (independent inputs; fixed random seed):

Statistic	Value per share (₹)
Mean	50.6
Median	49.5
Standard deviation	10.3
5th percentile	35.9
25th percentile	43.4
75th percentile	56.4
95th percentile	69.2
Probability value < ₹45	31%

The simulation reproduces what the scenarios implied: a central value near ₹50, with a plausible range of ₹36–₹69 (90% interval). The distribution is right-skewed (mean above median) because the terminal-value formula is convex in WACC and g.

15.6.3 Cautions

- **Garbage in, precision out.** The shape of the output depends entirely on the input distributions, which are themselves judgements.
- **Correlation matters.** Ignoring positive correlation between growth and margin narrows the spread falsely; ignoring negative correlation between WACC and growth in a rate shock widens it.
- **Don't over-interpret tails.** Rare combinations (very low growth with very low WACC) may be economically incoherent.
- **Distributions of inputs should not be "fat" beyond the limits of the economy** — e.g., g above Rf.

Simulation complements, not replaces, scenario analysis; use scenarios for narrative, simulation for quantifying spread.

15.7 Decision-tree and probability-weighting

For discrete events (regulatory approval, loss of a major customer, success of a clinical trial), use a decision tree: assign probabilities to branches, compute the value under each, and take the expected value. This is the standard approach for pharma pipeline valuation and project-finance contingencies (Chapter 20).

15.8 Presenting results

- **Football field chart:** horizontal bars showing value ranges from each method (DCF base, DCF sensitivity, DCF scenarios, multiples, precedent transactions, 52-week range) against the market price or offer price.
- **Tornado chart:** ranking of input sensitivities.
- **Probability distribution (histogram)** with the base case marked.
- **A written statement of the conclusion**: "We estimate fair value at ₹46–58 per share; our point estimate is ₹51."

15.9 Common errors

1. Reporting only the base case.
2. Sensitivities with unrealistic ranges (WACC ±3 points) that make everything look insensitive or wild.
3. Varying inputs that are *linked* (e.g., growth without reinvestment) independently.
4. Treating simulation outputs as "probabilities of being right" when inputs are subjective.
5. Reporting scenario-weighted values without disclosing the weights and their rationale.

Chapter 15 summary

- Value is uncertain; show it.
- Tornado analysis ranks drivers; two-way tables show interactions; break-evens ask what must be true.
- Scenarios give coherent stories; simulation quantifies spread.
- For Meridian: 90% interval ≈ ₹36–₹69, centre ≈ ₹50.

- WACC and terminal economics dominate; near-term growth matters least.

Review questions

1. Why is scenario analysis better than adding a risk premium to WACC for company-specific risk?
2. What is a tornado chart and what does it tell you?
3. List four pitfalls of Monte Carlo valuation.
4. Why is a simulated DCF distribution typically right-skewed?
5. Design three coherent scenarios for a cement company.

Chapter 16 — Cross-Checks: Multiples, Reverse DCF and Triangulation

No valuation should stand on a single method. DCF should be tested against what the market pays for similar assets, and against the market price itself. This chapter shows how to do both, and how to reconcile the answers.

16.1 Relative valuation in brief

Relative valuation estimates value by applying a multiple, derived from comparable companies or transactions, to a metric of the subject company.

Multiple	Formula	Best for	Pitfalls
P/E	Price ÷ EPS	Stable, profitable firms	Leverage-sensitive; distorted by one-offs
EV/EBITDA	EV ÷ EBITDA	Capital-intensive, differing leverage	Ignores capex; lease treatment; EBITDA quality
EV/EBIT	EV ÷ EBIT	When D&A differs by policy	
EV/Sales	EV ÷ Revenue	Loss-making, early stage	Ignores margin differences
P/B	Price ÷ Book equity	Banks, asset-heavy firms	Book value is accounting-based
PEG	P/E ÷ growth	Growth comparisons	Growth rate estimates are noisy
EV/Invested capital	EV ÷ capital	Capital-intensive; links to ROIC	
EV/Subscriber, EV/GMV	Sector-specific	Platforms/telecom	Weakly tied to profit

16.1.1 Process

1. **Select comparables** by business, size, growth, margin, risk, geography.
2. **Gather clean data:** adjust for one-offs and leases, align fiscal years (calendarisation).
3. **Compute multiples** (trailing and forward).
4. **Choose a representative multiple** (median, range) with adjustments for differences in growth and risk.
5. **Apply** to the subject's *normalised* metric and bridge to equity.
6. **Cross-check** that the resulting value is consistent with the subject's fundamentals.

16.1.2 Fundamentals behind multiples

A multiple is a compressed DCF. For a stable firm:

EV/NOPAT (forward) = (1 − g/ROIC) ÷ (WACC − g)

P/E = Payout ratio × (1 + g) ÷ (Ke − g) (for a stable-growth firm)

These show why P/E rises with growth, falls with risk, and rises with returns on new investment. Using these formulas, one can test whether a peer's multiple is "justified" by its growth and ROIC and whether it can be applied to the subject.

Illustration. Meridian's terminal-state forward EV/NOPAT = (1 − 0.3333) ÷ 0.06225 = 10.7×. A packaged-food leader trading at 40× forward NOPAT is priced for far higher growth or ROIC, or for a lower discount rate, than Meridian's terminal state — it is not a pure "comparable".

16.2 Precedent transactions

Multiples paid in acquisitions of similar companies, typically EV/EBITDA or EV/Sales. They reflect *control* premiums and *synergies* paid by those specific buyers, and market conditions at the time. Use as a cross-check for control valuations; adjust for deal size, timing, and synergy sharing.

16.3 Reverse DCF

A reverse DCF takes the **market price** as given and asks what the market is *implicitly* assuming.

16.3.1 Procedure

1. Build the DCF with base assumptions.
2. Set the equity value per share equal to the market price.
3. Solve, one at a time, for each key variable the value that reconciles the model to the price.
4. Judge whether the implied assumption is plausible.

16.3.2 Meridian reverse DCF

Suppose the stock were trading at ₹58 (about 13% above the DCF value). Holding other inputs at base, the market price implies *any one* of:

Implied variable	Base	Implied by ₹58
WACC	11.23%	10.62%
FY31 EBIT margin	14.5%	16.4%

Implied variable	Base	Implied by ₹58
Annual revenue growth, FY27–31	14→8%	+3.8 pts higher each year
Terminal growth	5.0%	6.5%

Reading: the market price is not outlandish. It could be justified by a 0.6-point lower discount rate, a 1.9-point higher terminal margin, or terminal growth near the risk-free rate. A price of ₹58 asks for either more optimism on margin than the plan assumes or a lower required return — judgements an investor can weigh.

16.3.3 Uses

- **Investing:** compare implied expectations with your own.
- **M&A:** what would we need to believe to pay the asking price?
- **Fairness opinions:** assess the reasonableness of a proposed exchange ratio.
- **IPO pricing:** test the issue price against underlying growth and margin needs.

16.4 Reconciling DCF with multiples

If DCF says ₹51 and peers' P/E implies ₹70, investigate rather than average.

Questions to ask

1. Does the DCF embed lower growth than the peer group's multiple embeds? Compare implied growth.
2. Does the DCF use a higher WACC than the market-implied one? Compare the market-implied cost of capital.
3. Are peers genuinely comparable? Check ROIC, growth, quality, scale.
4. Is the sector at a cyclical or sentiment peak?
5. Is the DCF terminal assumption conservative relative to the market's perception of durable moats?

The reconciliation itself is valuable: it identifies exactly which assumptions drive the difference.

16.5 Triangulation and weighting

A valuation report usually ends with several indications of value:

Method	Indication (₹/share)	Weight
DCF (base, WACC ±0.5, g ±0.5)	45–59 (point 51)	50%
Trading multiples (P/E, EV/EBITDA)	45–61	30%
Precedent transactions (illustrative data)	50–64	20%

The weights reflect reliability, data quality and the purpose; they are a judgement, and the weighted average is not "more accurate" than any individual method. Many valuers avoid mechanical averaging and instead conclude within the overlap of the most reliable ranges. Where one method is clearly more appropriate (e.g., DCF for a business with stable cash flows; NAV for a holding company) it should carry the conclusion.

16.6 The football field

A *football-field chart* plots each method's range as a horizontal bar on a common axis of value per share, with the market price or offer price as a vertical line. It makes overlaps and outliers visible at a glance. Typical bars: 52-week trading range; analyst target prices; trading multiples range; precedent transactions; DCF (WACC/g sensitivities); DCF (scenarios); LBO analysis (the maximum price a financial buyer could pay for a target IRR).

16.7 LBO analysis as a cross-check

An LBO analysis computes the price a financial sponsor could pay to achieve a required IRR (say 20–25%) over a 5-year hold with a specified leverage. It sets a *floor* on what a financial buyer would pay and cross-checks a strategic valuation. A simple version: forecast FCFF, assume debt at 4–5× EBITDA, repay with cash flow, exit at a multiple, and solve for the entry price giving the target IRR.

16.8 Sum-of-the-parts (SOTP)

For conglomerates, value each business separately by the most suitable method (DCF or multiples), sum them, subtract unallocated corporate costs and net debt, and add non-core assets. Then compare with the consolidated DCF. The *conglomerate discount* (the gap between SOTP and market value) often reflects capital-allocation inefficiency, governance concerns, or complexity — not necessarily mispricing.

16.9 Checklist: does the answer make sense?

1. Implied multiples in line with comparable ranges (or the difference explained).
2. ROIC path consistent with competitive advantage.
3. Growth ≤ market growth in the long run; terminal growth ≤ Rf.
4. Terminal value share of EV disclosed and reasoned.
5. Cash conversion consistent with history.
6. Bridge items complete.
7. Value within the football-field overlap or the gap explained.
8. Market-implied expectations (reverse DCF) compared and discussed.

Chapter 16 summary

- Multiples are compressed DCFs; understand what drives them before applying.
- Reverse DCF reveals what the market believes; compare it to your own beliefs.
- Reconcile — don't average — when methods disagree.
- Present ranges on a football field with the price marked.
- Meridian at ₹58 implies WACC 10.6%, margin 16.4% or g 6.5%.

Review questions

1. Derive the stable-growth P/E formula from the dividend discount model.
2. What is a reverse DCF, and why is it useful in negotiation?
3. Why should multiples from a growth leader not be applied directly to a slower-growing firm?
4. What is the difference between trading and transaction multiples?
5. What are the risks of mechanically weighting valuation methods?

Part V — Special Situations

Chapter 17 — Adjusted Present Value and Leveraged Transactions

WACC assumes that the capital structure stays roughly constant as a proportion of firm value. When that is untrue — an LBO that pays down debt quickly, a project financed with special-purpose debt, a recapitalisation — the **adjusted present value (APV)** method is cleaner and more transparent. APV separates the value of operations from the value of financing side effects.

17.1 The APV idea

APV = Value of the unlevered firm + PV of financing side effects

1. **Value the business as if financed entirely by equity**, discounting FCFF at the *unlevered cost of equity* (Ku).
2. **Add the PV of the interest tax shield**, discounted at a rate reflecting the risk of the tax shield (Kd if debt is fixed; Ku if debt adjusts with firm value).
3. **Subtract the expected cost of financial distress** (probability of distress × cost of distress).
4. **Add or subtract other financing side effects**: subsidised loans, issue costs, tax-advantaged financing.

This follows directly from Modigliani–Miller: leverage affects value only through taxes, distress costs and other frictions.

17.2 The unlevered cost of equity

Ku = Rf + βu × ERP

For Meridian: Ku = 7.0% + 0.75 × 6.0% = **11.5%**. (Higher than WACC of 11.225%, because WACC benefits from the after-tax debt.)

17.3 Meridian under APV

Step 1 — Unlevered value. Discount the same FCFF at Ku = 11.5% (terminal value recomputed at 11.5%):

Unlevered enterprise value = ₹2,780.4 crore.

Step 2 — PV of tax shields. If debt stays at ₹600 crore permanently (a constant perpetual debt policy) and the tax shield is as certain as the interest, discount at Kd:

PV(tax shield) = t × D = 0.25 × 600 = **₹150.0 crore**. (Interest = 9% × 600 = 54; tax shield = 13.5 a year; 13.5 ÷ 9% = 150.)

Step 3 — Distress costs. For an A/AA-rated firm with interest cover of 4.4×, expected distress costs are small; assume zero here.

APV (enterprise) = 2,780.4 + 150.0 = ₹2,930.4 crore.

Compare with the WACC-based EV of ₹2,913.3 crore — within 0.6%. The two methods agree because the WACC example assumes a stable 20% debt ratio, which is close to ₹600 crore of debt on a value of ~₹2,900–3,100 crore. The small difference arises because the WACC approach grows debt with firm value, whereas the APV example holds debt fixed.

Equity under APV = 2,930.4 − 600 + 150 + 100 = ₹2,580.4 crore → ₹51.61 per share.

17.4 Tax-shield assumptions

The treatment of the tax shield determines APV and must match the financing policy.

Debt policy	Tax-shield discount rate	PV of tax shield
Fixed amount, perpetual	Kd	t × D
Fixed schedule, repaid over time	Kd	Σ t × Interest$_t$ ÷ (1 + Kd)^t
Constant ratio to firm value (D/V)	Ku (Miles–Ezzell / Harris–Pringle)	t × Kd × D ÷ (Ku − g) approx.
Tax shield only if taxable income sufficient	Kd, with probability of usage	Expected value

Caveats: (i) the full shield is only valuable if the firm has enough taxable profit; (ii) interest-deductibility limits; (iii) personal taxes on interest income may offset part of the corporate benefit (Miller's argument).

17.5 Expected distress costs

Expected cost of distress = P(distress) × PV(costs of distress)

Costs of distress are *direct* (legal, administrative) and *indirect* (lost customers, suppliers' demands, inability to invest, employee flight): often 10–25% of firm value for non-financial firms. P(distress) can be estimated

from the bond rating: cumulative default probabilities over 10 years are on the order of: AAA ≈ 1–2%; AA ≈ 3–4%; A ≈ 5–6%; BBB ≈ 10–12%; BB ≈ 25–30%; B ≈ 45–50%.

Example. A BB-rated firm with unlevered value ₹1,000 crore, expected distress cost 20% of value and 10-year distress probability 25%: expected distress cost = 0.25 × 200 = ₹50 crore (undiscounted simplification), which offsets about a third of the tax shield of ₹150 crore on ₹600 crore of debt.

17.6 When APV is superior to WACC

1. **Leverage changes materially** (LBOs, recapitalisation, deleveraging after IPO).
2. **Different parts of the firm have different financing** (project finance, subsidised loans).
3. **Distress costs are material.**
4. **Interest deductibility is restricted** or tax losses prevent immediate use.
5. **Simplicity and transparency:** APV shows precisely how much value financing creates.

WACC remains the convenient default for stable-leverage corporates.

17.7 Leveraged buy-out (LBO) valuation

In an LBO, a financial sponsor acquires a company using substantial debt (typically 50–70% of the purchase price), repays debt from the target's cash flows, and exits after 3–7 years at a multiple.

17.7.1 LBO model structure

1. **Sources and uses:** purchase price, fees; funded by senior debt, mezzanine/subordinated debt, and sponsor equity.
2. **Operating forecast:** revenue, EBITDA, capex, working capital, taxes.
3. **Debt schedule:** mandatory amortisation, cash sweep, interest (including potential PIK interest).
4. **Exit:** EBITDA × exit multiple − net debt at exit = exit equity.
5. **Returns:** sponsor's IRR and multiple of money (MoM).

17.7.2 Worked illustration

A sponsor buys a business with EBITDA of ₹200 crore at 8× = ₹1,600 crore EV. Debt is 5× EBITDA = ₹1,000 crore at 11%; sponsor equity is ₹600 crore (fees ignored). EBITDA grows 10% a year. After interest, tax, capex and working capital, the business generates cash that repays debt of ₹60, ₹70, ₹80, ₹90 and ₹100 crore over five years (₹400 crore in total, leaving ₹600 crore). Year-5 EBITDA = 200×1.1^5 = ₹322.1 crore; exit at the same 8× = ₹2,577 crore; less remaining debt ₹600 crore gives exit equity of ₹1,977 crore. Money multiple = 1,977 ÷ 600 = 3.3×; IRR = $3.3^{1/5} - 1 \approx 27\%$.

Where did the return come from? Equity rose from 600 to 1,977, a gain of ₹1,377 crore. EBITDA growth (322.1 − 200 = 122.1 × 8×) contributes ₹977 crore; debt paydown contributes ₹400 crore; multiple expansion contributes **nothing**, because exit equals entry. A sponsor who assumes an exit multiple above the entry multiple is relying on a source of return nobody has promised.

17.7.3 LBO as a ceiling for the price

Solving for the entry price that yields a 20–25% IRR gives the maximum a financial buyer could pay; strategic buyers who capture synergies may pay more.

17.8 Practical complications

- **Circularity** between interest and cash sweep: use iterations or average balances.
- **Debt covenants** and capacity limits: test coverage and leverage ratios.
- **Management equity** (sweet equity, ESOP): dilution of sponsor returns.
- **Exit route risk:** IPO window, strategic buyer interest.

17.9 Capital structure changes inside a WACC model

If you still prefer WACC with a changing capital structure, compute **annual WACC** with year-specific debt weights, and use **cumulative discount factors**. Also adjust the levered beta by year. The terminal value should reflect the long-run target structure. This is feasible but laborious; APV typically offers a cleaner path.

Chapter 17 summary

- APV = unlevered value + PV(tax shield) − expected distress costs ± other side effects.
- Ku for operations; Kd (or Ku) for the tax shield depending on the debt policy.
- Meridian APV EV = ₹2,930 crore vs WACC EV ₹2,913 crore.
- APV excels when capital structure changes, in LBOs and in project finance.
- LBO returns come from EBITDA growth and debt paydown; multiple expansion is a risky assumption.

Review questions

1. Why is the unlevered cost of equity higher than WACC when taxes exist?
2. Compute the PV of the tax shield on perpetual debt of 400 at 10% with tax 30%.
3. Under which conditions should the tax shield be discounted at Ku rather than Kd?
4. List three sources of LBO returns.
5. Why can APV and WACC valuations differ and when is the difference large?

Chapter 18 — Valuing Banks, NBFCs and Insurers

Financial institutions are different. Debt is not a source of capital to be netted off; it is the raw material from which the business manufactures loans. Interest expense is an operating cost; working capital and capex are hard to define; and regulation caps leverage. FCFF and enterprise value, so useful elsewhere, are ill-defined. Valuation focuses on *equity* directly.

18.1 Why FCFF fails for banks

- Deposits and borrowings fund lending; separating "operating" from "financing" is impossible.
- Net interest income, not EBIT, is the heart of earnings.
- "Reinvestment" is growth in loans and the *regulatory capital* needed to support it, not capex and working capital.
- Cash balances are part of the business (liquidity reserves), not surplus.

Hence the main approaches are **dividend/FCFE models**, the **excess-return (residual-income) model** and **price-to-book (P/B) / P/E** multiples tied to ROE.

18.2 Cash flow to equity for a bank

FCFE = Net income − Increase in regulatory equity required

Regulatory equity required = (Regulatory capital ratio, e.g., CET1 + buffers) × Risk-weighted assets (RWA). When the loan book grows, RWA grows, and the bank must retain earnings to maintain capital ratios. Anything above that is distributable.

Example (illustrative bank, ₹ crore). Book equity ₹10,000; ROE 15%; growth in loans and equity 12% a year for five years, then 7% stable. Ke = 13%.

Year	Opening equity	Net income (15%)	Growth in equity needed	FCFE	PV @13%
1	10,000	1,500	1,200	300	265
2	11,200	1,680	1,344	336	263
3	12,544	1,882	1,505	376	261
4	14,049	2,107	1,686	421	258
5	15,735	2,360	1,888	472	256

PV of explicit FCFE ≈ ₹1,304 crore.

Terminal year (year 6): equity at the end of year 5 = 17,623; net income in year 6 = 15% × 17,623 = 2,644. At stable growth of 7% and ROE of 15%, the bank can distribute only the fraction 1 − g/ROE = 1 − 0.07/0.15 = 53.3% of earnings; the rest must be retained to fund growth. So **$FCFE_6$ = 2,644 × 0.5333 = 1,410**. Terminal value = 1,410 ÷ (0.13 − 0.07) = 23,498; PV = 23,498 ÷ 1.13^5 = **12,754**.

Equity value = 1,304 + 12,754 = ₹14,058 crore → P/B = 1.41×.

A common error is to assume a near-full payout in the terminal year; at 7% growth and 15% ROE a bank cannot pay out much more than half its earnings and still keep its capital ratios.

18.3 Excess-return model

Equity value = Book equity + PV of excess returns

Excess return in year t = (ROE_t − Ke) × Book $equity_{t-1}$

For a stable bank growing at g with a constant ROE:

P/B = (ROE − g) ÷ (Ke − g)

With ROE 15%, Ke 13%, g 7%: P/B = (0.15 − 0.07)/(0.13 − 0.07) = **1.33×**. The two-stage example above (1.41×) is higher because the first five years' 12% growth at ROE > Ke adds value.

Insights from P/B = (ROE − g)/(Ke − g):

- If ROE = Ke, P/B = 1 regardless of growth.
- If ROE > Ke, P/B > 1 and rises with growth.
- If ROE < Ke, P/B < 1 and falls with growth — growth destroys value.
- A bank with ROE 10% and Ke 13% should trade below book. Many public-sector banks with weak returns do.

18.4 Key drivers for banks and NBFCs

Driver	Meaning
Net interest margin (NIM)	(Interest income − interest expense) ÷ average earning assets
Credit cost	Provisions and write-offs ÷ average loans
Cost-to-income ratio	Operating expenses ÷ total income
Capital adequacy (CET1, Tier 1, total CRAR)	Regulatory capital ÷ RWA

Driver	Meaning
Loan growth and mix	Retail, corporate, MSME, unsecured
Asset quality: GNPA/NNPA, provision coverage	Predicts credit cost
CASA ratio and deposit franchise	Funding cost advantage
Return on assets (ROA) and ROE	Combine margin, cost, credit and leverage
Fee income share	Less capital-intensive earnings

DuPont for banks: ROE = ROA × Equity multiplier (Assets ÷ Equity). Regulation limits the multiplier, so ROA and credit cost determine ROE.

18.5 Cost of equity for financial firms

- **Beta:** bottom-up from listed banks; *do not un-lever* using debt/equity as for industrial firms, because debt is operating. Use observed equity betas of comparable banks (adjusting for differences in leverage via the regulatory capital ratio).
- **Regulatory capital** substitutes for the debt/equity ratio in determining equity risk.
- Typical Indian private banks: beta 0.9–1.2; PSU banks higher and volatile; NBFCs 1.0–1.5. These depend heavily on period.

18.6 NBFCs and housing finance companies

Value in the same way (excess-return/FCFE), recognising: (i) higher leverage than banks (but regulatory limits), (ii) funding-cost sensitivity (wholesale funding), (iii) asset-liability mismatch risk, (iv) asset-quality cycles. The *growth capital* constraint is often decisive: an NBFC that must raise equity to grow has lower FCFE — a high-growth lender can have low or negative FCFE even with high ROE.

18.7 Insurance companies

Insurers hold *float* — premiums collected before claims are paid. Value approaches:

- **Embedded value (EV) and appraised value** for life insurers: embedded value = adjusted net worth + value of in-force business (PV of future profits from existing policies); appraisal value = embedded value + value of new business (VNB × multiple).
- **Excess-return / P/B** for general insurers, with focus on combined ratio (< 100% means underwriting profit), investment yield and solvency

margin.

- Discount rates for embedded value reflect risk discount rates (RDR) disclosed by insurers; compare across companies.

18.8 Asset-management and broking firms

These are asset-light. Use DCF/FCFE (little reinvestment) or P/E tied to growth in AUM or volumes; operating leverage and market cycles drive volatility. Key value drivers: AUM growth, fee yield, cost ratio.

18.9 Pitfalls

1. Applying FCFF/WACC to a bank and subtracting deposits as debt.
2. Using ROE higher than sustainable (reflecting one-off recoveries) in the terminal period.
3. Ignoring regulatory capital needs when computing FCFE.
4. Ignoring asset-quality stress (credit costs rise in downturns).
5. Using P/B multiples without regard to ROE-versus-Ke.
6. Terminal ROE equal to current ROE for a bank whose margins will be competed away.

Chapter 18 summary

- For financial firms value *equity* directly: FCFE after regulatory capital retention, or the excess-return model.
- P/B = (ROE − g)/(Ke − g); ROE versus Ke determines whether growth adds value.
- Regulatory capital replaces working capital and capex as the "reinvestment".
- Insurers: embedded value and appraisal value; general insurers: P/B and combined ratio.

Review questions

1. Why is FCFF unsuitable for a bank?
2. Compute P/B for ROE 18%, Ke 14%, g 6%.
3. What is the effect on P/B of growth when ROE < Ke?
4. How does regulatory capital constrain FCFE?
5. List the key drivers of an NBFC's ROE.

Chapter 19 — Cyclical, Commodity, Infrastructure and Real-Estate Businesses

Some businesses defy the steady-growth template. Their earnings swing with the cycle, their assets have finite lives, or their value lies in real assets rather than operating margins. This chapter adapts DCF to those cases.

19.1 Cyclical businesses

A cyclical company's earnings and cash flows rise and fall with the economic cycle or industry cycle: steel, cement, auto, capital goods, shipping, chemicals, airlines, hotels. The challenge: *when* in the cycle is the valuation date, and what is a *normal* level of profit?

19.1.1 Principles

1. **Never extrapolate the current year.** Peak-year margins are not normal; trough margins are not permanent.
2. **Model the cycle or normalise it.** Either forecast the cycle explicitly for several years (expansion, peak, contraction, trough) or work with mid-cycle values throughout.
3. **Terminal value on mid-cycle economics.** Terminal margin, ROIC and capital intensity must be normalised.
4. **Link margins to the industry's cost curve.** In commodity businesses, the long-run price equals the marginal cost of the highest-cost producer needed to meet demand. A company with a low position on the cost curve earns a rent; one at the top earns only its cost of capital in the long run.
5. **Capital intensity is lumpy.** Capacity is added in blocks; model through capacity utilisation and expansion cycles rather than smooth reinvestment.

19.1.2 Normalisation methods

Method	How
Average margin	Average EBIT margin over a full cycle (7–10 years)
Mid-cycle price deck	Use long-run commodity price (from forward curves or cost-curve analysis) and company cost structure

Method	How
Average return on capital	Mid-cycle ROIC × invested capital = normalised NOPAT
Replacement-cost approach	Value = replacement cost of assets when ROIC ≈ WACC in the long run
Explicit cycle	Forecast price/volume for years 1–5 with turn, then mid-cycle

19.1.3 Illustration: a cement company

A cement company with capacity of 10 million tonnes, EBITDA per tonne historically ranging ₹600–₹1,800 (cycle average ₹1,100), replacement cost per tonne ₹6,500, WACC 11%. Mid-cycle EBITDA = 10 × 1,100 = ₹1,100 crore. Maintenance capex and tax reduce this to FCFF of roughly ₹600 crore. A Gordon valuation at g = 4%: 600 × 1.04 ÷ 0.07 ≈ ₹8,900 crore, i.e., ₹890 per tonne of capacity, compared with replacement cost of ₹6,500 per tonne = ₹6,500 crore for the whole. The value above replacement cost implies that the company (or the industry) earns returns above its cost of capital; if new capacity can be built at ₹6,500 per tonne, new entry would eventually cap the value near replacement cost (EV/tonne ≈ replacement cost). The valuer must decide whether barriers to entry (limestone reserves, location) justify the premium.

19.1.4 Beta of cyclicals

Cyclical firms tend to have high operating leverage and betas above 1. Bottom-up betas of peers over a full cycle are better than a regression over a single boom or bust.

19.2 Commodity and natural-resource businesses

Value depends on reserves, production profiles, prices and costs.

Steps:

1. **Reserves and production schedule** (proven, probable, possible) with probability-weights.
2. **Price deck:** consensus or forward-curve, converging to long-run price.
3. **Operating costs** by unit (cash cost per tonne/barrel) with inflation.
4. **Capex:** sustaining and expansion; decommissioning.
5. **Royalties, taxes and levies** (important in India for coal, minerals, hydrocarbons).
6. **Finite life:** cash flows end at depletion; no perpetuity. Terminal value is zero or salvage/closure.

7. **Discount rate** reflecting commodity-price systematic risk and country risk.

A mine with finite life cannot be valued with a perpetual-growth terminal value; use explicit cash flows to the end of reserve life. Alternatively, model reserve replacement for a going-concern producer and apply perpetual growth to a "sustaining" state.

19.3 Infrastructure and project-finance businesses

Roads (BOT/HAM), power plants, ports, airports, pipelines and telecom towers often have contractually defined lives (concessions) and regulated or contracted revenue.

19.3.1 Features

- **Finite concession:** cash flows end at concession expiry (with possible extension). Terminal value is 0 or the pre-agreed termination payment.
- **Revenue structure:** annuity (HAM), toll/traffic-based, or tariff-regulated (PPA, regulated return on equity).
- **High, often long-dated leverage:** 70–80% debt; **project IRR and equity IRR** are the key metrics.
- **Debt sizing:** governed by debt-service coverage ratio (DSCR).

19.3.2 Valuation approach

Use **FCFE** (cash flow to equity after debt service) or **FCFF with project WACC and APV**. The cash-flow waterfall is crucial: revenue → operating costs → taxes → debt service → reserve accounts → equity distributions. Key risks: traffic/demand, construction, regulatory change, refinancing.

19.3.3 Illustrative equity IRR

A toll road requires equity of ₹400 crore and is expected to distribute ₹40, ₹55, ₹65 and ₹75 crore in years 4–7 and then ₹90 crore a year in years 8–20 (nothing in years 1–3 while the road is built and ramps up). The total distribution is ₹1,405 crore, 3.5 times the equity invested — but the money arrives late. The equity IRR is about **11.6%**, and the NPV at a 14% cost of equity is about **−₹78 crore**. As structured, the investment does not meet the hurdle: a higher tariff, faster ramp-up, a longer concession or cheaper debt would be needed. The example shows why a high *multiple of money* can coexist with a poor *return*: timing matters.

19.4 Real estate and REITs

19.4.1 Net asset value (NAV) approach

NAV = Market value of investment properties + other assets − debt − other liabilities. The market value of each property is derived through:

- **Income capitalisation:** Value = Net operating income (NOI) ÷ Capitalisation rate (cap rate). For commercial property, cap rates are typically 7–9% in Indian metros, lower for prime Grade A assets.
- **DCF of NOI** with explicit lease expiries, re-leasing assumptions, vacancy, capex and an exit cap rate.
- **Comparable sales** per square foot.

19.4.2 Developers (residential)

Valuation is project-by-project: land cost, approvals, construction cost, sales velocity and price, collections schedule, RERA constraints. Use project-level DCF (inflows from customers, outflows for land/construction, taxes) and sum, then deduct corporate overheads and net debt. Land bank is valued at market with a probability-weighting for development timing.

19.4.3 REITs / InvITs

Distributable cash flow (NDCF) to unit-holders is the key; valuation by (i) NAV, (ii) DCF of NDCF at cost of equity, (iii) distribution yield versus risk-free rate, (iv) P/NAV. Mandatory distribution rules (90% of NDCF) make DDM-style models appropriate.

19.4.4 Cap-rate arithmetic

Cap rate = Rf + risk premium − expected growth. A property with NOI ₹10 crore, cap rate 8% → value ₹125 crore. A 50-bp rise in cap rate to 8.5% reduces value 5.9% to ₹117.6 crore — notice the sensitivity.

19.5 Asset-heavy and holding companies

Holding companies, investment firms, and companies whose main value is in listed or unlisted investments are valued on a **sum-of-the-parts** or **NAV** basis: value each holding (at market or by DCF), deduct holding-company debt and costs, and consider a **holding-company discount** (typically 15–30%) reflecting structural inefficiencies and tax leakage. The discount is an empirical regularity, not a theorem; it depends on governance and capital-allocation record.

19.6 Companies with a finite life or declining business

A business facing structural decline (print media, certain legacy telecoms) should be valued with *negative* terminal growth or finite cash flow; the Gordon formula works with negative g, though the value is smaller:

TV = $FCFF_{n+1}$ ÷ (WACC − g) with g = −3%: e.g., FCFF 100, WACC 10% → TV = 100 × 0.97 ÷ 0.13 = 746. The reinvestment rate in a declining business is *negative* (liquidating assets); consistency means FCFF can exceed NOPAT.

19.7 Summary of adaptations

Business type	Main adaptation
Cyclical	Mid-cycle margins; terminal on normalised economics; cost-curve logic
Commodity producers	Reserve life, price deck, finite cash flows
Infrastructure	Concession life, project IRR, FCFE/waterfall
Real estate	NOI/cap rate, NAV, project DCF
Holding company	SOTP/NAV with discount
Declining business	Negative g, liquidation-like reinvestment

Chapter 19 summary

- Cyclicals: normalise; avoid extrapolating peak or trough; terminal on mid-cycle.
- Resources and concessions: finite lives, no perpetuity.
- Real estate: NOI ÷ cap rate or DCF of NOI; NAV; project DCF for developers.
- Holding companies: NAV/SOTP with a justified discount.

Review questions

1. Why is a peak-year margin a poor basis for a terminal value?
2. How would you value a mine with 12 years of reserves?
3. A property earns NOI of ₹8 crore; cap rate 7.5%. What is its value? What if the cap rate rises to 8.5%?
4. Why is perpetual growth inappropriate for a toll road concession?
5. What drives a holding-company discount?

Chapter 20 — Start-ups, High-Growth Companies and Digital Platforms

Young companies break most of the assumptions that make DCF easy: there is no stable history, margins are negative, growth is explosive, survival is uncertain, and the capital structure is changing. Yet the logic of DCF still applies. This chapter adapts it — and describes the venture-oriented methods that practitioners use alongside it.

20.1 What is different?

Feature	Mature company	Start-up / high-growth
History	Long, stable	Short or none
Earnings	Positive, predictable	Negative; may be for years
Growth	Near economy's rate	50–200%+ initially
Survival	Near-certain	A real risk; many fail
Value source	Assets in place	Future growth options
Capital	Stable	Repeated funding rounds, dilution
Comparables	Many	Few, imperfect

20.2 A DCF for a young company

Damodaran-style "story to numbers" approach:

1. **Tell the story.** What is the market, how big can it be, how does the company win share, what will its economics look like at maturity?
2. **Convert to drivers.** Revenue in year 10 (market size × target share), target operating margin at maturity, sales-to-capital ratio, tax rate.
3. **Interpolate.** Create a path from today's revenue and (negative) margin to the year-10 targets; growth and margin converge year by year.
4. **Reinvestment.** From the sales-to-capital ratio: $\text{Reinvestment}_t = \Delta\text{Revenue}_t \div (\text{Sales-to-capital})$.
5. **Taxes.** Net operating losses (NOLs) build up and shelter future profits; model the tax asset.
6. **Discount rate.** Start with a high cost of capital (reflecting high beta, all-equity funding) and glide it to a mature level (e.g., 11–12%) by year 10.

7. **Terminal value** at year 10 with stable growth and ROIC near WACC (or modestly above for durable advantage).
8. **Survival probability.** Multiply the PV by the probability the firm survives — or, equivalently, add a distress/failure value.
9. **Equity bridge**, including *existing and future* dilution from options and funding.

20.2.1 Worked example: "NovaCart", an online grocery platform (illustrative)

Current revenue ₹50 crore; operating loss about ₹20 crore (a margin of roughly −40%). *Story:* reach ₹4,000 crore of revenue by year 10 (about 1.3% of a ₹3 lakh crore addressable market) at a mature EBIT margin of 14%, with a sales-to-capital ratio of 3.0 (an asset-light platform). Revenue grows at a compound rate of about 54% a year, margin improves linearly from −40% to +14%, opening tax losses of ₹30 crore build up with the early losses and shelter profits when they arrive, WACC glides from 15% in year 1 to 11.5% in year 10, terminal growth is 5% and terminal ROIC 14%.

Year	Revenue	EBIT	FCFF
1	77	(27)	(36)
2	120	(35)	(49)
3	186	(44)	(66)
4	289	(53)	(87)
5	447	(58)	(111)
6	693	(53)	(135)
7	1,074	(24)	(151)
8	1,665	53	(144)
9	2,581	222	(83)
10	4,000	560	(41)

Valuation. PV of the ten years of (negative) cash flows = **−₹441 crore** (these are the funding needs). Terminal value at year 10 = ₹4,362 crore, which discounts (cumulative factor 0.298) to **₹1,301 crore**. Value if the story succeeds = 1,301 − 441 = **₹860 crore**. Failure value is assumed at ₹5 crore (salvage of technology and brand) and the probability of survival at 70%, so the survival-weighted value is 0.7 × 860 + 0.3 × 5 = **≈ ₹604 crore**.

Convexity. Changing the story changes the answer enormously: year-10 revenue of ₹6,000 crore at a 15% margin gives ₹1,522 crore; ₹2,000 crore at 10% gives ₹140 crore; ₹1,000 crore at 8% gives **less than zero** (−₹42 crore) — a business that burns more cash than it can ever repay is worth zero to a limited-liability equity holder, not a negative amount. A margin of 10% instead of 14% cuts the successful-case value from ₹860 crore to ₹388 crore, and a lower capital efficiency (sales-to-capital of 2.0) to ₹617 crore. That *convexity* — large gains in success, a floor of zero in failure — is why growth companies are priced like options on success, and why a single point estimate is especially misleading for them.

20.3 Venture-capital method

Used by venture investors to determine the entry valuation.

1. Forecast the company's **exit value** at the horizon (typically 4–7 years), using a revenue/EBITDA multiple on the forecast results.
2. Choose a **target return** (IRR of 30–60% for early stage, reflecting high failure rates and illiquidity).
3. **Post-money valuation today = Exit value ÷ (1 + target IRR)^{n}**
4. **Pre-money = Post-money − Investment.**
5. **Ownership required = Investment ÷ Post-money**, adjusted for future dilution: Required ownership today = Required ownership at exit ÷ (1 − expected dilution).

Example. Exit value expected ₹600 crore in year 5. Investor target IRR 40%. Post-money = 600 ÷ 1.4^{5} = 600 ÷ 5.378 = **₹111.6 crore**. Investment of ₹20 crore → ownership at exit needed = 20 ÷ 111.6 = **17.9%**; pre-money = ₹91.6 crore. If the founders expect 30% dilution from later rounds before exit, the investor needs 17.9% ÷ (1 − 0.30) = **25.6%** today for the same exit ownership.

Why target IRRs are so high. The target IRR is *not* a cost of capital; it is an expected return that bakes in failure: if 3 out of 4 start-ups fail, the successful one must return enough for the portfolio. A 40% "required IRR" on a successful scenario might correspond to a 15–20% expected portfolio return. Mixing a venture-capital hurdle with a probability-weighted forecast double-counts risk; use one or the other.

20.4 Scenario and probability-weighted approaches (First Chicago method)

Build three scenarios — success, modest outcome (sideways), failure — with probabilities and exit values; discount each at a moderate rate and weight:

Scenario	Probability	Exit value in year 5 (₹ cr)	PV at 20%
Success	20%	1,200	482
Sideways	35%	250	100
Failure	45%	0–20 (say 10)	4
Weighted			**≈ 133**

(PV factor for 5 years at 20% = 0.4019: 1,200 × 0.4019 = 482; 250 × 0.4019 = 100; 10 × 0.4019 = 4. Weighted = 0.2 × 482 + 0.35 × 100 + 0.45 × 4 = 96.4 + 35 + 1.8 ≈ ₹133 crore.)

The advantage: probabilities and outcomes are explicit; the discount rate is a rate for the *systematic* risk of the successful business rather than the entire survival risk.

20.5 Other start-up methods

These are *heuristics* rather than valuation in the DCF sense, but common in early-stage markets.

Method	Description	Limitations
Berkus	Assign up to a fixed sum (e.g., ₹0.5 crore in an Indian adaptation of the original $500k) to each of five factors: sound idea, prototype, quality management team, strategic relationships, product rollout/sales	Caps value; ignores market size
Scorecard	Start from average pre-money valuation of comparable seed-stage companies; adjust by weighted factors (team 30%, market 25%, product 15%, competition 10%, marketing/sales 10%, need for funding 5%, other 5%)	Weights are conventions
Risk-factor summation	Start with a base pre-money and add/subtract ₹ amounts for 12 risk factors (management, stage, legislation, manufacturing, sales, funding, competition, technology, litigation, international, reputation, exit potential)	Subjective
Cost-to-duplicate	Sum of costs to rebuild the technology and team	Ignores future earnings; floor value
Comparable transactions	Price per user, price per GMV or revenue multiples from recent funding rounds	Reflects funding-market sentiment
Option pricing	Treat the equity as an option on enterprise value (Chapter 23)	Needs volatility and exercise-price estimates

20.6 Digital platforms and network-effect businesses

Marketplaces, social networks, payment platforms and SaaS products generate value from users, data and network effects. Specific modelling points:

1. **Unit economics:** customer acquisition cost (CAC), lifetime value (LTV), contribution margin, retention/churn. A healthy business has LTV/CAC above 3×.
2. **Cohort analysis:** revenue per cohort over time; revenue grows even without new users when retention is strong.
3. **Gross merchandise value (GMV) vs. net revenue:** take-rate (net revenue ÷ GMV) determines revenue; treat GMV carefully for accounting differences.
4. **Operating leverage:** high fixed (technology) costs, low marginal costs → margins expand dramatically beyond scale.
5. **Network effects and winner-takes-most:** market share targets in the terminal year should reflect whether network effects allow dominance; scenarios with winner-take-all vs fragmented outcomes diverge widely.
6. **Capitalise development and marketing:** much of the spending is investment (customer acquisition builds a customer base). Adjust ROIC.
7. **Stock-based compensation:** large; treat as a real expense.
8. **Valuation multiples:** EV/Revenue, EV/GMV, EV/active user — only as cross-checks; compare after adjusting for growth and margin differences (regression of EV/Sales on growth and margin among peers).

20.6.1 Top-down and bottom-up revenue

- *Top-down:* TAM → SAM → SOM → share. Easily inflated: demand a bottom-up check.
- *Bottom-up:* number of customers × ARPU × (1 − churn). More verifiable.

20.6.2 SaaS metrics

Annual recurring revenue (ARR), net revenue retention (NRR; > 110% is healthy), gross margin (typically 70–80%), "Rule of 40" (growth % + profit margin % ≥ 40%).

20.7 Discount rates for young firms

- **Beta** from bottom-up peers *at the start* is typically high (1.3–1.8); a high-growth firm with high operating leverage also has high business risk.
- **Debt**: none early on; the cost of capital is cost of equity.
- **Glide path:** as the firm matures, beta moves toward 1 (or the industry average) and cost of equity declines; by year 10 use the stable-state WACC.
- **No arbitrary add-ons** for failure risk: handle failure through survival probabilities.
- For *private* start-ups with undiversified investors, total beta (market beta ÷ correlation) is sometimes used to reflect the investor's actual exposure; this is investment value, not fair market value.

20.8 Dilution and the equity bridge

- **Convertible notes, SAFEs, preferred stock:** model liquidation preferences, participation and conversion. The equity "value per common share" differs from the price paid by preferred investors. Use the *waterfall* method or an option-pricing breakpoint analysis (OPM) to allocate equity value among classes (this is common in Ind AS 113 and in 409A-style valuations).
- **Future funding rounds** reduce current holders' percentage; either (i) include the cash raised and the resulting dilution in the model, or (ii) value pre-money and apply the dilution to ownership percentages.

20.9 A discipline for start-up valuation

1. Value the *story*, not the spreadsheet: ensure the numbers match the narrative.
2. Triangulate between DCF, VC method, scenarios and market comparables.
3. Test the convexity: valuations should vary hugely between failure and success.
4. Avoid mixing venture IRR hurdles with probability-weighted cash flows.
5. Present the valuation as a range with explicit probabilities.

Chapter 20 summary

- Young-firm DCF: story → drivers → year-10 targets → interpolate → survival-adjust.
- VC method: Post-money = exit ÷ $(1 + \text{IRR})^n$; adjust ownership for dilution.
- Scenario methods (First Chicago) separate probability of failure from the discount rate.
- Platforms: unit economics, retention, take-rate, network effects.
- Waterfalls and option-pricing allocate value among share classes.

Review questions

1. Why is a "40% target IRR" not a cost of capital?
2. Compute pre-money: exit ₹400 crore in 4 years, target IRR 50%, investment ₹15 crore.
3. How do you handle negative earnings in a DCF for a start-up?
4. What distinguishes a good platform's unit economics?
5. Why is survival probability better than a company-specific risk premium?

Chapter 21 — Valuing Distressed Companies

A distressed company is one that cannot comfortably meet its financial obligations: debt service is a struggle, covenants are breached, liquidity is thin. Valuing it is different, and the stakes are high — lenders, equity holders, restructuring advisers and (in India) resolution professionals and committees of creditors all depend on the numbers.

21.1 Sources and symptoms of distress

- **Financial distress:** too much debt relative to the cash flows. Interest cover below 1.5×, negative free cash flow, rating downgrades, covenant breaches, missed payments.
- **Operating distress:** losses even before interest; a structurally declining business, a failed product, cost inflation not passed on.
- **Liquidity distress:** maturities that cannot be refinanced; working-capital stretch.
- **Cyclical or one-off shocks:** a pandemic, a regulatory ban, a major customer loss.

The valuation task depends on the *cause*: a firm with a sound operating business and a bad balance sheet is valued differently from a firm with a structurally broken business.

21.2 Why standard DCF breaks down

1. **Going-concern assumption.** Standard DCF assumes survival forever. A distressed firm may not survive.
2. **Negative or erratic cash flows.** The explicit forecast may contain years of negative FCFF; Gordon growth on a negative terminal cash flow is meaningless.
3. **Discount rate.** Observed betas and cost of capital reflect the *expected* distress; adding distress risk to both the discount rate *and* the cash flows double-counts.
4. **Capital structure not stable.** Debt will be restructured; WACC with current weights is meaningless.
5. **Equity as an option.** Equity in a highly levered firm behaves like a call option on the enterprise: limited liability gives it value even when the

firm's assets are worth less than debt.

21.3 Approaches

21.3.1 Modified DCF: probability-weighted cash flows

Estimate expected cash flows that include the probability of distress.

Expected $FCFF_t$ = $FCFF_t$(survive) × P(survive to t) + Distress-case cash flow × P(distress)

or, simply, **Expected $FCFF_t$ = $FCFF_t$ × (1 − Probability of distress$_t$)** if distress implies zero further cash flow to the claim being valued. Then discount at a cost of capital that reflects only the *systematic* risk of the cash flows.

21.3.2 Going-concern DCF + distress sale value

Value the firm as if it recovers (the going-concern value), then blend with the value it would fetch if distress occurs:

Value of equity = DCF equity value × (1 − P) + Distress-sale equity value × P

with P the probability of distress over the forecast. The distress-sale value is typically estimated as a percentage of book value or of the going-concern value (e.g., 50–70% of book for tangible assets; much lower for intangibles).

Example. Going-concern enterprise value ₹2,000 crore; distress-sale EV ₹900 crore; P = 30%. Blended EV = 2,000 × 0.70 + 900 × 0.30 = ₹1,670 crore. If debt is ₹1,400 crore: *going-concern* equity = 600; *blended* equity = 270 (a 55% reduction — leverage magnifies the effect of distress risk on equity).

21.3.3 Adjusted present value (APV)

Firm value = Unlevered firm value + PV(tax shield) − Expected bankruptcy costs

Expected bankruptcy costs = (Unlevered firm value − Distress-sale value) × P(distress). APV is especially good here: the tax shield may be unusable (losses), bankruptcy costs are explicit, and capital structure is changing.

21.3.4 Equity as a call option

Treat equity as a call option on the firm's assets with strike equal to the face value of debt:

Equity = Call(Assets value V, Strike = Debt face value D, time to debt maturity T, volatility σ)

Even when V < D, the equity has positive value because assets might recover before maturity. Black–Scholes inputs: V = enterprise value (from DCF); D = face value of debt; T = weighted debt maturity; σ = asset volatility; r = risk-free rate. *Example:* V = 1,000, D = 1,200, T = 3 years, σ = 40%, r = 7%. Black–Scholes gives a call value of about ₹281 crore. So equity is worth about 281 even though V − D = −200; the debt is worth about 1,000 − 281 = 719 (roughly 60% of its face value of 1,200).

21.3.5 Relative valuation of distressed firms

Multiples of revenue and EBITDA are used more often than P/E (which is meaningless when earnings are negative). Adjust peer multiples downward for distress: if healthy telecom peers trade at 2× revenue, apply perhaps 1.25× to a distressed one.

21.3.6 Liquidation value

When going-concern value is lower than the liquidation value, creditors will prefer to liquidate (or the law forces it). Estimate realisable value asset-by-asset:

Asset	Typical recovery
Cash	100%
Receivables	60–90% (less for aged and disputed)
Inventory	40–70% (raw materials higher than WIP/finished goods of specialised products)
Land	60–90% of market (depending on title/approvals)
Plant & machinery	20–50% of book (specialised plant lower)
Intangibles/goodwill	0–10%

Subtract liquidation costs (legal, broker, wind-down staff) and priority claims (employee dues, statutory dues, secured creditors) to find the amount available to unsecured creditors and equity. Liquidation value establishes the *floor* for a distressed valuation.

21.4 The Indian insolvency context (IBC)

Under the Insolvency and Bankruptcy Code, 2016, the corporate insolvency resolution process (CIRP) values the corporate debtor using two registered valuers: the **fair value** (going-concern, arm's-length) and the **liquidation value** (net realisable in liquidation), used by the committee of creditors in evaluating resolution plans. Resolution applicants offer a plan whose value to creditors is compared with these benchmarks. Valuation in this context includes: asset classes' realisable values, contingent claims, avoidance transactions, and the effect of the waterfall (Section 53 priority). Practitioners should consult the current IBC regulations and ICAI valuation standards for requirements.

21.5 Valuing the debt of a distressed firm

The market value of debt is below face value; its expected return is below the promised yield. To get the *expected* cost of debt: **Expected return = (1 − P(default)) × Promised yield + P(default) × (Recovery rate − 1)**, in simplified form for a one-year bond. For debt valuation, discount *expected* cash flows (including recoveries) at a rate for systematic risk. When deducting debt in the bridge for a distressed firm, use *market or fair value* of debt (not face value) only if the question is the value to *equity holders post-restructuring*; for the value of the enterprise to all claimants, the question of debt face vs market value does not arise.

21.6 Restructuring scenarios and post-restructuring value

Many valuations of distressed firms are *plan-based*: what will the business be worth if a plan (debt-for-equity swap, haircuts, asset sales, operational turnaround) succeeds?

1. Build the *post-restructuring* operating forecast (cost cuts, asset disposals).
2. Estimate the post-restructuring capital structure and WACC.
3. Compute post-restructuring enterprise value.
4. Allocate value across claim classes by seniority (the **waterfall**): secured → priority statutory → unsecured → subordinated → equity.
5. Compare each class's recovery under the plan with its liquidation recovery; creditors accept a plan only if it beats liquidation.

21.7 Special considerations

- **Management's forecasts are biased upward** in distress. Insist on stress testing; compare to history.
- **Contingent liabilities** and litigation can dominate.
- **Cash trapped in subsidiaries**; guarantees and cross-defaults among group companies.
- **Control rights shift** to creditors; shareholders' interest may be out-of-the-money though legally intact.
- **Tax attributes** (NOLs) may be limited by change-of-control rules.
- **Time to resolution** affects value: legal delays erode the going-concern.

21.8 A compact worked example

Cobalt Steel Ltd. Normal-state enterprise value: ₹3,000 crore (explicit DCF). Debt face value ₹2,800 crore; 3-year maturity. Volatility of assets 35%; risk-free 7%.

1. Equity as an option: call with V = 3,000, K = 2,800, T = 3, σ = 35%, r = 7%. Black–Scholes gives about ₹1,056 crore. So equity ≈ 1,056 (compared with 3,000 − 2,800 = 200 on a simple subtraction): with three years to maturity and 35% asset volatility, most of the equity value is option (time) value.
2. DCF + distress: P(distress) = 25%, distress sale EV = 1,500. Blended EV = 3,000 × 0.75 + 1,500 × 0.25 = 2,625. Equity = 2,625 − 2,800 < 0, so on a blended-EV subtraction the *equity is worth zero*; yet the option approach assigns it a substantial value because the firm has three years in which its assets might recover.

The gap between the two shows that equity in a highly levered firm is better described as an option than as a residual (and that the two methods answer slightly different questions: the option model assumes the asset value follows a volatile path with no forced earlier liquidation, whereas the blend assumes a one-off distress event). The conclusion: *equity has positive value because of optionality; its value is highly sensitive to volatility and the time to resolve.* A valuation report must state the approach and its limitations.

Chapter 21 summary

- Standard DCF fails for distressed firms (going-concern assumption, negative flows, discount-rate and capital-structure issues).
- Tools: probability-weighted flows; DCF + distress value; APV with bankruptcy costs; option pricing; liquidation values; plan-based waterfall.
- Equity in a distressed firm has option value even when $V < D$.
- Under the IBC both fair value and liquidation value are determined by registered valuers.

Review questions

1. Why does adding a distress premium to WACC *and* reducing cash flows double-count risk?
2. Compute blended equity: DCF equity 800, distress equity 100, P = 35%.
3. Explain why equity can be worth something when assets are worth less than debt.
4. What recovery rates would you assume for plant and machinery in a forced liquidation?
5. What is the role of valuation in the Indian CIRP process?

Chapter 22 — Valuation in India and Other Emerging Markets

Emerging markets add layers to valuation: higher and more volatile inflation, currency risk, country risk, thinner data, less liquid markets, and a regulatory framework that prescribes valuation methods for many transactions. This chapter explains how to adapt DCF, with special attention to India.

Note on regulation. Valuation-related provisions of the Companies Act, SEBI regulations, FEMA, the Income-tax Act and Ind AS are amended often. This chapter describes the structure and the principles; before relying on any rule for a real engagement, check the current text and any latest circulars.

22.1 Inflation, currency and consistency

22.1.1 Nominal in rupees

The cleanest approach for an Indian company with predominantly rupee revenues is to forecast **nominal rupee cash flows** and discount at a **nominal rupee** cost of capital built from the rupee risk-free rate (G-Sec yield). Inflation is then implicitly in both numerator and denominator.

22.1.2 Real model

Alternatively forecast in constant prices and discount at a *real* rate: **(1 + real) = (1 + nominal) ÷ (1 + inflation)**. Real models are less common because working capital, taxes and depreciation are nominal (based on historical cost) and inflation affects them in non-neutral ways.

22.1.3 Choosing the currency

- **Rupee cash flows → rupee discount rate.**
- **Dollar cash flows (an exporter's contracts; a company reporting in USD)** → dollar discount rate.
- For a company with mixed revenues, model each currency's cash flows and convert at *expected* exchange rates, or model in a single currency with forward-rate-consistent conversion.

22.1.4 Converting a discount rate between currencies

By interest-rate parity, the expected depreciation of the rupee ≈ the inflation (or risk-free rate) differential:

$$(1 + WACC_{inr}) = (1 + WACC_{usD}) \times (1 + Rf_{inr})/(1 + Rf_{usD})$$

Example. Rupee Rf 7.0%; dollar Rf 4.3%; rupee WACC 11.225%. Dollar WACC = 1.11225 ÷ (1.07 ÷ 1.043) − 1 = **8.4%**. Value computed in dollars at 8.4% from dollar cash flows (obtained by converting rupee flows at forward rates) equals the rupee value converted at spot. Any discrepancy means an inconsistent assumption.

> **Pitfall.** Taking a rupee cash-flow forecast and discounting at a dollar WACC — a "cheap" 8–9% rate — to get a higher value. This is one of the most common errors in cross-border valuations. It overvalues the business by ignoring rupee depreciation.

22.2 Country risk

The risk-free rate and ERP for India must reflect the risk of investing there.

22.2.1 Approaches to the equity risk premium

1. **Total ERP for India, estimated directly** from implied premiums or long-run local returns.
2. **Mature-market ERP + country risk premium (CRP):**

$$ERP(India) = ERP(mature) + CRP$$

and CRP = Sovereign default spread × (σ equity ÷ σ government bond). If the mature-market ERP is 4.5%, India's sovereign default spread is 1.5% and the relative equity/bond volatility is about 1.5, then CRP ≈ 2.25% and total ERP ≈ 6.75%. (The numbers are illustrations; observe current data.)

3. **Exposure-weighted CRP for multinationals:** weight CRP by the share of revenues (or operations) from each country: an Indian company with 30% revenue from the USA/Europe has a lower weighted CRP than a purely domestic firm.

22.2.2 The risk-free rate

G-Sec yield in INR includes the sovereign default risk and expected inflation. For a *dollar* valuation of an Indian company: Rf(USD) = US Treasury yield; ERP includes the CRP.

22.2.3 Beta

Regression against the local index (Nifty) gives betas relative to the Indian market. For comparability with global peers, use a global index and a global ERP — but do not mix.

22.3 Practical data challenges

Issue	Practical response
Short or discontinuous listing history	Use peer betas from a broader set (including global peers); bottom-up
Thin trading in small caps	Adjust regression betas (Dimson); rely on bottom-up
Unlisted peers	Use industry data; transaction databases; adjust for lack of marketability
Related-party transactions and promoter-driven decisions	Normalise; scrutinise governance; consider a governance-risk adjustment via scenarios
Unreliable segment or financial data	Increase scenario breadth; widen the range; use more conservative terminal assumptions
Rapid structural change (formalisation, digital shifts)	Shorter forecast horizons with explicit scenarios

22.4 The Indian regulatory framework for valuation (overview)

Context	Governing framework	What it says about valuation
Preferential allotment, rights issues to some persons, non-cash consideration, schemes of arrangement (unlisted companies)	**Companies Act, 2013** and rules (e.g., sections 62, 192, 230–232)	Valuation by a **registered valuer** in the relevant asset class (securities or financial assets; land and building; plant and machinery) under the **Companies (Registered Valuers and Valuation) Rules, 2017**
Pricing of shares for non-residents (FDI, transfers)	**FEMA** and RBI/FEMA non-debt instruments rules	Pricing at or above/below fair value determined using an *internationally accepted pricing methodology* on an arm's-length basis, certified by a chartered accountant, a SEBI-registered merchant banker or a cost accountant — DCF is widely used
Preferential issues by listed companies, takeovers, delisting, buybacks	**SEBI** regulations (ICDR, SAST, delisting)	Formula-based minimum prices (volume-weighted averages) for frequently traded shares; independent valuations in specified cases
Taxation of share issues/transfers at below fair value	**Income-tax Act and Rules (e.g., rule 11UA)**	Prescribes methods for fair market value of unquoted shares, including DCF for certain purposes

Context	Governing framework	What it says about valuation
Financial reporting	**Ind AS 113 (fair value), Ind AS 36 (impairment), Ind AS 103 (business combinations), Ind AS 102 (share-based payment)**	Fair-value hierarchy; market-participant assumptions; impairment testing of CGUs and goodwill; PPA
Insolvency	**IBC, 2016** and CIRP regulations	Two registered valuers; fair value and liquidation value
Professional standards	**ICAI Valuation Standards** (and the International Valuation Standards, IVS)	Definitions, bases, approaches, scope of work, reporting, business-valuation standard

22.5 Ind AS and valuation

22.5.1 Ind AS 113 fair value

Fair value is an *exit price* from the perspective of *market participants*. Inputs are classified: **Level 1** (quoted prices in active markets), **Level 2** (observable inputs other than Level 1), **Level 3** (unobservable inputs — most DCF-based valuations). Where DCF is used, the cash flows and the discount rate should reflect assumptions *market participants* would use, not entity-specific synergies unless market participants would also benefit.

22.5.2 Ind AS 36: value in use for impairment

The recoverable amount of an asset or CGU = higher of fair value less costs of disposal and *value in use* (VIU). VIU is a DCF with specific rules:

- Cash flows based on reasonable and supportable assumptions, normally within a maximum five-year budget with extrapolation using a steady or declining growth rate (unless a longer period can be justified).
- Exclude cash flows from future restructuring not yet committed and from improving or enhancing the asset's performance.
- Exclude financing cash flows and income-tax receipts/payments (in principle, VIU is *pre-tax*, using a **pre-tax discount rate**).
- Discount rate reflects the current market assessment of the time value of money and the risks specific to the asset.

Converting post-tax WACC to a pre-tax rate: the pre-tax rate is the rate that, applied to pre-tax cash flows, gives the same value as the post-tax rate applied to post-tax flows — usually found by iteration. A rough shortcut: pre-tax rate ≈ post-tax rate ÷ (1 − t), but the exact rate depends on the cash-flow profile.

22.5.3 Ind AS 103 purchase price allocation

Acquired identifiable assets (customer relationships, brands, technology) and liabilities are measured at fair value; the residual is goodwill. Common intangible-valuation techniques: **relief-from-royalty** (brands), **multi-period excess earnings** (customer relationships), **cost approach** (software), **with-and-without** (non-competes). Contributory-asset charges ensure that the intangible is credited only with its own share of cash flow.

22.6 Control, minority and marketability in Indian practice

- **Open offers (SAST):** minimum price formulas based on market prices and negotiated price; valuation is relevant to fairness opinions.
- **Delisting:** discovered price through reverse book-building; valuation reports provide a floor.
- **Schemes of arrangement** (mergers/demergers): share-exchange ratio supported by valuation reports from registered valuers and a fairness opinion from a merchant banker; the ratio considers asset, income and market approaches with weights.
- **Discounts for marketability** (typically 10–30%) are applied in some contexts (e.g., unlisted minority stakes in corporate-governance disputes) and *not* in others (fair value for buyouts of oppressed minorities). The standard of value and the purpose determine the answer.

22.7 Valuation under uncertain governance

Promoter-controlled companies, related-party dealings, group cross-holdings, pledge of shares, and litigation are material influences on value in many Indian companies. Ways to handle:

1. Reflect expected leakage in cash flows (scenario with related-party transactions at non-arm's-length costs).
2. Incorporate a probability-weighted contingency for litigation or regulatory action.
3. Avoid layering a blanket "governance discount" onto the discount rate; instead state the specific risk and its cash-flow effect.
4. Cross-check with market price (the market usually discounts perceived governance risk).

22.8 Illustrating the effect of the country-risk adjustment on Meridian

Meridian is a domestic business; its WACC of 11.225% already embeds an India-wide ERP (6.0%). Suppose the same business were valued by a foreign investor in dollars. The foreign investor builds a dollar WACC: Rf 4.3% (US), mature ERP 4.5%, CRP 2.25%, unlevered beta 0.75, D/E 25% → levered beta 0.891, Ke = 4.3% + 0.891 × (4.5% + 2.25%) = 10.3%; Kd (USD, Indian credit) 6.3%; WACC = 0.8 × 10.3% + 0.2 × 6.3% × 0.75 = 9.2%. Converting this to rupees using the Rf differential: (1.092) × (1.07/1.043) − 1 = 12.0% — *higher* than the 11.225% rupee WACC we used. The gap (0.8 point) reflects differences in how the premium was assembled; reconcile the two approaches, which also shows why the choice of ERP matters. A careful valuer chooses one internally consistent set and documents it.

22.9 Emerging-market checklist

1. Currency of cash flows = currency of discount rate.
2. Terminal growth ≤ long-run nominal growth of the economy and ≤ the risk-free rate in that currency.
3. Country risk consistent with the ERP approach and with the revenue geography.
4. Inflation pass-through assumptions explicit (margins in inflationary periods).
5. Tax rates: statutory vs effective; incentives expiring.
6. Regulatory and statutory requirements for the purpose of the valuation.
7. Governance, related-party and legal risks translated into scenario cash flows or documented adjustments.
8. Liquidity: marketability discount only where the standard of value requires it.

Chapter 22 summary

- Match currency and inflation across cash flows and discount rates; convert rates using parity.
- Country risk enters through Rf and ERP; avoid double-counting.
- India's regulatory framework prescribes registered valuers and methods for many transactions; Ind AS 113/36/103 govern reporting

valuations.

- Governance and legal risks belong in cash-flow scenarios.

Review questions

1. Convert a 12% rupee WACC to a dollar WACC given Rf(INR) 7.2% and Rf(USD) 4.0%.
2. What is wrong with discounting rupee cash flows at a dollar WACC?
3. How does VIU under Ind AS 36 differ from a normal DCF?
4. When might a marketability discount *not* be applied?
5. How should related-party leakage be reflected in valuation?

Chapter 23 — Real Options, Intangibles and ESG in DCF

DCF treats the future as a single expected path. But managers can *change* course — expand if things go well, abandon if they go badly, wait before investing. That flexibility has value that a static DCF misses. This chapter introduces real options; then looks at how intangible assets and environmental, social and governance (ESG) factors are reflected in a DCF.

23.1 The idea of real options

A **real option** is the right, not the obligation, to take a business action at a future date at a predetermined cost. Financial options give rights over securities; real options give rights over real activities.

Real option	Analogous to	Example
Option to expand	Call option	Pilot plant → full-scale plant; first-generation product → follow-ons
Option to delay (wait)	Call option	Mining or oil lease: develop when prices justify
Option to abandon	Put option	Exit a project for salvage value if it fails
Option to switch	Exchange option	Flexible plant that can produce either product
Option to contract	Put option	Scale down in a downturn
Growth option	Call on future projects	A start-up's right to enter adjacent markets

Why a DCF ignores them. A static DCF assumes a fixed plan. When uncertainty is large and flexibility is genuine, the expected value of the plan with flexibility exceeds the value of the plan without — the difference is the option value.

23.2 Valuing an option with Black–Scholes (or a binomial tree)

Map the real option to a financial option:

Financial option parameter	Real-option equivalent
Stock price S	PV of expected cash flows from the project/asset if undertaken now
Strike K	Investment cost required to exercise
Time T	Time until the decision must be made (lease expiry, patent life)

Financial option parameter	Real-option equivalent
Volatility σ	Volatility of the project's value (from comparable firms or scenario spread)
Risk-free r	Risk-free rate
Dividend yield	Cash-flow leakage from delay (competitive erosion)

23.2.1 Worked example

A company can build a new plant for ₹400 crore. If built today the PV of cash flows is ₹300 crore, so NPV = −₹100 crore and the project is rejected. But the company holds a licence valid for 5 years allowing it to build any time. Project value volatility 35%; r = 7%.

Black–Scholes: S = 300, K = 400, T = 5, σ = 35%, r = 7% → option value ≈ **₹98 crore**. The licence is worth nearly ₹100 crore even though the project has negative NPV today — it gives the right to build if demand or prices rise. A static DCF would assign zero.

23.2.2 Cautions

- Real-option valuation is only meaningful when the option is *real* (the firm truly has the right, exclusive or protected) and *flexibility is valuable* (uncertainty is high).
- Volatility is hard to estimate; ranges of option values are wide.
- Options that can be exercised by competitors (shared options) are worth less.
- Don't double-count: if the DCF already includes probability-weighted scenarios with managerial responses, adding an option premium overstates value.

23.3 Decision trees as a practical alternative

A decision tree with explicit probabilities and management responses is often easier to explain and audit than Black–Scholes.

Example. A pharma project: Phase II costs ₹50 crore; success probability 40%. If successful, Phase III costs ₹150 crore with 60% success; if approved, PV of launch cash flows ₹1,500 crore. Work backwards: Value at Phase III start (success) = 0.6 × 1,500 − 150 = ₹750 crore. Value at Phase II start = 0.4 × 750 − 50 = ₹250 crore (before discounting for time). Flexibility to abandon after a failed Phase II avoids the Phase III cost; a static DCF that assumes all phases are undertaken would have computed

$0.4 \times 0.6 \times 1{,}500 - 50 - 150 = 360 - 200 = 160$ — understating value by ₹90 crore because it forces spending on Phase III even after Phase II failure.

23.4 Valuing intangible assets

In many businesses the value is in brands, customer relationships, technology, patents and data. They appear in a DCF through the cash flows they enable, not separately — except when they are valued individually for transactions, purchase price allocation, impairment or transfer pricing.

Technique	Concept	Typical use
Relief-from-royalty	Value = PV of royalties saved by owning rather than licensing the asset	Brands, trademarks, technology
Multi-period excess earnings (MPEEM)	Cash flows attributable to the asset after deducting charges for contributory assets	Customer relationships, leading intangible
With-and-without	Difference in business value with and without the asset	Non-compete agreements, key contracts
Greenfield	Value of a start-up business with the asset as the only one in place	Licences
Cost approach	Cost to recreate	Software, workforce
Market approach	Comparable transactions	Patents, domain names

Relief-from-royalty example. A brand supports revenue of ₹1,000 crore growing 8% for 5 years then 5%. A market royalty of 3% of revenue is considered arm's length; tax 25%; discount rate 13%; perpetual life. Royalty after tax in year 1 = $1{,}000 \times 1.08 \times 3\% \times 0.75$ = ₹24.3 crore. A rough value: ₹24.3 crore growing at ~7% over a long horizon at 13% → $24.3 \div (0.13 - 0.07) \approx$ ₹405 crore (an approximation; an explicit schedule would adjust the growth fade and add a tax-amortisation benefit of ~10–15%).

23.5 ESG and valuation

ESG — environmental, social and governance — matters for valuation through three channels:

1. **Cash flows:** revenues (green products, carbon pricing, boycotts), costs (compliance, energy), capex (decarbonisation), fines and litigation.
2. **Discount rate:** exposure to systematic ESG risk (climate transition, regulatory shifts) may affect cost of capital; green and sustainability-linked financing may reduce cost of debt.

3. **Terminal value and growth:** durability of the business model under climate and regulatory scenarios.

23.5.1 Incorporating ESG into cash flows (preferred)

Pillar	Examples	How to reflect
E	Carbon price on emissions; energy transition capex; physical climate risk (flooding, heat)	Scenario analysis: carbon-price paths (e.g., ₹ per tonne CO_2e), 2°C scenarios; add capex for decarbonisation; reduce revenue where assets are at physical risk
S	Labour standards, health and safety, community relations, product safety	Cost of compliance and welfare; churn and productivity; fines; reputational revenue effects
G	Board independence, related-party dealings, accounting quality, anti-corruption	Probability-weighted penalties and fines; leakage to promoters; cost-of-capital effect for poor governance in some contexts

23.5.2 Worked example: carbon cost for a cement producer

A cement producer emits 0.8 tonnes of CO_2 per tonne of cement and produces 10 million tonnes: 8 million tonnes CO_2 a year. Suppose a regulatory carbon price of ₹1,000 per tonne from year 5, rising 10% a year. Cost in year 5 = 8 × 1,000 = ₹800 crore *if all emissions are priced and not abatable*; with abatement capex that cuts emissions by 30% by year 8 at ₹1,500 crore, the net after-tax effect on FCFF can be modelled. Without such modelling the valuation overstates value in a transition scenario.

23.5.3 Discount-rate approaches

Some practitioners add an ESG risk premium (e.g., 25–100 bp) for firms with poor ESG scores; others reduce the cost of debt for sustainability-linked loans. Because the evidence on a systematic ESG risk premium is mixed and an add-on is opaque, *cash-flow scenarios are more transparent*: they show *what* is risked and *when*.

23.5.4 Reporting frameworks

Global frameworks (IFRS Sustainability Disclosure Standards, ISSB) and the Indian SEBI's BRSR (Business Responsibility and Sustainability Report, with a "core" set for large listed companies) provide data inputs. ESG ratings differ widely between providers; use primary data over scores where possible.

23.5.5 ESG valuation checklist

1. Identify material ESG issues by sector (materiality maps).

2. Quantify exposure (emissions, water, safety incidents, governance flags).
3. Build scenarios for regulation, carbon price and demand shifts.
4. Put costs and revenues in the cash flows with timing.
5. Document discount-rate choices; avoid double-counting.
6. Reflect the *opportunity* side as well: green products, lower financing cost, supply-chain resilience.

Chapter 23 summary

- Real options capture flexibility that a static DCF misses: expand, delay, abandon, switch.
- Black–Scholes and decision trees are the main tools; both need credible inputs.
- Intangible assets are valued through royalty, excess-earnings, with-and-without and cost techniques.
- ESG is best reflected in explicit cash-flow scenarios (carbon price, capex, fines, welfare cost) rather than in an opaque discount-rate premium.

Review questions

1. Why can a project with negative NPV be worth pursuing?
2. Map the inputs of the Black–Scholes model to a mining lease option.
3. Compute the pharma pipeline value if Phase II costs 40, p(II) = 50%, Phase III costs 120, p(III) = 50%, launch value 1,200.
4. Why is the relief-from-royalty method sensitive to the royalty rate?
5. How would you incorporate a future carbon price into FCFF?

Chapter 24 — Pitfalls, Governance and Professional Practice

Technique is necessary but not sufficient. This final chapter collects the most common errors, describes how to review and audit a DCF model, and sets out the conventions of a professional valuation report — including the ethical dimension of an exercise that is routinely pressured to deliver a particular answer.

24.1 The twenty most common DCF errors

Cash-flow errors

1. **Growth without reinvestment.** Forecast revenue grows; capex and working capital do not.
2. **Terminal reinvestment inconsistent with terminal growth and ROIC** (Chapter 12).
3. **Using accounting profit instead of free cash flow** (ignoring working capital, capex).
4. **Double-counting the tax shield** — in FCFF *and* in WACC.
5. **Treating stock compensation as free** (adding back to FCFF).
6. **Mismatched lease treatment** across EBITDA, FCFF, WACC and bridge.
7. **Including non-operating income in FCFF and valuing the asset again in the bridge.**
8. **Peak-year or trough-year base** for a cyclical business.

Discount-rate errors

9. **Discounting FCFE at WACC or FCFF at the cost of equity.**
10. **Currency mismatch** — rupee flows at a dollar rate.
11. **Beta not re-levered** to the target capital structure.
12. **Book-value weights** in WACC.
13. **Inconsistent Rf and ERP** (different currencies, maturities or dates).
14. **A "conservative" buffer added to WACC** on top of conservative cash flows.

Terminal-value errors

15. **Terminal growth greater than the risk-free rate or long-run nominal GDP growth.**
16. **Terminal margins or ROIC at peak levels with no competitive justification.**
17. **Exit multiple taken from an inflated market without cross-checking the implied growth.**

Bridge and reporting errors

18. **Omitted debt-like items** (pensions, leases, deferred consideration, guarantees).
19. **Using the wrong share count** (basic vs diluted; stale; ignoring buybacks, new issues).
20. **No range, no sensitivities, no statement of what drives the value.**

24.2 Model governance

A DCF model is a decision tool and, in many settings, a legal and audit document. It should be built and controlled like software.

24.2.1 Structure

- **One input sheet;** no hard-coded numbers in formulas.
- **Clear flow:** inputs → calculations → outputs; consistent column structure (time runs left to right).
- **Colour coding:** blue for inputs, black for formulas, green for links to other sheets.
- **Separate scenarios** with a switch; keep a documented base case.
- **Version control** and change log.

24.2.2 Checks

- Balance sheet balances (if a three-statement model is built).
- Cash reconciles across statements.
- FCFF reconciles using two methods (EBIT-based, net-income-based).
- Sources and uses of cash tie.
- Growth and ROIC within expected bounds in each year; flags for outliers.
- Terminal-year checks: $g \leq Rf$, $WACC - g \geq 3$ points, reinvestment ≥ 0, capex $\geq$ depreciation.

- Sensitivity-table base cell equals model output.
- Circularity handled and documented (iteration switch).

24.2.3 Independent review

Where stakes are high, a second person should rebuild key calculations independently, test extreme inputs (zero growth, high WACC) and review the logic and assumptions, not only the spreadsheet formulas.

24.3 The valuation report

A professional valuation report is judged by whether another competent valuer could follow, reproduce and challenge it. Typical structure:

1. **Executive summary:** conclusion of value, standard and premise of value, date, purpose, key assumptions, range.
2. **Engagement terms:** subject, purpose, valuation date, scope, sources, limitations, independence declaration.
3. **Business and industry description.**
4. **Financial analysis:** historical performance and normalisation adjustments.
5. **Methodology:** approaches considered and rejected, those adopted and why.
6. **Assumptions:** forecast, discount rate, terminal value — each with a source or rationale.
7. **Calculations and results:** DCF exhibits, bridge, multiples.
8. **Sensitivity and scenario analysis.**
9. **Cross-checks and reconciliation.**
10. **Conclusion** and range.
11. **Appendices:** documents, data sources, calculations, glossary.
12. **Limiting conditions** and reliance statements.

Under the ICAI Valuation Standards and the registered-valuer rules, specified disclosures about the valuer's independence, the information relied on, the inspection or management representations obtained, and the limitations are required. Check current requirements.

24.4 Independence, bias and ethics

Valuation is often commissioned by a party with a preferred answer: a promoter who wants a high value for a share issue, a buyer who wants a low value for a purchase, a company that wants to avoid impairment. Structural pressure produces predictable biases.

Types of bias

- **Client bias:** unconsciously adopting assumptions that favour the engagement sponsor.
- **Anchoring:** settling on a number early and adjusting only slightly.
- **Confirmation bias:** seeking evidence for the expected value.
- **Optimism bias in forecasts:** especially in management plans.
- **Precision illusion:** reporting spurious decimals.

Safeguards

1. State assumptions *before* computing the answer, and document why.
2. Obtain independent evidence for key inputs (peer data, industry reports, historical test).
3. Use ranges and scenario weighting.
4. Disclose conflicts and the reliance placed on management.
5. Retain the ability to say no. A valuer whose conclusion can be dictated is not a valuer.
6. Test the plan against history (hit-rate of management budgets; margin vs peers).

24.5 Management plans and forecasts: how to challenge them

1. **Compare with history.** Ask for the last 5 years' budgets and outcomes; compute the average over-optimism.
2. **Test against industry.** Does the plan imply growth or margin outside the peer range?
3. **Test the drivers.** Capacity, headcount, distribution reach: do they support the revenue?
4. **Reconcile the cash.** Does the plan's cash flow tie to the funding plan?
5. **Examine the first two years carefully;** the later years are less reliable, but their error sits mainly in the terminal assumptions.

6. **Understand incentives:** budgets prepared for fundraising or covenant negotiation are biased.

24.6 Quick diagnostics: is this DCF credible?

Ask these ten questions of any DCF you receive.

1. What is the standard and premise of value, and is the method suitable?
2. What share of value is in the terminal value, and how was it derived?
3. What are the implied terminal growth, ROIC and multiple?
4. Are growth and reinvestment linked? What ROIC path does the forecast imply?
5. Are margins consistent with the competitive position and peers?
6. Is the WACC built bottom-up with consistent inputs? Does it match market evidence?
7. Is the bridge complete?
8. Do the sensitivities show a range, and which inputs drive it?
9. Is the result consistent with market multiples, and if not, why?
10. Is the report transparent about limitations?

24.7 Valuation in negotiations and disputes

- **For buyers:** use the DCF to set a walk-away price: the value with realistic synergies shared with the seller. Calculate the *maximum* you can pay while earning your cost of capital.
- **For sellers:** use the DCF to articulate the story; defend it with evidence; know the buyer's likely synergies.
- **For tribunals and courts:** the valuer's duty is to the court. Courts scrutinise independence, methodology, internal consistency and reliance on market evidence; unexplained "adjustments" and results that move with the client's interests are damaging.

24.8 Keeping skills sharp

- **Back-test:** compare past valuations to what happened (growth, margins) and to later market prices.
- **Study market-implied expectations** regularly.
- **Learn from errors:** keep a log of assumptions that proved wrong and why.

- **Update data** (risk-free rates, ERP, betas, multiples) at each engagement; never recycle stale inputs.

24.9 A final word

DCF is not a machine that produces truth; it is a *discipline for thinking about value*. It forces an analyst to say what she believes about the future and to carry those beliefs through to a number, consistently. When it is done well, the valuation is less a pronouncement than an argument: *here is what must be true for this business to be worth this much, and here is how confident we are.* The reader may disagree with the assumptions — and the model should make that disagreement easy to express and to test. That is its real value.

Chapter 24 summary

- Know the twenty most common errors; most are consistency violations.
- Build the model like software; test it; have it independently reviewed.
- Report assumptions, ranges and limitations; keep independence.
- Challenge management plans against history and peers.
- DCF is a discipline for thinking, not a source of truth.

Review questions

1. List five consistency checks that can be built into a DCF model.
2. How can a valuer guard against client bias?
3. What should an executive summary of a valuation report contain?
4. How would you test the credibility of a management plan?
5. Choose three errors from Section 24.1 and explain how each would distort value.

Part VI — Applications and Extensions

Chapter 25 — Valuation in Mergers and Acquisitions

The valuation techniques of the preceding chapters are most often put to work in a transaction. M&A adds three layers: the value of the target *on its own*, the value of *synergies* that the combination creates, and the question of *who captures* that value. This chapter shows how to separate them, how to price a deal and how to test its effect on the buyer's shareholders.

25.1 Three values in every deal

1. **Standalone value** of the target: what the business is worth under its current owner and strategy (the DCF of Chapter 14).
2. **Value to the acquirer:** standalone value *plus* the present value of synergies and of any improvements the acquirer can make, *minus* integration and transaction costs.
3. **Price:** what is actually paid, somewhere between the seller's reservation value (at least the standalone value, or the next-best alternative) and the buyer's maximum.

Value created for the buyer = Value to the acquirer − Price paid
Value captured by the seller = Price paid − Standalone value

Where the price falls between standalone value and value-to-acquirer is a matter of negotiation, competition and information. A buyer who pays the *whole* of the synergy value to the seller earns exactly its cost of capital on the deal — no more.

> **Key idea.** Synergies are not free. The seller usually demands a share; competition among bidders pushes the whole of it to the seller; and many synergies never materialise. The disciplined buyer determines a *walk-away price* equal to standalone value plus a share of *risk-adjusted* synergies.

25.2 Types of synergy

Type	Source	Reliability	Typical treatment
Cost synergies	Overlapping overheads, procurement scale, plant rationalisation, logistics	Highest	Quantify run-rate savings, ramp-up and one-off costs to achieve

Type	Source	Reliability	Typical treatment
Revenue synergies	Cross-selling, wider distribution, joint product development, pricing power	Lowest	Probability-weight; include only with specific evidence; include the margin and the reinvestment needed
Financial synergies	Lower cost of debt from a larger balance sheet, use of tax losses, debt capacity	Medium	Value via WACC/APV; check that the lower cost of capital is real
Operating improvements ("control" value)	Better management of the target's assets: margin, working capital, capex	Medium	Value as a separate scenario of the standalone DCF, labelled as buyer-specific
Dis-synergies	Customer loss, talent attrition, cultural friction, distribution-channel conflict	Often under-estimated	Deduct; many deals fail here

25.3 Valuing synergies by DCF

Treat synergies as a separate project: forecast incremental after-tax cash flows, subtract the cost to achieve and discount at a rate that reflects their risk.

1. **Run-rate benefit** (₹ crore a year, pre-tax) and **ramp-up profile** (e.g., 33% in year 1, 66% in year 2, 100% from year 3).
2. **One-off costs to achieve** (severance, systems, plant closure).
3. **Tax** on net synergy.
4. **Reinvestment** — revenue synergies need working capital and capex.
5. **Discount rate:** usually the target's WACC *plus* a premium (1–3 points) for execution risk; cost synergies are less risky than revenue synergies, so some practitioners use different rates for each.
6. **Terminal value:** cost savings may persist with modest growth (2–3%) or erode under competition; revenue synergies should fade more quickly.

25.3.1 Example: Crestline Brands buys Meridian

Crestline, a larger FMCG group, studies buying 100% of Meridian. Its analysis (₹ crore):

- Cost synergies at run-rate of **1.5% of Meridian's revenue** (procurement, logistics, overheads), reached in three years: 33% in year 1, 66% in year 2, 100% thereafter.
- One-off costs of ₹30 crore in each of years 1 and 2.

- Tax 25%; discount rate 12.2% (Meridian's WACC 11.225% plus about one point for execution risk); synergies grow 3% after year 5.

₹ crore	Y1	Y2	Y3	Y4	Y5
Pre-tax run-rate savings (1.5% × revenue × ramp)	11.3	25.3	42.1	45.9	49.6
Integration cost	(30.0)	(30.0)	—	—	—
Net after-tax synergy cash flow	(14.0)	(3.5)	31.6	34.4	37.2

PV of years 1–5 at 12.2% = **₹49.7 crore**. Terminal value at year 5 = 37.2 × 1.03 ÷ (0.122 − 0.03) ≈ **₹416.5 crore**; PV = **₹234.2 crore**. **Total PV of synergies = ₹283.9 crore, or ₹5.68 per Meridian share** — about 11% of standalone value.

25.3.2 Pricing

- Standalone intrinsic value: **₹51.27** per share.
- Value to Crestline = 51.27 + 5.68 = **₹56.95** per share. This is the *maximum* it should pay (the walk-away price).
- Suppose Meridian's unaffected market price is ₹42, and Crestline offers a **25% premium: ₹52.50** per share (₹2,625 crore for 50 crore shares).
- Seller's gain over the market price: ₹10.50 (25%); over intrinsic value: ₹1.23.
- Buyer's value creation = 56.95 − 52.50 = **₹4.45 per share (₹222 crore)** — about 78% of synergy value retained by the buyer.

If a rival bidder were to push the price to ₹56.95, the buyer's NPV would be zero. A *winner's curse* arises when the winning bid is the one with the most optimistic estimate of synergy.

25.4 Accretion and dilution analysis

Boards also look at the effect of the deal on the acquirer's earnings per share (EPS). It is a *market-optics* test, not a value test: a deal can be EPS-accretive and value-destroying (buying a low-quality business with cheap debt), or EPS-dilutive and value-creating.

25.4.1 Example

Crestline: net income ₹4,000 crore; 200 crore shares; EPS ₹20.00; P/E 30× → share price ₹600. Meridian's FY26 net income: ₹139.5 crore (EPS ₹2.79); the offer of ₹52.50 implies P/E of 18.8×.

Scenario	Combined net income	Shares	EPS	vs ₹20.00
All-cash (new debt of ₹2,625 crore at 9%: after-tax interest ₹177.2 crore); no synergies	4,000 + 139.5 − 177.2 = 3,962.3	200	19.81	−0.9%
All-cash, with year-5 synergies (₹37.2 crore post-tax)	3,999.5	200	20.00	0.0%
All-stock (2,625 ÷ 600 = 4.375 crore new shares); no synergies	4,139.5	204.375	20.25	+1.3%

Why does the stock deal accrete when the cash deal does not? Because Crestline's P/E (30×) is higher than the price paid for Meridian (18.8×): the acquirer is "buying earnings with expensive paper". The earnings yield on Crestline stock (3.3%) is lower than the after-tax cost of debt (6.75%), so paying with stock is cheaper *in EPS terms*. That says nothing about value: stock issued at a high multiple shares the combined firm's risks and rewards with the seller's holders.

25.4.2 Exchange ratio

In a stock deal the exchange ratio = Offer price per target share ÷ Acquirer share price = 52.50 ÷ 600 = **0.0875 Crestline share per Meridian share**. In practice the ratio is fixed by *relative* valuations of the two companies on the same basis (DCF, multiples, NAV), not by market prices alone; fairness opinions test it.

25.5 Cash, stock or mixed

Consideration	Seller's risk	Buyer's advantage	Considerations
Cash	None after closing	Keeps all upside and synergy; no dilution	Needs financing; tax on seller; leverage rises
Stock	Shares post-merger risk and synergy	Conserves cash; shares risk of overpaying	Dilution; relies on acquirer's valuation being fair
Mixed / earn-out	Partial	Balances	Contingent payments align incentives; need careful valuation

In stock deals, the fairness of the exchange ratio depends on *both* companies' valuations; mis-valuing the acquirer's own shares can transfer wealth to the seller.

25.6 Control premium and its sources

Observed premiums over unaffected prices are often 20–40% but vary enormously by market and sector. A control premium may reflect: (i) synergies, (ii) better management, (iii) removal of a governance discount, (iv) the market undervaluing the target, (v) competitive bidding, and (vi) the buyer's overpayment (hubris). Of these, only (i)–(iii) are fundamentals; (iv) justifies a lower offer if the DCF supports it; (v) and (vi) are traps.

25.7 Transaction mechanics that affect value

- **Debt-free, cash-free pricing:** headline price quoted on an enterprise-value basis; adjustments for net debt, working-capital peg, and debt-like items at closing.
- **Working-capital peg:** an agreed "normal" level of working capital; any shortfall or excess adjusts price.
- **Escrow and indemnities:** protect the buyer from undisclosed liabilities; affect the effective price.
- **Earn-outs:** part of consideration contingent on performance; value as a contingent claim (probability-weighted PV).
- **Tax structuring:** stock vs asset purchase, step-up in tax basis, share-swap relief, section 79/72A-type loss carry-forward limits.
- **Regulatory approvals** (competition, sectoral, FEMA pricing, SEBI takeover code) may require minimum prices and open offers.

25.8 Purchase price allocation (PPA)

After closing, Ind AS 103 requires the price paid to be allocated to identifiable assets and liabilities at fair value, with the residual recorded as goodwill. The PPA valuation uses the intangible-valuation methods in Chapter 23 and is reviewed by auditors. Goodwill is tested annually for impairment (Ind AS 36); a goodwill impairment is the accounting recognition that the price paid exceeded value.

25.9 Fairness opinions

A fairness opinion, typically by an investment bank or independent valuer, states whether the consideration is *fair, from a financial point of view*, to a specified party. It usually contains DCF, trading comparables, precedent transactions and a premium analysis, presented as ranges on a football

field. A fairness opinion is not a valuation *report* in the statutory sense and does not recommend the transaction; its limitations (reliance on management data; no independent verification) are stated up front.

25.10 Checklist for a buyer

1. Standalone DCF built independently of management's pitch; reconcile with market price.
2. Synergies itemised, owned (who delivers?), costed and risk-adjusted.
3. Walk-away price = standalone + a share of risk-weighted synergies.
4. Sensitivity of value to the three biggest drivers (typically margin, growth and synergy delivery).
5. Funding plan and leverage after closing, including covenant headroom.
6. Accretion/dilution as a secondary, optical test.
7. Integration plan with milestones and accountability.
8. Contingency: what if synergies are half and delayed by a year?

Chapter 25 summary

- Separate standalone value, synergy value and price.
- Value synergies as a project with ramp-up, costs and a risk-adjusted discount rate.
- Buyer's maximum = standalone + risk-weighted synergies; every rupee above destroys value.
- Accretion/dilution is an optical test; stock deals accrete when the acquirer's P/E exceeds the price paid.
- Crestline/Meridian: synergy value ₹284 crore (₹5.68 per share); max price ₹56.95; offer ₹52.50 gives the buyer ₹4.45 per share of value.

Review questions

1. Why is a revenue synergy riskier than a cost synergy?
2. Compute the exchange ratio if the offer price is ₹70 and the acquirer's price is ₹560.
3. Why can a deal be accretive and still destroy value?
4. What is the winner's curse in an auction?
5. How is an earn-out valued?

Chapter 26 — Private Companies, Minority Stakes and Disputes

Most businesses are not listed. Valuing a private company brings extra issues: no market price, no observable beta, owner-managers whose personal finances are entangled with the business, illiquid shares and often an adversarial setting — a family settlement, a shareholder dispute, a tax assessment. This chapter collects the techniques that adapt DCF to these situations.

26.1 What changes for a private company

Issue	Listed company	Private company
Price signal	Daily market price	None; transactions rare
Beta	Regression available	Bottom-up from peers only
Information	Audited, regulated disclosures	Variable quality; management accounts
Owner involvement	Professional management	Owner-managers; key-person risk
Liquidity	High	Low; marketability discount
Control	Dispersed ownership	Often concentrated; control matters
Governance	Board, regulations	Informal; related-party dealings
Capital structure	Policy-driven	Ad hoc; sometimes all-equity or family loans

26.2 Normalising a private company's financials

Before forecasting, adjust history for the distortions common in owner-run businesses (see Chapter 8):

1. **Owner compensation** — replace actual with market-rate salary for the role(s) performed. If the owner works 70 hours a week for ₹12 lakh and a hired CEO would cost ₹60 lakh, deduct the difference.
2. **Personal and discretionary expenses** run through the company (vehicles, travel, family salaries not working in the business) — add back.
3. **Related-party transactions** at non-market terms (rent paid to the owner's other entity, purchases from a family firm) — restate at market prices.
4. **Non-operating assets** (land, personal investments, loans to relatives) — remove income and value separately.

5. **Revenue recognition and accounting practice** — cash-basis or tax-driven accounting may need conversion to accrual.
6. **Unrecorded liabilities** — leave encashment, gratuity, deferred taxes, disputed statutory dues.
7. **Under-investment** — maintenance capex deferred to inflate profits.

26.2.1 Key-person dependence

If revenue depends on a few relationships held by the owner, value depends on whether they will stay and whether they can be transferred. Reflect this in scenarios (e.g., 20% revenue loss over two years if the key person leaves) or in a transition arrangement priced into the deal (earn-out, non-compete).

26.3 The discount rate for private companies

- **Risk-free rate and ERP:** as for public companies.
- **Beta:** bottom-up from listed peers, un-levered, re-levered at the target (industry-average) capital structure of the private firm — not its current, possibly idiosyncratic one.
- **Size premium:** applied if justified by data (small private firms are riskier than large listed peers).
- **Company-specific risk:** prefer scenarios; if an add-on is used, quantify and justify (key-person, concentration).
- **Cost of debt:** the firm's actual borrowing rate on a normalised basis (a family loan at 0% is not a market rate).

Total beta. For an *undiversified* owner, the relevant risk is total risk, not just market risk. *Total beta = Market beta ÷ Correlation of the firm with the market.* If market beta is 0.9 and the correlation is 0.5, total beta = 1.8 — double. This gives an **investment value to that specific owner**, not a fair market value. Fair market value assumes a hypothetical willing buyer who is diversified.

26.4 Discounts and premiums: a coherent framework

Value measurements pass through several levels:

Level	What it values	Adjustment to move to next level
1. Control, marketable	100% of equity, freely tradable, controller can change policies	− Discount for lack of control (DLOC) if valuing a minority

Level	What it values	Adjustment to move to next level
2. Minority, marketable	A non-controlling stake that could be sold on an active market	− Discount for lack of marketability (DLOM) if the shares are illiquid
3. Minority, non-marketable	A non-controlling stake in a private company	

Which level did the DCF produce? If the forecast is the management plan under current policies (no restructuring upside) and the discount rate is a public-market rate, the result is close to **Level 2** (minority, marketable). If it includes buyer-specific improvements, it moves toward Level 1. The valuer must state the level and apply only the adjustments that bridge to the level of the interest being valued.

26.4.1 DLOC

The reciprocal of the control premium: **DLOC = 1 − 1/(1 + control premium)**. A 30% control premium → 23.1%. Premiums derived from public-company takeovers include synergy and are not pure "control". Use them with caution; the actual value of control depends on how much the controller can improve the business and how easily.

26.4.2 DLOM: methods

Several approaches estimate how much less an unlisted stake is worth because it cannot be sold quickly.

1. **Restricted-stock studies.** Compare the price of restricted shares of listed companies (which cannot be sold for a period) with the market price of freely tradable shares of the same company. Historical discounts were often 15–35% for older studies, lower in recent periods as holding periods shortened.
2. **Pre-IPO studies.** Compare prices at which private shares changed hands before an IPO with the IPO price. Discounts of 30–50% have been reported in older studies, but they conflate marketability with expected growth and information.
3. **Option-pricing models.** A holder who cannot sell for T years is like a holder who *buys a protective put* with maturity T: the put's cost approximates the loss of marketability. For an at-the-money put on a stock with 35% volatility, risk-free 7% and 3-year restriction, the Black–Scholes put value is about **13%** of the stock price (see the table).
4. **Discounted expected holding costs and liquidity premium.** A model-based approach; less common.

Put value as % of stock price (at-the-money, r = 7%)	1 year	2 years	3 years	5 years
Volatility 25%	6.6%	7.6%	7.8%	7.4%
Volatility 35%	10.3%	12.4%	13.3%	13.6%
Volatility 45%	14.0%	17.3%	18.9%	19.8%

The protective-put model gives the *upper* end for an investor with certainty of no sale; many practitioners cross-check it with empirical studies.

26.4.3 Factors that affect DLOM

Expected holding period; volatility of the business; dividend payout (a high payout reduces the need to sell); size of the block; transfer restrictions; prospects of an IPO or sale; information quality; whether the interest has put rights or tag-along/drag-along rights.

26.4.4 Worked example

A DCF gives equity value of ₹1,000 crore for a closely held company (Level 2: minority, marketable). A promoter holds 10%; the valuation is for a gift of that stake.

- Pro-rata value: ₹100 crore.
- DLOM selected at 20% (between the restricted-stock and option-model indications) → **₹80 crore**.

If instead a strategic buyer acquires 100% in an arm's-length negotiated deal, the price is based on the *control* value, with no DLOM; synergies enter via the strategic premium.

26.5 Applying standards in disputes

Courts and tribunals approach valuation differently from commercial buyers.

- **Oppression and mismanagement; buy-out of a minority:** fair value is typically computed *without* a minority discount, because the minority is being compelled out. A marketability discount may also be rejected.
- **Dissenting shareholders in mergers:** fair value in appraisal proceedings; method and inputs scrutinised.
- **Family settlements:** often negotiated; valuation seeks a neutral reference with documented method.

- **Tax matters:** the statutory method may be prescribed (for example, rules for valuing unquoted shares); where a choice exists, consistency and documentation matter.
- **Matrimonial and succession cases:** valuation of a business interest as part of a division of property; standard-of-value rules depend on jurisdiction.

In each context the valuer's duty runs to the tribunal, and the court will examine independence, internal consistency and reliance on market evidence.

26.6 Employee stock option plans (ESOPs) in private companies

Ind AS 102 requires the grant-date **fair value** of employee options to be expensed over the vesting period. Steps:

1. Determine the **fair value of the underlying share** at grant date (DCF/market approach for a private company).
2. Choose the **option-pricing model** (Black–Scholes or binomial/lattice).
3. Inputs: share price, exercise price, expected term, risk-free rate, expected volatility (from peers for a private company), dividend yield.
4. Adjust for forfeitures and service/performance conditions.

Example. Share price ₹100, exercise price ₹100, expected term 4 years, volatility 40%, risk-free 7%: Black–Scholes call ≈ ₹41 (about 41% of the share price; a long term and high volatility make options valuable). Granting 10 lakh options implies a fair-value expense of about ₹4.1 crore over the vesting period.

26.7 Valuation date and subsequent events

The valuation date fixes the information set: only information known or knowable at that date should be used. Events after the date are relevant only if they provide evidence of conditions that existed at the date; otherwise they should be disclosed but not reflected. Pay attention in fast-moving situations (a regulatory announcement a week after the date, a customer loss) and in tax or legal settings where dates are prescribed.

26.8 A private-company valuation checklist

1. Standard and premise of value stated; interest (100%, minority, specific shares) defined.
2. Financials normalised, with owner and related-party adjustments supported.
3. Peers selected on business, size and risk; betas bottom-up.
4. Cash-flow forecast with key-person and concentration risks in scenarios.
5. Level of value identified; DLOC/DLOM applied only if appropriate, with support.
6. Cross-check with market multiples (adjusted for size and liquidity) and transaction data.
7. Waterfall for multiple share classes.
8. Subsequent events and valuation-date discipline.
9. Complete documentation and a transparent statement of limitations.

Chapter 26 summary

- Normalise owner-driven distortions; scrutinise key-person risk.
- Build beta bottom-up; the cost of capital should reflect a hypothetical diversified buyer for fair market value.
- Distinguish control vs minority and marketable vs non-marketable levels; apply discounts only as needed.
- DLOM tools: restricted-stock studies, pre-IPO studies, protective-put models.
- Dispute valuations often forbid minority and marketability discounts.

Review questions

1. What is total beta, and when would it be used?
2. How do you estimate DLOM using an option model?
3. A DCF gives ₹500 crore on a minority-marketable basis. What is the value of a 20% unlisted stake with a 15% DLOM?
4. What normalisations would you make for an owner-managed distribution business?
5. Why do courts often disallow minority discounts in oppression cases?

Chapter 27 — Second Case Study: A Listed Pharmaceutical Company

Chapter 14 valued a simple company with a clean balance sheet. This chapter works through a more realistic one: **Aster Pharma Ltd**, a listed company with heavier working capital, mid-year cash flows, leases, a minority interest in a subsidiary, a listed investment, an unfunded pension obligation and outstanding employee options. The purpose is to show how the equity bridge, the mid-year convention and the terminal value are handled when the real-world complications appear. Aster is fictional; its numbers are for teaching.

27.1 Engagement

- **Subject:** Aster Pharma Ltd (listed on NSE/BSE).
- **Valuation date:** 31 March 2026.
- **Purpose:** Estimate fair value of equity per share, as input to a board decision on a buy-back.
- **Standard of value:** Fair value (market-participant basis); going concern; no control premium (minority, marketable).
- **Approach:** FCFF DCF with mid-year discounting; cross-checks with trading multiples and sensitivities.
- **Units:** ₹ crore, nominal.

27.2 Business and base-year data

Aster develops and sells branded generics in India (about 60% of revenue) and exports (40%). It has strong chronic-therapy brands, a growing US generics business and a contract-manufacturing unit held through a 90%-owned subsidiary.

Base year FY2026	₹ crore
Revenue	4,000
EBITDA (after Ind AS 116: rent excluded, ROU depreciation in D&A)	960 (24.0%)
D&A (including right-of-use depreciation)	200 (5.0%)
EBIT	760
NOPAT at 25%	570
Operating invested capital	4,300
ROIC (base)	13.3%

Base year FY2026	₹ crore
Operating NWC	880 (22% of revenue)

Items for the bridge (at 31 March 2026):

Item	₹ crore
Borrowings	700
Lease liabilities (Ind AS 116)	150
Cash and bank balances (surplus)	400
Listed investment (non-operating, at market)	200
Non-controlling interest — 10% of the subsidiary, at estimated value	250
Unfunded pension/gratuity deficit (pre-tax 60)	45 post-tax
Value of outstanding employee options (option-pricing model)	80
Shares outstanding (basic)	100 crore

27.3 Forecast assumptions

Driver	FY27	FY28	FY29	FY30	FY31	Terminal
Revenue growth	12%	11%	10%	9%	8%	5%
EBITDA margin	24.5%	25.0%	25.5%	26.0%	26.0%	26.0%
D&A (% of revenue)	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%
Capex (% of revenue)	7.0%	7.0%	6.5%	6.5%	6.0%	via ROIC
NWC (% of revenue)	22%	22%	22%	22%	22%	via ROIC
Tax rate	25%	25%	25%	25%	25%	25%

Rationale: Domestic growth ahead of market on chronic-therapy share gains; export growth driven by US generics launches; margin expansion from mix and scale, held at 26% from FY30 (below the 28–30% of best-in-class peers); capex heavy in the first years for capacity and a new R&D facility (R&D expensed); working-capital intensity constant.

27.4 Free cash flow forecast

₹ crore	FY27	FY28	FY29	FY30	FY31
Revenue	4,480.0	4,972.8	5,470.1	5,962.4	6,439.4
EBITDA	1,097.6	1,243.2	1,394.9	1,550.2	1,674.2
D&A	(224.0)	(248.6)	(273.5)	(298.1)	(322.0)
EBIT	873.6	994.6	1,121.4	1,252.1	1,352.3
NOPAT	655.2	745.9	841.0	939.1	1,014.2
Add: D&A	224.0	248.6	273.5	298.1	322.0
Less: capex	(313.6)	(348.1)	(355.6)	(387.6)	(386.4)

₹ crore	FY27	FY28	FY29	FY30	FY31
Less: increase in NWC	(105.6)	(108.4)	(109.4)	(108.3)	(104.9)
FCFF	**460.0**	**538.0**	**649.6**	**741.3**	**844.9**

Reconciliation of ROIC. Invested capital rises from 4,300 to 4,495, 4,703, 4,895, 5,092 and 5,262; ROIC on opening capital: 15.2%, 16.6%, 17.9%, 19.2%, 19.9%. This is well above WACC and rises across the forecast, which must be justified by the margin expansion (the main driver) and by the capital efficiency of the export business. We assume it *fades* to a terminal ROIC of **16%**, not the 20% reached in year 5.

27.5 The cost of capital

Input	Value	Source
Risk-free rate	7.0%	10-year G-Sec at the valuation date
Equity risk premium	6.0%	Total India ERP (as in the Meridian case)
Unlevered beta	0.80	Bottom-up, pharma peers
Target D/E	15% (D/V = 13.04%)	Peer average; leases included in debt
Levered beta	0.80 × [1 + 0.75 × 0.15] = 0.89	
Cost of equity	7.0% + 0.89 × 6.0% = **12.34%**	
Pre-tax cost of debt	8.5%	AA-rated; lease rate similar
After-tax cost of debt	6.375%	
WACC	0.8696 × 12.34% + 0.1304 × 6.375% = **11.56%**	

27.6 Mid-year discounting

Pharma cash flows accrue evenly through the year, so the discount factors use the mid-year convention: year t is discounted at $(1 + \text{WACC})^{t - 0.5}$.

	FY27	FY28	FY29	FY30	FY31
Time (years)	0.5	1.5	2.5	3.5	4.5
Discount factor at 11.56%	0.9468	0.8486	0.7607	0.6819	0.6112
FCFF	460.0	538.0	649.6	741.3	844.9
PV	435.5	456.5	494.2	505.5	516.4

Sum of PV (explicit) = **₹2,408.1 crore**.

27.7 Terminal value

- Terminal-year revenue = 6,439.4 × 1.05 = 6,761.4.
- Terminal EBIT margin = EBITDA 26.0% − D&A 5.0% = **21.0%** → EBIT 1,419.9; NOPAT = **1,064.9**.
- Reinvestment rate = g/ROIC = 5% ÷ 16% = **31.25%** → terminal FCFF = 1,064.9 × 0.6875 = **732.1**.
- TV = 732.1 ÷ (0.1156 − 0.05) = **₹11,157.1 crore** at FY31.
- Because the TV is a Gordon perpetuity of mid-year flows, it is discounted with the *same mid-year factor as the last explicit flow* (4.5 years): PV = 11,157.1 × 0.6112 = **₹6,819.2 crore**.

Enterprise value = 2,408.1 + 6,819.2 = ₹9,227.3 crore. Terminal value is **73.9%** of EV.

27.8 The equity bridge

₹ crore	
Enterprise value	9,227.3
Less: borrowings	(700.0)
Less: lease liabilities	(150.0)
Add: surplus cash	400.0
Add: listed investment (market value)	200.0
Less: non-controlling interest (10% of the subsidiary)	(250.0)
Less: pension/gratuity deficit (post-tax)	(45.0)
Less: value of outstanding employee options	(80.0)
Equity value to common shareholders	**8,602.3**
Shares outstanding (crore)	100.0
Value per share	**₹86.02**

27.8.1 Notes on each item

- **Leases.** Because EBITDA and FCFF are *before* lease payments (Ind AS 116), the lease liability is debt and is deducted; the cost of debt and the debt weight in WACC include it.
- **Listed investment.** Its dividends are not in EBIT; add at market value. If a sale would attract capital-gains tax, deduct the tax.
- **NCI.** The subsidiary is 90% owned; the consolidated FCFF includes 100% of its cash flows, so the 10% claim is deducted. The ₹250 crore is 10% of an *estimated equity value of ₹2,500 crore* (from a separate DCF/multiple), not the book value of NCI on the balance sheet.

- **Pension deficit.** An actuarial deficit of ₹60 crore pre-tax is deducted at ₹45 crore after the tax relief on contributions.
- **Options.** Outstanding options are valued with an option-pricing model (₹80 crore); *future* option grants are in the ESOP expense already deducted in EBITDA.

27.9 Sensitivities

Case	EV (₹ cr)	Value per share (₹)	vs base
Base (mid-year, WACC 11.56%, g 5%, ROIC 16%)	9,227.3	86.02	—
End-year discounting instead of mid-year	8,736.0	81.11	−5.7%
WACC +1 point (12.56%)	8,037.8	74.13	−13.8%
Terminal ROIC = WACC (no excess returns)	8,037.5	74.13	−13.8%
WACC −1 point (10.56%)	10,842.9	102.18	+18.8%
Terminal ROIC 20% (year-5 level sustained)	9,847.2	92.22	+7.2%
Terminal growth 4% / 6%	8,802.0 / 9,791.6	81.77 / 91.67	−4.9% / +6.6%

The mid-year convention adds about 5–6% — a *methodological* choice that is as large as a half-point change in the discount rate, so it must be stated. A one-point increase in WACC and an assumption that excess returns vanish after FY31 each remove about 14%; a one-point *decrease* in WACC adds almost 19% (the asymmetry is the convexity of the terminal-value formula).

27.10 Cross-checks

Measure	Aster at DCF value	Comment
EV / FY26 EBITDA (960)	9.6×	Assumed peer range 11–16× (illustrative) → market might pay more
EV / FY27E EBITDA (1,097.6)	8.4×	
EV / NOPAT (FY26)	16.2×	
Terminal EV / EBITDA (FY31)	6.7× on TV 11,157 ÷ EBITDA 1,674	Conservative

The DCF implies a lower multiple than the trading range assumed for the peers; the reverse DCF (Chapter 16) would show what extra growth or margin the market is assuming. For a buy-back decision the question is

whether the actual market price is below ₹86, with a margin of safety for model uncertainty.

27.11 Conclusion

Fair value of Aster's equity at 31 March 2026: ₹7,400–₹9,200 crore (₹74–₹92 per share), point estimate ₹8,600 crore (₹86), on a minority, marketable basis. The range runs from the downside cases (WACC +1 point, or no excess returns after FY31) to the upside case of a sustained 20% terminal ROIC. Key judgement areas: margin expansion to 26% EBITDA, a terminal ROIC of 16%, and the assumption of a stable 5% terminal growth.

27.12 What this case adds to Chapter 14

1. **Mid-year convention** and consistent treatment of the terminal value.
2. **A fuller bridge**: leases, investments, NCI, pension, options.
3. **Working-capital-heavy** business: ΔNWC of about ₹105 crore a year is a large claim on cash.
4. **ROIC path scrutiny:** the forecast produces 20% ROIC by year 5; the terminal fade to 16% is a conscious conservative decision.
5. **A larger share of value from near-term cash** (26% of EV in the explicit period vs 18% for Meridian) because of faster early cash generation and higher margins.

Chapter 27 summary

- Aster Pharma: EV ₹9,227 crore; equity ₹8,602 crore; ₹86.02 per share.
- The bridge removes leases, NCI, pension and options; adds cash and the listed investment.
- Mid-year discounting adds 5–6% relative to end-year; state the convention.
- Terminal ROIC and WACC are the dominant sensitivities again.

Review questions

1. Why is the lease liability deducted from EV when EBITDA is after Ind AS 116?
2. Why is the TV discounted at 4.5 years in this model?
3. How would a 5% sale-tax on the listed investment change the bridge?

4. Reconcile the ROIC path in the forecast with the terminal ROIC assumption.
5. What additional diligence would you do on the 10% NCI value?

Chapter 28 — A Library of Compact Valuations

This chapter presents six short valuations of different kinds of business. Each follows the same pattern — facts, approach, numbers, result, lesson — and illustrates one theme from the book. They are fictional and simplified; treat them as worked exercises that can be rebuilt in a spreadsheet in half an hour each.

Case A — An IT-services company (asset-light, high ROIC)

Facts. *IndiaTech Services Ltd*: revenue ₹6,000 crore; EBIT margin 22% (EBIT ₹1,320 crore); no debt; net cash ₹1,800 crore (surplus); 100 crore shares. Revenue growth 10%, 9%, 8%, 7%, 6%; margin steady at 22%. Marginal sales-to-capital 4.0 (very asset-light). Tax 25%.

Approach. All-equity firm: WACC = cost of equity = 7.0% + 0.90 × 6.0% = **12.4%**. Mid-year discounting. Terminal growth 5%, terminal ROIC 25% (strong, durable client relationships and brand, but fading from a very high current level).

Results (₹ crore).

Component	Value
PV of FCFF, years 1–5	4,213.2
PV of terminal value	9,753.6
Enterprise value	**13,966.9**
Add: net cash	1,800.0
Equity value	**15,766.9**
Per share	**₹157.67**

TV share of EV: 69.8%. EV/FY26 EBIT = 10.6×.

Sensitivities (₹ per share). Terminal ROIC 18%: 148.2; terminal ROIC = Ke (12.4%): 132.9; margin 20% instead of 22%: 144.5; WACC 13.4% (+1 point): 141.8.

Lessons.

1. An asset-light business has low reinvestment, so FCFF is close to NOPAT and growth costs little.

2. Because reinvestment is small, terminal ROIC matters less than for capital-intensive firms: even at ROIC = Ke the value falls "only" 16%.
3. The cash is surplus: it is added in the bridge and *not* inside the WACC.
4. Margin is the dominant driver: a 2-point fall cuts the value by about 8%.

Case B — A cement producer (cyclical, mid-cycle terminal)

Facts. *Rajmandir Cement*: current volume 8.0 million tonnes growing 5% a year; EBITDA per tonne follows the cycle ₹1,300, ₹1,000, ₹800, ₹1,100, ₹1,300 over the next five years (cycle average ≈ ₹1,100); D&A ₹300 crore; capex ₹520 crore a year; tax 25%; WACC 11.5%; terminal growth 4%; terminal ROIC 13%. Terminal volume = year-5 volume × 1.04.

Explicit cash flows (₹ crore).

Year	Volume (Mt)	EBITDA/t (₹)	EBITDA	EBIT	FCFF
1	8.40	1,300	1,092.0	792.0	374.0
2	8.82	1,000	882.0	582.0	216.5
3	9.26	800	740.9	440.9	110.7
4	9.72	1,100	1,069.6	769.6	357.2
5	10.21	1,300	1,327.3	1,027.3	550.5

PV of explicit FCFF (mid-year at 11.5%) = **₹1,203.7 crore**.

Terminal value under three views of "normal".

Terminal EBITDA/t	Terminal value (₹ cr)	PV of TV	EV	vs mid-cycle
₹1,100 (mid-cycle)	5,926.5	3,631.3	**4,835.0**	—
₹1,300 (year-5 peak, "naive")	7,396.8	4,532.2	5,735.9	**+18.6%**
₹800 (trough)	3,721.1	2,280.0	3,483.7	**−27.9%**

Lessons.

1. Terminal margins must reflect the *cycle*, not the year in which the forecast ends; here the forecast ends at a peak, so naive extrapolation overstates EV by almost a fifth.
2. The explicit period includes the downturn (years 2–3): cash flow falls 80% from year 1 to year 3 and recovers — a pattern lost if one used a single average margin from the start.
3. Debt covenants matter in the trough years: a leveraged cyclical may breach covenants even if intrinsic value is positive.

Case C — A non-banking finance company (excess-return model)

Facts. *Kaveri Finance Ltd*: book equity ₹6,000 crore; ROE 17%; equity (and loans) grow 18% a year for five years, then 8% in stable growth with terminal ROE 16%; Ke 14%.

Approach. FCFE = net income − growth in book equity (the retention needed to keep leverage constant). The five-year FCFEs are small or negative because growth absorbs all earnings: PV of explicit FCFE = **−₹282 crore**. Book equity at year 5 = ₹13,726 crore; terminal net income = 16% × 13,726 = 2,196; $FCFE_6$ = 2,196 × (1 − 0.08/0.16) = 1,098; TV = 1,098 ÷ (0.14 − 0.08) = ₹18,302 crore; PV = **₹9,505 crore**.

Equity value = −282 + 9,505 = ₹9,223 crore → P/B = 1.54×.

Sensitivities (P/B).

Terminal ROE	P/B
12%	0.75
14% (= Ke)	1.14
16% (base)	1.54
18%	1.93
Base ROE but Ke 15%	1.25

Lessons.

1. The value is entirely in the terminal period; the explicit years consume cash.
2. The terminal ROE relative to Ke is decisive. If the competitive position erodes ROE to 12%, P/B falls below 1 — consistent with theory.
3. Even at ROE = Ke the firm is worth 1.14× book because the five years of growth at ROE 17% add value.

Case D — An early-stage software company (First Chicago method)

Facts. *CloudLedger*, an accounting-software start-up raising its Series A. Exit values in year 5 under three scenarios; discount rate 25% (a high rate reflecting the systematic risk of early-stage equity, with *failure risk handled in the probabilities*).

Scenario	Probability	Exit equity value (₹ cr)	PV at 25% (× 0.3277)	Weighted PV
Success	20%	1,500	491.5	98.3
Sideways	35%	300	98.3	34.4
Failure	45%	15	4.9	2.2
Expected value today (post-money-equivalent)				**₹134.9 crore**

Cross-check with the VC method. The investor proposes ₹30 crore for 20% of the company: implied post-money ₹150 crore, pre-money ₹120 crore. Expected value (₹134.9 crore) is below ₹150 crore: the investor would be paying about 11% above the expected value. Pre-money consistent with the First Chicago value: 134.9 − 30 = ₹104.9 crore, implying an ownership stake of 30 ÷ 134.9 = 22.2% for ₹30 crore.

Lessons.

1. The expected value is dominated by the success scenario (73% of the total), yet only 20% likely — typical of venture investments.
2. The result is only as good as the probabilities and exit values; test them with a market comparison (multiples of revenue at exit).
3. Do not double-count failure risk: using both a 25% rate and a 45% failure probability together is deliberate here (the 25% is a risk premium for systematic risk and illiquidity), but a 50% venture-hurdle rate on top would be excessive.

Case E — A listed holding company (sum-of-the-parts with discount)

Facts. *Sagar Holdings Ltd* owns: a 40% stake in a listed company (market value of its stake ₹4,500 crore); 100% of an unlisted manufacturing subsidiary (separate DCF: ₹3,000 crore); a property portfolio (₹800 crore); cash ₹300 crore. Holding-company debt: ₹1,400 crore. Capitalised value of holding-company costs (PV of overhead): ₹200 crore.

Component	₹ crore
Listed stake (market value)	4,500
Unlisted subsidiary (DCF)	3,000
Property portfolio	800
Cash	300
Less: holding-company debt	(1,400)
Less: PV of holding-company costs	(200)

Component	₹ crore
Net asset value (NAV)	**7,000**
Less: holding-company discount (25%)	(1,750)
Value of equity	**5,250**

Lessons.

1. A holding company is a portfolio; each part is valued by the most appropriate method.
2. The *holding-company discount* reflects tax leakage on distributions, overhead, lack of control over the underlying cash flows, and capital-allocation scepticism. It is empirical and variable (10–40%); explain the basis.
3. Avoid double-counting: if the subsidiary's value already reflects the group's cash, do not also add the same cash at holding level.
4. Check the *listed-stake value* net of tax on a realistic exit; a very large block may not sell at the quoted market price.

Case F — A toll-road concession (finite life, equity IRR)

(See Chapter 19.) Equity of ₹400 crore; no distributions in years 1–3; ₹40, 55, 65, 75 crore in years 4–7; ₹90 crore a year in years 8–20. Total distributions ₹1,405 crore (3.5× equity); equity IRR **11.6%**; NPV at 14% = **−₹78 crore**.

Lessons.

1. A finite concession has no terminal value; all value is explicit.
2. A strong multiple of money (3.5×) can coexist with a below-hurdle return because the cash arrives late.
3. The levers (tariff, ramp-up, debt cost, concession extension) can be tested one at a time; for instance, a one-year acceleration of every distribution (a faster ramp-up) would raise the equity IRR from 11.6% to about 13.2%.

Pulling the cases together

Case	Business type	Value driver that dominates	Method adaptation
A	IT services	Margin	All-equity WACC; surplus cash in the bridge
B	Cement	Terminal mid-cycle EBITDA/t	Explicit cycle + normalised terminal

Case	Business type	Value driver that dominates	Method adaptation
C	NBFC	Terminal ROE vs Ke	FCFE and excess-return model
D	Start-up	Probability and size of success	Scenario-weighted with moderate rate
E	Holding company	Value of underlying stakes and the discount	SOTP/NAV
F	Infrastructure	Timing of distributions	Finite-life FCFE; IRR vs hurdle

The common thread: *identify what drives value in this particular business, build the model so that driver is explicit, and test it.*

Chapter 28 summary

- Six compact valuations show how method and emphasis change by business type.
- Terminal assumptions drive most of the value in A–D; in E the underlying values and the discount do; in F timing does.
- Every adaptation is a case of *consistency*: the model's structure follows the business's economics.

Review questions

1. In Case A, why is the value less sensitive to terminal ROIC than in a capital-intensive business?
2. In Case B, what would you do if the forecast period ended at a trough instead of a peak?
3. In Case C, why is the explicit-period PV negative?
4. In Case D, why is the discount rate 25% and not 50%?
5. In Case E, what could justify a holding-company discount of only 10%?

Chapter 29 — The Integrated Model: Three Statements, FCFF, FCFE and Reconciliation

The valuations so far used a *driver-based* forecast: revenue, margin and reinvestment, from which FCFF follows. Many professional models go one step further and build a full **three-statement model** — income statement, balance sheet and cash-flow statement that articulate with one another. The integrated model adds discipline (the balance sheet must balance, cash must reconcile), allows the debt schedule and interest to be modelled properly, and permits an FCFE valuation as a cross-check. This chapter builds it for Meridian and shows how FCFF and FCFE valuations reconcile.

29.1 Why build an integrated model?

1. **Error-catching.** An unbalanced balance sheet reveals an error that a driver-only model would hide.
2. **Financing visibility.** The debt schedule, interest, coverage ratios, dividends and cash needs can be tested — useful for lenders, covenants and capital-structure decisions.
3. **FCFE and ROE.** The model produces net income, dividends and ROE, necessary for an equity-side valuation.
4. **Scenario consistency.** A change in one assumption flows through all three statements.
5. **Credibility.** Reviewers can trace every number.

The cost is complexity. For a quick valuation, the driver-based approach is enough; for a transaction, a financing decision or a fairness opinion, the integrated model is standard.

29.2 Linking the statements

The three statements link as follows.

From	To	Link
Income statement net income	Balance sheet equity	Retained earnings += net income − dividends
Income statement D&A	Balance sheet fixed assets	Net fixed assets = opening + capex − D&A

From	To	Link
Revenue and cost days	Balance sheet working capital	NWC = f(revenue, days or %)
Balance sheet debt	Income statement interest	Interest = rate × (opening or average) debt
Cash-flow statement closing cash	Balance sheet cash	Balance sheet cash = closing cash
Cash-flow items	Balance sheet changes	ΔNWC, capex, Δdebt, dividends

The key identity: *ΔAssets = ΔLiabilities + ΔEquity.* If the cash-flow statement captures every non-cash movement, the balance sheet will balance automatically with cash as the balancing item (or, as here, with dividends set so that cash stays at a chosen level).

29.3 Assumptions for Meridian's integrated model

We keep the Chapter 14 operating assumptions and add financing assumptions.

Item	Assumption
Revenue growth, EBIT margin	As Chapter 14
D&A	5% of revenue
NWC	10% of revenue
Net reinvestment	ΔRevenue ÷ 2.0 (sales-to-capital)
Net fixed assets	Opening + net reinvestment − ΔNWC (i.e., net capex)
Capex	Net capex + D&A
Intangibles, other operating assets, non-operating investments, cash	Held constant (100, 250, 100, 150)
Debt	Increases each year by 20% of net reinvestment
Interest	9% on the opening debt balance
Tax	25% of profit before tax
Dividends	Equal to FCFE, so that cash stays at ₹150 crore

29.4 Income statement (₹ crore)

	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E
Revenue	2,000.0	2,280.0	2,553.6	2,809.0	3,061.8	3,306.7
EBITDA	340.0	399.0	459.6	519.7	581.7	644.8
D&A	(100.0)	(114.0)	(127.7)	(140.4)	(153.1)	(165.3)
EBIT	**240.0**	**285.0**	**332.0**	**379.2**	**428.6**	**479.5**
Interest	(54.0)	(54.0)	(56.5)	(59.0)	(61.3)	(63.6)

	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E
PBT	186.0	231.0	275.4	320.2	367.4	415.9
Tax @ 25%	(46.5)	(57.8)	(68.9)	(80.1)	(91.8)	(104.0)
Net income	**139.5**	**173.3**	**206.6**	**240.2**	**275.5**	**311.9**
Interest cover (EBIT ÷ interest)	4.4×	5.3×	5.9×	6.4×	7.0×	7.5×
ROE (on opening equity)		15.8%	17.0%	18.2%	19.4%	20.5%

29.5 Balance sheet (₹ crore)

	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E
Net fixed assets	900.0	1,012.0	1,121.4	1,223.6	1,324.7	1,422.7
Operating NWC	200.0	228.0	255.4	280.9	306.2	330.7
Intangibles & goodwill	100.0	100.0	100.0	100.0	100.0	100.0
Other operating assets	250.0	250.0	250.0	250.0	250.0	250.0
Non-operating investments	100.0	100.0	100.0	100.0	100.0	100.0
Cash	150.0	150.0	150.0	150.0	150.0	150.0
Total assets	**1,700.0**	**1,840.0**	**1,976.8**	**2,104.5**	**2,230.9**	**2,353.4**
Borrowings	600.0	628.0	655.4	680.9	706.2	730.7
Equity	1,100.0	1,212.0	1,321.4	1,423.6	1,524.7	1,622.7
Total liabilities & equity	**1,700.0**	**1,840.0**	**1,976.8**	**2,104.5**	**2,230.9**	**2,353.4**

Balance check: assets equal liabilities plus equity in every year. ✔

29.6 Cash-flow statement and reconciliation to FCFF and FCFE

₹ crore	FY27E	FY28E	FY29E	FY30E	FY31E
Net income	173.3	206.6	240.2	275.5	311.9
Add: D&A	114.0	127.7	140.4	153.1	165.3
Less: increase in NWC	(28.0)	(27.4)	(25.5)	(25.3)	(24.5)
Cash from operations	259.3	306.9	355.1	403.3	452.7
Less: capex	(226.0)	(237.1)	(242.6)	(254.2)	(263.3)
Free cash flow after capex	33.3	69.8	112.5	149.1	189.4
Add: net borrowing	28.0	27.4	25.5	25.3	24.5
FCFE (= dividends paid)	**61.2**	**97.1**	**138.0**	**174.4**	**214.0**
Reconciliation: FCFF	73.7	112.2	156.7	195.1	237.1
Less: after-tax interest	(40.5)	(42.4)	(44.3)	(46.0)	(47.7)
Add: net borrowing	28.0	27.4	25.5	25.3	24.5
= FCFE	61.2	97.1	138.0	174.4	214.0

The FCFE computed from the cash-flow statement equals the one derived from FCFF: the two approaches reconcile year by year. Closing cash stays at ₹150 crore, so the balance sheet balances.

29.7 FCFE valuation: first attempt (debt tied to reinvestment)

Discount FCFE at the cost of equity of 12.34%. For the terminal value, assume stable growth of 5% with terminal ROIC 15% (reinvestment 33.3% of NOPAT), debt still funding 20% of reinvestment:

- Terminal NOPAT = 377.6; net income$_6$ = (EBIT$_6$ 503.4 − interest 65.8) × 0.75 = 328.3; FCFE$_6$ = 328.3 − 0.80 × 125.9 = **227.6**.
- TV = 227.6 ÷ (0.1234 − 0.05) = **3,098.9**; PV = 1,731.6.
- PV of FCFE FY27–31 = 457.9.
- Equity value of operations = 457.9 + 1,731.6 = **2,189.5**; add cash and investments of 250 → **2,439.5**, or ₹48.79 per share.

This is **4.8% below** the FCFF-based equity value of ₹2,563.3 crore (₹51.27 per share). Why?

29.8 Reconciling the difference

The FCFF valuation used WACC computed at a *constant* D/V of 20% of market value. The integrated model instead lets debt grow only by 20% of net reinvestment, while firm value is growing by more than that, so the *market-value* debt ratio drifts down (to roughly 18% by year 5) and the book debt-to-invested-capital ratio falls from 41% to 35%. Meanwhile the FCFE model keeps the cost of equity at 12.34%, which assumes constant leverage. The two valuations therefore describe *different financing policies*.

Test: align the policy. Suppose debt is instead held at **20% of the market value of the enterprise** each year (so debt rises as value rises). The enterprise value at the end of each year, from the FCFF model, is: ₹2,913 (today), 3,167, 3,410, 3,636, 3,849, 4,044. Debt follows: 600, 633, 682, 727, 770, 809. Net borrowing is then 33, 49, 45, 43, 39 (not 28, 27, 26, 25, 25), interest is higher, and FCFE is: **66.5, 118.1, 155.9, 188.6, 224.1**. Terminal: debt grows 5% a year at 20% of EV, so terminal net

borrowing = 0.05 × 809 = 40.4; $FCFE_6$ = 237.6; TV = 237.6 ÷ 0.0734 = 3,235; PV = 1,808. PV of FCFE = 506.4; equity of operations = 506.4 + 1,807.7 = 2,314.1; plus 250 → **₹2,564.1 crore (₹51.28 per share)**.

The FCFE valuation now equals the FCFF valuation (₹2,563.3 crore) to within rounding. **Lesson:** FCFF/WACC and FCFE/Ke are two views of the same economics *only if the financing policy assumed in the discount rate matches the policy modelled in the cash flows.*

29.9 What the integrated model teaches that the driver model does not

1. **Leverage and coverage.** Interest cover rises from 4.4× to 7.5×: Meridian de-levers as it grows, so a lender would be comfortable and a financial engineer would see unused debt capacity.
2. **ROE.** ROE climbs from 15.8% to 20.5%, driven by margin expansion and *falling* leverage — a reminder that ROE can rise without increasing leverage when ROIC rises sharply.
3. **Dividend capacity.** FCFE of ₹61 crore in year 1 rising to ₹214 crore in year 5 gives the dividend capacity; the payout ratio (FCFE ÷ NI) rises from 35% to 69%.
4. **Funding gap.** In year 1 free cash flow after capex is only ₹33 crore against interest of ₹54 crore: the company needs the extra borrowing (₹28 crore) to pay dividends of ₹61 crore. If management wanted to pay out *more*, it would need either more debt or external equity.
5. **Cash is a policy variable.** We held cash at ₹150 crore; a different dividend policy would accumulate cash, whose interest income belongs outside FCFF and whose *value* belongs in the bridge.

29.10 Building the integrated model: a step-by-step recipe

1. **Opening balance sheet** (from the reformulated statements, Chapter 4): reconcile invested capital.
2. **Revenue and operating items**: revenue, EBITDA, D&A.
3. **Working-capital schedule**: receivable, inventory and payable days or % revenue.
4. **Fixed-asset schedule**: opening NFA + capex − depreciation = closing NFA.

5. **Debt schedule**: opening, drawdowns, repayments, closing; interest on opening or average balances.
6. **Equity schedule**: opening + net income − dividends (± issues/buybacks) = closing.
7. **Tax schedule**: taxable profit, tax, deferred tax if modelled.
8. **Cash-flow statement**: derived from the changes.
9. **Balance sheet**: cash from the cash-flow statement; check the balance.
10. **FCFF, FCFE and valuation** (DCF sheet).
11. **Checks**: balance sheet, cash, debt, FCFF/FCFE reconciliation.

Handling circularity

If interest is computed on *average* balances and the cash sweep depends on net income, a circular reference arises. Options: (i) compute interest on *opening* balances (as here) — simple and avoids circularity at the cost of small inaccuracy; (ii) use iteration with a switch; (iii) use a macro to copy-paste converged values.

29.11 Common pitfalls in integrated models

- **Plugs.** Forcing the balance sheet to balance with an unexplained plug hides errors.
- **Hard-coded balances** in forecast columns.
- **Inconsistent timing** between interest and balances.
- **Cash sweep and minimum cash** not modelled, leading to negative cash.
- **Tax loss and deferred-tax items** omitted.
- **Not reconciling FCFF with the cash-flow statement.**
- **Discounting FCFE at a cost of equity inconsistent with the leverage path.**

Chapter 29 summary

- An integrated model links income statement, balance sheet and cash-flow statement and must balance.
- FCFF and FCFE can both be reconciled to the cash-flow statement year by year.
- FCFF/WACC and FCFE/Ke give the same equity value *only if* the financing policy is consistent in both.

- Meridian: FCFE valuation with debt tied to reinvestment = ₹48.79 per share; with debt at 20% of market EV = ₹51.28, matching the FCFF value.

Review questions

1. List the six links between the three statements.
2. Why does the FCFE valuation differ from the FCFF valuation when debt is tied to reinvestment?
3. Why can ROE rise while leverage falls?
4. How can circularity be avoided in the interest calculation?
5. What does it mean that dividends equal FCFE in the model?

Chapter 30 — Taxes, Tax Losses and Deferred Taxes in DCF

Tax is the single largest cash outflow of most profitable companies after operating costs, and it is often treated carelessly in DCF models: one rate, applied everywhere, forever. This chapter looks at the tax inputs a careful valuation needs: the right rate, the treatment of tax losses and credits, timing differences and deferred taxes, the tax shield on debt, and international and structural issues.

Note. Tax law changes frequently and differs by jurisdiction; this chapter describes concepts, with Indian references for orientation only. Always check the current law for a real engagement.

30.1 Which tax rate?

Rate	Definition	Use in DCF
Statutory (headline) rate	Rate in the law, plus surcharge and cess	Starting point
Marginal rate	Rate on the next rupee of taxable profit (relevant when the firm is profitable)	Terminal value; tax shield on interest; value of additional deductions
Effective rate	Tax expense ÷ pre-tax profit (book)	Historical analysis; sanity check
Cash tax rate	Taxes paid ÷ pre-tax profit	Near-term forecasts, where deferral and credits matter

Indian illustration. Domestic companies can choose between the older regime (a headline rate of 25% or 30% plus surcharge and cess, with various deductions and incentives, and MAT) and a concessional regime of about 22% before surcharge and cess (about 25.17% all-in) with fewer deductions and no MAT credit. New manufacturing companies may qualify for a lower regime. The correct near-term rate depends on the regime elected and the incentives the company can actually use. The book uses 25% as a rounded marginal rate.

Rule of thumb. Use the *expected cash tax rate* in the explicit period, converging to the *marginal rate* in the terminal period. Do not carry a low effective rate (from a tax holiday or accumulated credits) into perpetuity.

30.2 Tax on operating profit versus actual tax

FCFF uses **tax on EBIT** (the "unlevered" tax), not the actual tax paid. The interest tax shield is recognised in the after-tax cost of debt in WACC (or in APV). Why does the distinction matter?

- If you use actual taxes (which are reduced by interest deductions) in FCFF *and* use an after-tax cost of debt, you count the tax shield twice.
- If the firm's actual leverage differs from the target, actual tax differs from the target-leverage tax; the unlevered tax avoids contamination from the firm's current financing.

30.3 Tax losses (NOLs) and unabsorbed depreciation

A company with accumulated losses can set them against future profits, within the limits of the law (for example, time limits for business losses and conditions on change of ownership). The present value of future tax savings is an asset.

30.3.1 Two ways to treat NOLs

1. **Inside FCFF:** forecast taxable income after set-off and compute taxes accordingly; cash taxes in the early years are lower; the benefit is in the PV of cash flows.
2. **As a separate asset:** compute FCFF with the full tax on EBIT, then add the PV of the NOL shield in the bridge.

Use *one* approach. Method 2 is cleaner for transparency.

30.3.2 Worked example

A company has opening unabsorbed losses of ₹200 crore and expects EBIT of −100, 40, 150, 250 and 300 over five years; tax rate 25%; discount rate 11%.

Year	EBIT	Opening NOL	Loss added	Loss used	Taxable income	Tax with NOL	Tax without NOL	Saving
1	(100)	200	100	—	0	0	0	0
2	40	300	—	40	0	0	10.0	10.0
3	150	260	—	150	0	0	37.5	37.5
4	250	110	—	110	140	35.0	62.5	27.5
5	300	0	—	—	300	75.0	75.0	0

(In this simple example, year-1's loss is added to the pool and the whole pool of ₹300 crore is used over years 2–4.) The savings of ₹75 crore (= 25% of ₹300 crore) have a present value at 11% of **₹53.7 crore** — 72% of the undiscounted amount. A valuer adds ₹53.7 crore to EV (Method 2) or builds the lower taxes into FCFF (Method 1).

Caveats. The shield is only as valuable as the firm's ability to generate taxable profit before the losses expire; the shield may be lost on a change of control if the law restricts it; and a probability of non-use should reduce the value.

30.4 Minimum alternate tax (MAT) and credits

Where a minimum tax applies on book profits and the excess over normal tax can be carried forward as a credit, the credit is an asset whose value depends on when it will be utilised. Model the utilisation schedule: credits are used when normal tax exceeds MAT, subject to the carry-forward period. Value as the PV of expected utilisation. Companies in the concessional regime generally cannot use or earn MAT credit.

30.5 Timing differences and deferred taxes

Book depreciation and tax depreciation often differ. Accelerated tax depreciation defers tax: cash taxes are lower early and higher later; the deferred tax liability (DTL) in the balance sheet records the difference.

30.5.1 Worked example

An asset costs ₹1,000 crore. Book depreciation is straight-line over 10 years (₹100 crore a year). Tax depreciation is accelerated: ₹300, 250, 200, 150 and 100 crore in years 1–5. Tax rate 25%; discount rate 11%.

- PV of tax shield with tax depreciation = **₹194.4 crore**.
- PV of the shield if tax depreciation matched the book schedule = **₹147.2 crore**.
- The *value of acceleration* = **₹47.2 crore**.

Deferred tax liability: end of year 1: (300 − 100) × 25% = ₹50 crore; year 3: (750 − 300) × 25% = ₹112.5 crore; year 5: (1,000 − 500) × 25% = ₹125 crore (the peak); then it reverses to zero by year 10.

30.5.2 How to treat DTLs in valuation

- **Perpetually-growing capex** means that the DTL on the pool of assets *does not reverse* in aggregate: new assets create new deferrals that offset the reversals of old ones. In that case, using *cash* taxes in FCFF (i.e., recognising the deferral as lower cash tax) and *not* deducting the DTL in the bridge is consistent.
- **If the firm will stop investing** (liquidation or run-off), the DTL reverses; deduct the PV of the reversal or model it in FCFF.
- **Never both:** if cash taxes are used in FCFF, do not deduct the DTL; if book taxes are used, deduct the PV of the DTL's reversal.

A **deferred tax asset (DTA)** from losses or provisions is valued in the same spirit: only if it will actually reduce future cash taxes.

30.6 The tax shield on debt

- The shield = t × interest, valuable only if the firm has taxable profit to absorb the deduction.
- **Interest-deductibility limits** (such as a cap on interest relative to EBITDA for related-party debt) reduce the shield.
- **Personal taxes:** if investors pay more tax on interest than on equity returns, part of the corporate benefit is offset (Miller). Valuation at the company level typically ignores it, but a distinction should be made when valuing for a particular investor.
- In **APV** the tax shield is discounted at the cost of debt (fixed debt) or at the unlevered cost of equity (debt adjusted to value). In **WACC** the shield is in the after-tax Kd.

30.7 Group structure, transfer pricing and international tax

- **Subsidiaries in different jurisdictions** have different rates; compute taxes by entity or use a weighted rate by profit.
- **Repatriation taxes** (withholding on dividends) reduce the value of cash trapped overseas.
- **Transfer pricing:** profit allocations between group entities affect taxes; adjustments or disputes can create liabilities.
- **Permanent establishment risks** and **general anti-avoidance rules** are legal risks to be treated as contingent liabilities if material.

30.8 Tax in the terminal value

1. Use the **marginal statutory rate** unless a lasting lower rate is justified (e.g., a permanently lower regime).
2. **Incentives** with expiry dates must not be extrapolated.
3. **DTL growth** in perpetuity: be consistent with the choice in 30.5.2.
4. If tax law is *expected to change* (rate cuts, regime changes), consider scenarios rather than a single guess.

30.9 Checklist

1. Which regime applies; which incentives are available and when do they expire?
2. Near-term cash tax rate vs terminal marginal rate.
3. NOLs, unabsorbed depreciation, MAT credits: how much, when usable, probability of loss on change of control.
4. DTL/DTA treatment consistent between FCFF and bridge.
5. Tax shield on debt only to the extent usable.
6. Entity-level taxes and repatriation costs for multi-jurisdiction groups.
7. Contingent tax liabilities and disputed assessments.

Chapter 30 summary

- Use cash tax rates near term, marginal in the terminal period; tax on EBIT in FCFF.
- NOLs and credits are assets: value by PV of savings, once.
- Accelerated tax depreciation defers tax; treatment of DTL must match the taxes in FCFF.
- The tax shield on debt exists only if there is taxable profit to absorb it.

Review questions

1. Why is tax on EBIT, not actual tax, used in FCFF?
2. Compute the PV of the tax saving from a loss pool of ₹100 crore used evenly over 4 years at 25% tax and 10% discount rate.
3. When is it appropriate to ignore a deferred tax liability in the bridge?
4. How should tax-holiday benefits be treated in the terminal period?
5. What happens to the value of NOLs when control changes?

Chapter 31 — Sector Primers: What Drives Value in Fourteen Industries

A DCF is a template; each industry fills it differently. This chapter gives a short primer for fourteen sectors: what drives revenue, margins and reinvestment; which valuation method is natural; and what traps to avoid. The remarks are qualitative orientation, not data; for any real engagement, replace them with current industry research and company disclosures.

How to read these primers

For each sector the primer answers five questions:

1. **Economics:** how does the business make money?
2. **Key value drivers:** what determines growth, margin and capital intensity?
3. **Preferred methods:** which valuation approach fits best, and which cross-checks?
4. **Cash-flow and discount-rate specifics.**
5. **Common traps.**

31.1 Fast-moving consumer goods (FMCG) and packaged foods

- **Economics:** branded, repeat-purchase products sold through distributors and retail; high gross margins, heavy advertising and distribution spend; modest capex.
- **Value drivers:** volume growth (penetration, new categories), pricing and mix (premiumisation), input costs (agricultural commodities, packaging, energy), advertising intensity, distribution reach, working-capital discipline (negative or low NWC for leaders).
- **Methods:** FCFF DCF with long fade period for strong brands; P/E and EV/EBITDA as cross-checks (leaders trade at high multiples reflecting durable ROIC).
- **Specifics:** low beta; high terminal ROIC defensible only with a proven brand moat; treat brand spending as an investment when analysing ROIC.

- **Traps:** extrapolating margin expansion without commodity cycles; ignoring distributor-credit risk; assuming perpetual pricing power; mixing leaders' multiples with a regional player.

Case reference: Meridian Foods (Chapters 3–16).

31.2 Pharmaceuticals

- **Economics:** branded and generic drugs; high R&D and regulatory costs; patents and exclusivities create time-limited advantages.
- **Value drivers:** product pipeline and launches, patent expiries ("cliffs"), regulatory inspections (plant compliance), US generic price erosion, domestic chronic-therapy share, R&D productivity, working capital (receivables, inventory).
- **Methods:** FCFF DCF with product-level or segment-level forecasts; risk-adjusted NPV (rNPV) for pipeline assets with probabilities of success by stage; relative valuation on EV/EBITDA.
- **Specifics:** capitalise R&D for ROIC analysis; model patent cliffs explicitly rather than smooth growth; mid-year discounting.
- **Traps:** assuming a pipeline succeeds without probability weighting; ignoring regulatory risk (warning letters, import alerts); extrapolating a launch year's earnings.

Case reference: Aster Pharma (Chapter 27).

31.3 IT services and software

- **Economics:** people-based services (billing rates, utilisation, attrition) or products/platforms (subscriptions, licences).
- **Value drivers:** headcount growth, pricing, utilisation, offshore mix, deal wins and renewals, currency (USD revenue vs INR costs), wage inflation.
- **Methods:** FCFF or FCFE DCF (asset-light; often net cash), P/E and EV/EBIT; for SaaS, EV/ARR and rule-of-40 analysis.
- **Specifics:** very low reinvestment; stock-based compensation is real cost; currency assumptions drive reported margins; treat cash as surplus.
- **Traps:** forgetting that high ROIC attracts competition; ignoring technology disruption; ignoring wage inflation; using USD growth with INR costs without a currency view.

Case reference: IndiaTech Services (Chapter 28, Case A).

31.4 Banks and NBFCs

- **Economics:** interest-rate spread on loans over funding, fees, credit costs, regulatory capital limits growth.
- **Value drivers:** loan growth, net interest margin, credit cost, cost-to-income, capital adequacy, funding mix (CASA), asset quality (GNPA/NNPA), ROE.
- **Methods:** excess-return model and P/B tied to (ROE − g)/(Ke − g); DDM/FCFE after regulatory capital; P/E.
- **Specifics:** do not un-lever beta; terminal ROE vs Ke is the key judgement; stress test credit cost.
- **Traps:** FCFF; ignoring the capital that growth consumes; peak ROE in terminal.

Case reference: Chapter 18; Kaveri Finance (Chapter 28, Case C).

31.5 Insurance

- **Economics:** collect premiums, invest the float, pay claims; life insurers sell long-term policies with embedded profit emerging over years; general insurers earn underwriting result plus investment income.
- **Value drivers:** new business volumes and margins (VNB), persistency, combined ratio, investment yield, solvency margin, mortality/claims experience.
- **Methods:** embedded value (life), appraisal value, P/B and ROE, DDM; for general insurers P/B and combined ratio analysis.
- **Specifics:** discount rate (risk discount rate) assumptions drive embedded value; regulatory capital restricts distributions.
- **Traps:** treating policyholder funds as equity; ignoring persistency; mixing embedded-value bases.

31.6 Automobiles and ancillaries

- **Economics:** cyclical demand tied to income, interest rates and fuel prices; high fixed costs; capital-intensive; model-cycle capex.
- **Value drivers:** volumes by segment, realisations, commodity costs, product launches, market share, operating leverage, working capital, electrification transition costs.

- **Methods:** DCF on mid-cycle margins; EV/EBITDA and P/E across the cycle; SOTP for conglomerates with financing arms.
- **Specifics:** explicit cycle for 3–5 years then mid-cycle; capex lumpiness; technology transition capex and stranded asset risks.
- **Traps:** valuing at peak-cycle margins; ignoring the cost of electrification; neglecting finance-subsidiary leverage.

31.7 Capital goods, engineering and construction

- **Economics:** order-book driven revenue; project execution; working capital heavy (receivables, retention money, advances).
- **Value drivers:** order inflow and backlog conversion, margin on orders, execution, working-capital cycle, government capex.
- **Methods:** DCF built from order book to revenue; EV/EBITDA; P/E; order-book-to-sales.
- **Specifics:** model working capital explicitly (large swings); guarantee and contingent liabilities; arbitration receivables.
- **Traps:** treating the order book as revenue without execution risk; ignoring stuck receivables and contingent claims.

31.8 Cement, steel, metals and chemicals (commodity cyclicals)

- **Economics:** price-taking producers; cost position on the industry cost curve determines through-cycle returns.
- **Value drivers:** commodity price and spread, volumes, capacity utilisation, energy and raw material costs, expansions.
- **Methods:** DCF with explicit cycle and mid-cycle terminal; EV/tonne of capacity vs replacement cost; EV/EBITDA on mid-cycle EBITDA.
- **Specifics:** beta over a full cycle; leverage amplifies cycle risk.
- **Traps:** extrapolating the current spread; ignoring capacity additions by competitors; ignoring covenant risk at the trough.

Case reference: Rajmandir Cement (Chapter 28, Case B).

31.9 Power, utilities and renewable energy

- **Economics:** regulated or contracted tariffs; high capital intensity; long-lived assets; leverage-friendly cash flows.

- **Value drivers:** regulated equity base and allowed return, PPA tariffs, plant load factor, fuel pass-through, receivables from distribution companies, capex pipeline.
- **Methods:** FCFE/DDM for regulated returns; project-level DCF for contracted plants; EV/RAB (regulatory asset base) for regulated utilities; SOTP for portfolios.
- **Specifics:** finite concession or asset life; counterparty receivable risk; regulatory change; for renewables, tariff, curtailment, resource risk and refinancing.
- **Traps:** perpetual growth on a finite-life asset; ignoring receivable delays; assuming returns above the regulated allowed ROE.

31.10 Telecommunications and digital infrastructure

- **Economics:** subscriber-based revenue, heavy spectrum and network capex, high operating leverage, regulatory and licence cost.
- **Value drivers:** ARPU, subscriber base and mix, churn, spectrum payments, network capex, tariff actions, data growth.
- **Methods:** DCF with explicit spectrum and capex cycles; EV/EBITDA; EV/subscriber (cross-check only); SOTP for towers and fibre.
- **Specifics:** large lease and licence liabilities as debt-like items; regulatory dues; consolidation structure.
- **Traps:** ignoring deferred spectrum payments and adjusted gross revenue (AGR)-type dues; extrapolating ARPU growth.

31.11 Real estate and infrastructure developers

- **Economics:** project-based cash flows; land, approvals and sales velocity dominate; long cycles; regulatory compliance (RERA).
- **Value drivers:** pricing, absorption, construction cost, land bank value, approvals, financing cost.
- **Methods:** NAV by project DCF; cap-rate valuation for rental assets; P/NAV.
- **Specifics:** collections vs recognised revenue; customer advances; land at market not cost.
- **Traps:** valuing land at book; ignoring approvals timing; double-counting customer advances.

31.12 Retail and consumer internet

- **Economics:** retail: store economics (sales per square foot, rent, labour); e-commerce: unit economics (contribution per order), CAC, retention.
- **Value drivers:** same-store growth, store rollouts, gross margin, lease costs (Ind AS 116 treatment), inventory turns, delivery costs, take-rate.
- **Methods:** DCF by cohort/store; EV/EBITDA post-lease consistency; EV/GMV or EV/revenue for early-stage platforms; scenario analysis.
- **Specifics:** leases are debt-like — be consistent; store-level maturity curves; negative working capital for some.
- **Traps:** mixing pre- and post-Ind AS 116 metrics; assuming new stores mature as fast as old ones; ignoring cash burn to reach scale.

31.13 Healthcare delivery (hospitals and diagnostics)

- **Economics:** capacity-based (beds, centres), occupancy and ARPOB (average revenue per occupied bed), payor mix; long gestation for new facilities.
- **Value drivers:** occupancy, ARPOB, new-bed additions, maturity profile, doctor retention, regulatory price controls.
- **Methods:** DCF by facility cohort (mature vs ramping); EV/bed; EV/EBITDA.
- **Specifics:** ramp-up losses on new facilities; land and building ownership vs lease; doctors' compensation.
- **Traps:** applying mature-hospital margins to new hospitals immediately; ignoring capital intensity.

31.14 Airlines, hotels and hospitality

- **Economics:** high fixed costs and operating leverage, volatile demand, capital-intensive (aircraft, properties), leases.
- **Value drivers:** load factor/occupancy, yield/RevPAR, fuel, currency, fleet/room additions, seasonality.
- **Methods:** DCF on mid-cycle; EV/EBITDAR (before rent) when leases are material; NAV for owned hotel real estate.
- **Specifics:** leases/aircraft financing as debt; asset-light vs asset-heavy models in hotels; shocks (pandemics, fuel spikes).

- **Traps:** peak-year extrapolation; ignoring lease obligations; underestimating shock frequency.

31.15 A one-page comparison

Sector	Capital intensity	Cyclicality	Natural primary method	Beta tendency
FMCG	Low–moderate	Low	FCFF DCF, long fade	Low
Pharma	Moderate	Low–moderate	FCFF + rNPV	Low–moderate
IT services	Very low	Moderate	FCFF/FCFE	Moderate
Banks/NBFCs	n/a (regulatory capital)	High	Excess return/P/B	Moderate–high
Insurance	n/a	Moderate	Embedded value/P/B	Moderate
Autos	High	High	Mid-cycle DCF	High
Capital goods	Moderate	High	Order-book DCF	High
Cement/metals	Very high	Very high	Mid-cycle DCF; EV/capacity	High
Power/utilities	Very high	Low	FCFE/regulated base	Low
Telecom	Very high	Low–moderate	DCF; SOTP	Moderate
Real estate	High	High	NAV	High
Retail/e-commerce	Moderate	Moderate	Cohort DCF	Moderate–high
Healthcare delivery	High	Low	Cohort DCF	Low–moderate
Airlines/hotels	Very high	Very high	Mid-cycle DCF; EV/EBITDAR	High

Chapter 31 summary

- Method follows business model: FCFF for operating firms, FCFE/excess return for financials, NAV for real-estate and holdings, cohort/unit-economics DCF for growing platforms and facilities.
- The characteristic traps are sector-specific: peak margins in cyclicals, unprobabilised pipelines in pharma, perpetual growth on finite assets, lease inconsistency in retail and airlines.
- Beta tendencies and capital intensity are starting points for bottom-up analysis, never substitutes.

Review questions

1. Why is FCFF unsuitable for insurers and banks while natural for FMCG?
2. How does a patent cliff change the DCF structure for a pharma company?
3. What distinguishes the treatment of leases in retail from that in an IT-services firm?
4. Why does a finite concession change the terminal-value approach?
5. Choose a sector and list the three assumptions that would most affect its DCF.

Chapter 32 — The Limits of DCF: Critiques, Behaviour and Judgement

A book on DCF should end with an honest account of what DCF cannot do. The method is powerful, but it is not an oracle. This chapter reviews the standard criticisms, the behavioural traps that afflict even careful practitioners, the contexts where DCF is the wrong primary tool, and the habits that distinguish good valuers from merely competent ones.

32.1 The standard criticisms

32.1.1 "Garbage in, garbage out"

The criticism: a DCF can be tuned to produce any answer. Response: true — which is why a DCF must be *disciplined* (growth tied to reinvestment, ROIC fades, TV share disclosed, scenarios and reverse-DCF cross-checks). A DCF that is documented, consistent and tested produces a *defensible range*, not an arbitrary number. The remedy for tunable models is transparency, not abandoning the method.

32.1.2 "Terminal value dominates"

True: for most going concerns, 60–85% of value is in the TV. The remedy is to treat TV as a hypothesis to be tested (Chapter 12) and to examine implied multiples, ROIC and growth. If a valuer cannot defend the terminal state, they cannot defend the valuation.

32.1.3 "The discount rate is unknowable"

Estimates of the ERP and beta carry wide error bands. But a precise but wrong discount rate is worse than an approximate right one; show sensitivity to the rate and avoid false precision. Where the discount rate is deeply uncertain, express value as a function of it (as in the WACC × g tables).

32.1.4 "The future is not predictable"

Forecasts are uncertain; valuation is a statement about expectations. The method does not require perfect forecasts, only *unbiased expected values* and honest ranges. Scenario analysis and simulation represent the uncertainty rather than ignoring it.

32.1.5 "Markets know better"

Market prices aggregate many opinions; a DCF represents one. Where market prices are the only observable signal (liquid, widely followed stocks), the burden is on the DCF to explain differences (reverse DCF). But markets also display bubbles and panics, so an estimated intrinsic value is a useful anchor and warning.

32.1.6 "DCF ignores flexibility and strategy"

A static DCF misses managerial options (expand, abandon). Real-options thinking (Chapter 23) addresses that, with care.

32.1.7 "DCF is too slow and complicated"

For quick decisions a multiple is fine. The point of DCF is *understanding*: forcing explicit assumptions. Even a simple one-stage DCF using the value-driver formula (EV = NOPAT × (1 − g/ROIC)/(WACC − g)) reveals more about a business than a P/E.

32.2 Behavioural traps for the valuer

Bias	How it appears	Counter-measure
Anchoring	The first number (the asking price, the market price, last year's value) pulls the estimate toward it	Build the valuation independently before looking at the price; note the first estimate and the reasons for changes
Confirmation	Seek evidence supporting a favoured conclusion	Seek disconfirming evidence; assign a colleague to argue the other side
Overconfidence	Ranges too narrow; point estimates presented as precise	Widen ranges; backtest past valuations
Optimism / planning fallacy	Management plans, and valuers who adopt them, are too rosy	Compare plans with history; apply base rates
Narrative bias	A compelling story overrides numerical discipline	Convert the story to numbers and test each; identify what must be true
Recency	Extrapolating the last few years	Use full-cycle data; study base rates
Herding	Adopting the consensus discount rate or multiple without thought	Derive independently; explain differences
Sunk-cost / commitment	Defending a model because of effort invested	Review fresh; invite challenge
Client pressure	Drifting toward a desired answer	Pre-commit to assumptions and methods; disclose conflicts

32.2.1 Base rates

Few companies sustain high growth for long. A forecast of 20% growth for ten years should be compared with how often companies of that size have achieved it. Historical studies of growth persistence consistently show strong *mean reversion*: high growth fades, and above-normal margins and returns attract competition. A DCF that assumes persistence must argue *why this firm is exceptional*, and should say so explicitly.

32.2.2 The outside view

Ask: "Among companies that look like this one, what happened?" Combine the inside view (the specific story) with the outside view (reference class). If the inside view says 25% margin but 90% of the reference class never exceeded 18%, require strong evidence.

32.3 When DCF is not the right primary tool

Situation	Better primary tool	Why
Liquidation or run-off	Liquidation/asset value	No going concern
Holding companies and funds	NAV/SOTP	Value is in the underlying assets
Highly speculative early-stage with no model	Comparable rounds, VC method, scenario analysis	Cash-flow forecast arbitrary
Financial institutions	Excess-return/DDM	Enterprise cash flow undefined
Short-horizon trading or merger arbitrage	Market prices, deal terms	Intrinsic value less relevant
Real estate	Cap-rate and comparable sales	Market-evidence strong
Illiquid pooled assets	NAV with adjustments	

In many of these, DCF still appears as a cross-check; the point is that it should not be the *only* anchor.

32.4 Where DCF is evolving

- **Data and automation.** Spreadsheet models are increasingly generated or supported by software and data providers; the risk is uncritical acceptance of mechanically produced numbers. The analyst's value lies in judgement: choosing the story, challenging assumptions.
- **Scenario and probabilistic thinking.** Wider use of simulations and probability-weighted outcomes, especially for climate and regulatory risks.

- **Intangibles.** Better ways of capitalising R&D, brands and data in invested capital and ROIC.
- **ESG and climate integration.** More explicit scenario modelling of carbon pricing, physical risk and transition capex (Chapter 23).
- **Standardisation of professional practice.** Valuation standards, registered-valuer regimes and reporting requirements push toward documented, reviewable models.

None of these change the principle: value is the present value of expected cash flows, adjusted for risk.

32.5 Habits of excellent valuers

1. **They ask "what must be true?"** for any value, and check whether it is plausible.
2. **They look at the base rate** before the story.
3. **They triangulate** — DCF, multiples, transactions, reverse DCF — and reconcile.
4. **They own their assumptions.** Every key input has a source, an owner and a date.
5. **They present ranges** and say which assumptions drive them.
6. **They are comfortable saying "I don't know"** for inputs that cannot be estimated, and show the sensitivity instead.
7. **They keep a record** of past valuations and review them against what happened.
8. **They separate the analysis from advocacy**: advisory roles may involve advocacy; valuation does not.
9. **They keep it simple where possible** — a model that nobody can follow is not a better model.
10. **They update** — risk-free rate, ERP and peers change; the same company is worth different amounts in different markets.

32.6 Closing summary of the book's argument

1. **Value is the present value of the cash a business will generate for its owners, discounted for time and risk.**
2. **Cash flow is operating profit after tax less reinvestment.** Growth must be paid for.

3. **Return on invested capital relative to the cost of capital determines whether growth creates value.**
4. **Consistency is the main discipline:** cash flow, discount rate, currency, claim and financing policy must match.
5. **Terminal value is where most of the value and most of the error live.**
6. **The bridge from enterprise to equity must be complete and non-duplicative.**
7. **Uncertainty should be shown, not hidden:** sensitivities, scenarios, reverse DCF.
8. **Different businesses need different adaptations,** but the same principles.
9. **Professional practice** requires documentation, independence and humility.

Chapter 32 summary

- The standard criticisms of DCF are partly valid; the response is discipline and transparency, not abandonment.
- Behavioural biases (anchoring, optimism, narrative) are the practitioner's main enemies; counter them with base rates and independent challenge.
- DCF is not always the right primary tool; know when to lead with another method.
- Excellent valuation is judgement supported by consistent arithmetic.

Review questions

1. Respond to the criticism that "DCF can produce any number you want".
2. What is the outside view, and how would you apply it to a forecast of 25% growth for ten years?
3. List three situations where DCF should not be the primary method and name the alternative in each.
4. How can an analyst guard against anchoring on the market price?
5. Which of the nine closing principles do you consider most commonly violated in practice, and why?

Chapter 33 — Growth, Competitive Advantage and the Value of ROIC

Every chapter of this book has returned to the same relationship: *growth creates value only when new investment earns more than its cost of capital.* This closing chapter treats that relationship directly. It shows how to measure the value of growth, how long excess returns can last, how to recognise a competitive advantage, and how to translate qualitative judgements about moats into the numbers a DCF needs.

33.1 The value of one rupee of reinvestment

Suppose a firm reinvests ₹1 in new capital that earns a perpetual return of ROIC, while investors require WACC. The present value of that perpetual operating profit is ROIC ÷ WACC rupees (in steady state, ignoring growth in the new project itself). So:

- ROIC = WACC → ₹1 invested creates ₹1 of value; **net value created = 0**.
- ROIC = 1.5 × WACC → ₹1 creates ₹1.50; **net ₹0.50**.
- ROIC = 0.7 × WACC → ₹1 creates ₹0.70; **net −₹0.30**.

ROIC (WACC = 11.225%)	Value per ₹1 reinvested	Net value created
8.0%	0.71	−0.29
11.2%	1.00	0.00
15.0%	1.34	+0.34
20.0%	1.78	+0.78
30.0%	2.67	+1.67

This is why a firm earning 30% on new capital should grow as fast as it can (each rupee reinvested creates ₹1.67 of surplus), and a firm earning 8% should *shrink* — return capital to investors — unless it can improve returns.

33.2 The value-driver formula and the "value of growth"

For a steady-state firm, **EV = NOPAT₁ × (1 − g/ROIC) ÷ (WACC − g)**. Divide by NOPAT₁ to obtain the forward EV/NOPAT multiple:

EV/NOPAT₁ multiple	g = 0%	g = 2%	g = 4%	g = 5%	g = 6%
ROIC = WACC (11.2%)	8.9×	8.9×	8.9×	8.9×	8.9×
ROIC = 13%	8.9×	9.2×	9.6×	9.9×	10.3×
ROIC = 15%	8.9×	9.4×	10.2×	10.7×	11.5×
ROIC = 20%	8.9×	9.8×	11.1×	12.1×	13.4×
ROIC = 30%	8.9×	10.1×	12.0×	13.4×	15.3×

Read the table across: with ROIC = WACC the multiple is flat at 1/WACC = 8.9× whatever the growth. Read down: higher ROIC lifts the multiple, but only when there is growth to apply it to. At g = 0, ROIC is irrelevant because nothing is reinvested.

Value of growth. The "earnings power value" (EPV) is $NOPAT_1 \div WACC$ — the value if the firm never grew (and never reinvested). The difference between DCF value and EPV is the **value of growth**. For $NOPAT_1 = 100$, ROIC 15%, g 5%, WACC 11.225%: EV = 100 × (1 − 0.3333) ÷ 0.06225 = 1,071; EPV = 890.9; **value of growth = 180** (17% of EV). This quantifies how much of a valuation rests on the assumption that growth will earn above its cost: only ₹180 of ₹1,071 in this case, so a moderately growing firm with a decent ROIC is mostly valued on its *current* earning power. For a high-growth, high-ROIC firm the proportion is much higher, and the valuation is correspondingly speculative.

33.3 How long can excess returns last?

Because ROIC tends to be competed toward WACC, the *duration* of excess returns — the **competitive-advantage period (CAP)** — is as important as their level. A simple model: new investment earns ROIC_high for N years and ROIC = WACC thereafter; growth is 6% throughout; WACC 11%; $NOPAT_1 = 100$.

CAP (years of ROIC 20% on new capital)	EV (₹)	Increase vs no excess returns
0 (ROIC = WACC)	909	—
5	1,010	+11%
10	1,090	+20%
15	1,154	+27%
20	1,205	+33%
40	1,322	+45%
Perpetual	1,400	+54%

Observations:

1. **Early years matter most, but a long moat keeps adding.** About a fifth of the maximum increase (11 of 54 points) arrives in the first five years, over a third (20 of 54) in ten and about 60% in twenty. Because of discounting, each additional decade of excess returns adds less than the one before.
2. **The perpetual-moat assumption is expensive.** It adds 54% to value versus none; a ten-year moat adds 20%. A valuer who assumes the terminal ROIC stays far above WACC forever should be able to say what protects it for that long.
3. **Judging duration is judging a moat.** The next sections list the sources.

33.4 Sources of competitive advantage

Source	Mechanism	Evidence in the numbers	Typical durability
Brand	Customers pay more or choose repeatedly	High gross margin, stable share, price increases without volume loss	Long, but erodes if quality or relevance slips
Switching costs	Costly for customers to change (software, banking relationships)	High retention, low churn, pricing power over time	Long; falls with technological change
Network effects	Value rises with users	Rapid share consolidation; rising margins with scale	Potentially very long; vulnerable to multi-homing and disruption
Cost advantage / scale	Lower unit costs from scale, process, location, proprietary resource	Low position on the industry cost curve; margin stable through cycles	Medium–long
Intangible rights (patents, licences, regulation)	Legal protection or scarce rights	Excess returns during the protected period	Finite and datable (patent life; licence term)
Efficient scale in a local market	Market only supports one or two players	Few entrants; stable returns	Medium; depends on market growth
Distribution / ecosystem	Control of channels or standards	Shelf presence, supplier terms	Medium

Test the claim. For each asserted advantage, ask: (i) what *evidence* shows it exists (historical ROIC above WACC through a cycle; stable or rising market share); (ii) what *threat* could erode it (new technology, regulatory change, a better-funded competitor); (iii) how *long* it should last given those threats. Convert answers to a CAP and to terminal ROIC.

33.5 Reading ROIC properly

A high reported ROIC can be misleading if the denominator is understated or the numerator inflated. Adjustments:

1. **Capitalise intangible investment.** R&D, brand-building and customer-acquisition spending is expensed; capitalising and amortising it increases invested capital and lowers ROIC for intangible-intensive firms while raising NOPAT for growing firms.
2. **Include acquired goodwill** when assessing returns on *total* capital deployed. ROIC *excluding* goodwill measures the underlying operation; ROIC *including* goodwill measures whether acquisitions earned their keep.
3. **Use average or opening capital** consistently.
4. **Adjust for leases** (include right-of-use assets and lease debt under Ind AS 116).
5. **Normalise for cyclical profits** (use mid-cycle margins).
6. **Compare with peers** on the same definition.

33.5.1 Decomposition

ROIC = NOPAT margin × Capital turnover. Two firms with the same ROIC can have very different strategies: a high-margin, low-turnover luxury brand and a low-margin, high-turnover discount retailer. The sustainability of each source of ROIC differs: price premium (brand) versus volume efficiency (scale).

33.5.2 Incremental ROIC

The ROIC that matters for value is the return on *new* investment — **incremental ROIC** = ΔNOPAT ÷ Net investment (lagged). It can be higher or lower than average ROIC. A mature firm often has incremental ROIC *below* average ROIC (the best opportunities have been taken); a firm entering a new, profitable market may have the reverse. In a forecast, specify the marginal sales-to-capital and the margin trajectory, and compute the implied incremental ROIC as a check.

33.6 From story to numbers: translating a moat into a forecast

Qualitative judgement	Translation
"Strong brand; pricing power"	Gross margin stable or rising; pass-through of cost inflation; terminal ROIC substantially above WACC; CAP 10–20 years
"Commodity business with no moat"	Margin reverts to the industry average (mid-cycle); terminal ROIC ≈ WACC; CAP short
"Scale advantage in a growing market"	Margin rises with scale to a cap; reinvestment high early; ROIC rising to plateau
"Regulated returns"	ROIC ≈ allowed return; growth = growth in regulated asset base
"Disruption risk"	Scenario with early fade; probability-weighted; lower terminal growth
"Patent-protected product"	High margins to patent expiry; sharp fall after (patent cliff); not perpetual

33.7 Growth, discipline and capital allocation

A company's capital-allocation record is evidence for ROIC and for how much reinvestment will earn:

- **Organic reinvestment** — track returns on incremental capex and working capital.
- **Acquisitions** — compare the price paid with the value created; track goodwill impairments.
- **Dividends and buybacks** — returning cash is correct when the firm lacks opportunities above WACC. A buyback at a price above intrinsic value transfers wealth to selling shareholders.
- **Leverage policy** — affects the discount rate and risk but not the operating ROIC.

A caution about growth for its own sake. Management teams often prefer growth (size, prestige, compensation tied to revenue). A valuer should ask whether proposed growth earns its cost of capital; if not, the DCF should show *negative* value from the incremental growth — and a prudent analyst might value the business as if the growth were *not* pursued.

33.8 Worked illustration: three companies, same growth, different values

All three have $NOPAT_1$ = ₹100 crore, WACC 11.225% and expect 5% growth forever.

Company	Terminal ROIC	Reinvestment rate	$FCFF_1$	EV	EV/NOPAT
A — "Commodity"	11.225%	44.5%	55.5	891	8.9×
B — "Solid brand"	15%	33.3%	66.7	1,071	10.7×
C — "Dominant platform"	30%	16.7%	83.3	1,338	13.4×

The same growth rate gives EVs that differ by 50% purely because of the return on reinvested capital. Notice the *cash* difference: A pays out 55.5% of its NOPAT and C pays out 83.3%, even though both grow at 5%. C can fund the same growth with less than half the reinvestment. **That** is what a moat buys.

33.9 Practical guidance for DCF builders

1. **Make terminal ROIC an explicit input.** Document its value, its basis (evidence of advantage) and its sensitivity.
2. **Fade ROIC** toward the terminal level across the explicit period or in a fade period; avoid step changes.
3. **Report the value of growth** and the share of EV derived from excess returns.
4. **Choose CAP deliberately:** 5, 10, 15 or 20 years, linked to the source of advantage.
5. **Cross-check against base rates:** what fraction of firms with ROIC of 25% today still have it in ten years? (Typically a minority: persistence is lower than most stories assume.)
6. **When in doubt, be conservative about duration, not about the present.** An overstated near-term margin is visible and testable; an overstated moat is invisible until too late.

Chapter 33 summary

- Value per rupee reinvested = ROIC ÷ WACC; growth creates value only when ROIC > WACC.
- EV/NOPAT depends on growth *and* ROIC; at ROIC = WACC growth is irrelevant.
- The value of growth is the DCF value minus earnings power value; it can be a small or large part of EV.
- The competitive-advantage period matters; most value from a moat arrives in the first ten years.

- Translate qualitative moats into ROIC paths and a CAP, with evidence and threats.

Review questions

1. Why should a firm with ROIC below WACC not grow?
2. Compute $EV/NOPAT_1$ for ROIC 18%, g 4%, WACC 10%.
3. What is the "value of growth" and how is it measured?
4. How would you estimate the CAP for a company whose advantage is a 12-year patent?
5. Distinguish average ROIC from incremental ROIC.

Appendices

Appendix A — Formula Sheet

A.1 Time value of money

Item	Formula
Future value	$FV = PV \times (1 + r)^n$
Present value	$PV = FV \div (1 + r)^n$
Effective annual rate	$(1 + r/m)^m - 1$
NPV	$\Sigma\, CF_t \div (1 + r)^t$
Perpetuity	$PV = C \div r$
Growing perpetuity	$PV = C_1 \div (r - g)$
Annuity	$PV = C \times [1 - (1 + r)^{-n}] \div r$
Growing annuity	$PV = C_1 \times [1 - ((1 + g)/(1 + r))^n] \div (r - g)$
Fisher	$(1 + \text{nominal}) = (1 + \text{real})(1 + \text{inflation})$
Mid-year factor	$1 \div (1 + r)^{t - 0.5}$

A.2 Operating performance

Item	Formula
NOPAT	$EBIT \times (1 - t)$
Invested capital (operating)	Net fixed assets + operating NWC + intangibles + other operating assets − other operating liabilities
Invested capital (financing)	Equity + debt + preference + minorities − cash − non-operating investments
ROIC	NOPAT ÷ Invested capital (opening or average)
ROIC decomposition	(NOPAT ÷ Revenue) × (Revenue ÷ Invested capital)
Reinvestment rate	Net investment ÷ NOPAT
Growth identity	g = Reinvestment rate × ROIC
Net investment	Capex − D&A + ΔNWC (+ other operating assets)
Reinvestment (sales-to-capital)	ΔRevenue ÷ (Sales-to-capital)
EVA	NOPAT − WACC × Invested capital
MVA	Market value of firm − Invested capital

A.3 Free cash flows

Item	Formula
FCFF (from EBIT)	$EBIT(1 - t) + D\&A - Capex - \Delta NWC$
FCFF (from EBITDA)	$EBITDA(1 - t) + t \times D\&A - Capex - \Delta NWC$
FCFF (from net income)	$NI + Interest(1 - t) + D\&A - Capex - \Delta NWC$
FCFF (from CFO)	$CFO + Interest(1 - t) - Capex$

Item	Formula
FCFE	NI + D&A − Capex − ΔNWC + Net borrowing
FCFE (from FCFF)	FCFF − Interest(1 − t) + Net borrowing
FCFE with constant debt ratio δ	NI − (1 − δ) × Net reinvestment
Stable-growth FCFE	NI × (1 − g/ROE)
Stable-growth FCFF	NOPAT × (1 − g/ROIC)

A.4 Discount rates

Item	Formula
CAPM	$Ke = Rf + \beta \times ERP$
APT	$Ke = Rf + \Sigma\, \beta_i\, \lambda_i$
Blume-adjusted beta	0.67 × raw β + 0.33
Unlevered beta (Hamada)	$\beta_u = \beta_l \div [1 + (1 - t)\, D/E]$
Re-levered beta	$\beta_l = \beta_u \times [1 + (1 - t)\, D/E]$
Asset beta with debt beta	$\beta_a = \beta_e \times E/[E + D(1 - t)] + \beta_d \times D(1 - t)/[E + D(1 - t)]$
Cash-adjusted beta	β_operating = β_unlevered ÷ (1 − Cash/Firm value)
Cost of debt	Kd = Rf + default spread
WACC	$E/V \times Ke + D/V \times Kd(1 - t)$
Unlevered cost of equity	$Ku = Rf + \beta_u \times ERP$
ERP with country risk	ERP(mature) + CRP; CRP = default spread × σ_equity/σ_bond
Currency conversion of rates	$(1 + r_{inr}) = (1 + r_{usD})(1 + Rf_{inr})/(1 + Rf_{usD})$

A.5 Valuation

Item	Formula
Enterprise value (DCF)	$\Sigma\, FCFF_t \div (1 + WACC)^t + TV_n \div (1 + WACC)^n$
Terminal value (Gordon)	$FCFF_{n+1} \div (WACC - g)$
Terminal value (value driver)	$NOPAT_{n+1} \times (1 - g/ROIC) \div (WACC - g)$
TV when ROIC = WACC	$NOPAT_{n+1} \div WACC$
TV (exit multiple)	$Multiple \times Metric_n$
H-model	$FCFF_n\, [(1 + g_s) + H(g_h - g_s)/2] \div (WACC - g_s)$
Equity bridge	EV − debt − preference − NCI + cash + non-operating assets
Value per share	Equity to common ÷ diluted shares
DDM (Gordon)	$P_0 = D_1 \div (Ke - g)$
Stable P/E	Payout × (1 + g) ÷ (Ke − g)
Stable P/B	(ROE − g) ÷ (Ke − g)
Stable EV/NOPAT	(1 − g/ROIC) ÷ (WACC − g)
Excess-return model	Book equity + PV of (ROE − Ke) × opening book equity
APV	Unlevered value + PV(tax shield) − expected distress cost

Item	Formula
PV tax shield (perpetual debt)	$t \times D$
Distressed equity	DCF equity × (1 − P) + Distress equity × P
VC post-money	Exit value ÷ $(1 + \text{IRR})^n$
Control premium ↔ DLOC	$\text{DLOC} = 1 - 1/(1 + \text{CP})$
Cap-rate value	NOI ÷ Cap rate
Black–Scholes call	$S \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$, $d_1 = [\ln(S/K) + (r + \sigma^2/2)T] \div (\sigma\sqrt{T})$, $d_2 = d_1 - \sigma\sqrt{T}$

A.6 Meridian Foods Ltd — summary of the running case (₹ crore)

Item	Value
Base revenue / EBIT / NOPAT (FY26)	2,000 / 240 / 180
Invested capital (FY26)	1,450
WACC (Ke 12.34%, Kd 9%, D/V 20%)	11.225%
FCFF FY27–FY31	73.7, 112.2, 156.7, 195.1, 237.1
Terminal g / ROIC	5% / 15%
PV explicit / PV TV	537.7 / 2,375.6
Enterprise value	2,913.3
Equity value	2,563.3
Value per share	₹51.27
Sensitivity range (WACC ±0.5, g 4.5–5.5%)	₹45.2 – ₹58.8
Scenario range (bear–bull)	₹30.0 – ₹67.3
Monte Carlo 90% interval	₹36 – ₹69

Appendix B — Practice Problems with Solutions

Attempt each problem before reading the solution. Where tables of discount factors are needed, compute directly or use Appendix D. Amounts are in ₹ crore unless stated.

Problems

1. NPV and IRR. A project costs ₹800 and returns ₹250, ₹300, ₹350 and ₹400 at the ends of years 1–4. Cost of capital 11%. Compute the NPV and the approximate IRR.

2. Gordon growth. A share is expected to pay a dividend of ₹12 next year, growing at 5% for ever. Ke = 13%. (a) Value the share. (b) If the market price is ₹180, what growth rate is the market implying?

3. FCFF. EBIT ₹600; tax rate 25%; depreciation ₹150; capex ₹260; operating NWC rises from ₹400 to ₹448. Compute FCFF.

4. FCFE. Using the data in Problem 3, net income is ₹380, new debt raised ₹100 and debt repaid ₹40. Compute FCFE.

5. Beta. A peer has equity beta 1.15 and D/E 40% (tax 25%). Compute its unlevered beta; re-lever to D/E 30%; compute the resulting cost of equity at Rf 7%, ERP 6%.

6. WACC. Equity (market) ₹1,400; debt ₹600; Ke 14%; pre-tax Kd 10%; tax 25%. Compute WACC.

7. Two-stage DCF. FCFF for the next five years: 100, 120, 140, 160, 180. WACC 10%; terminal growth 4% (assume $FCFF_6 = FCFF_5 \times 1.04$). Debt 500; cash 80; shares 20 crore. Compute EV, equity value and value per share.

8. Terminal value consistency. NOPAT in the first terminal year is ₹250; WACC 10.5%; g = 5%. Compute TV when terminal ROIC is (a) 12.5% and (b) equal to WACC.

9. EVA. EBIT ₹420; tax 25%; invested capital ₹2,000; WACC 10.5%. Compute NOPAT, ROIC and EVA.

10. Mid-year. Re-value Problem 7 using the mid-year convention (TV by Gordon discounted at 4.5 years). By what percentage does EV change?

11. APV. Unlevered FCFF is ₹150 next year, growing 5% for ever. Ku = 12%. The firm will keep ₹800 of debt permanently at 9%; tax rate 25%. Compute APV enterprise value.

12. Bank. A bank earns a stable ROE of 16%, has Ke 13% and grows at 6%. Compute the justified P/B. What would P/B be if ROE were 13%?

13. Venture capital. A start-up is expected to be worth ₹400 crore in 4 years. An investor wants a 50% IRR and will invest ₹15 crore. Compute post-money, pre-money and the ownership needed at exit; adjust for 20% dilution before exit.

14. Distress. Going-concern equity value is ₹800; distress-sale equity value ₹100; probability of distress 35%. Compute the blended equity value.

15. Currency. A rupee WACC is 12%; Rf(INR) 7.2%; Rf(USD) 4.0%. Compute the equivalent dollar WACC.

16. Mini-DCF (three-year explicit period). A company has revenue of ₹1,000, growing 15%, 12%, 10% in years 1–3; EBIT margins 14%, 15%, 16%; tax 25%; marginal sales-to-capital 2.5. WACC 10%; terminal g 4%; terminal ROIC 12%, terminal margin 16%. Debt ₹300; cash ₹50; 25 crore shares. Compute value per share.

17. Stub period. Valuation date 30 September 2026, fiscal year-end 31 March. FY27 FCFF is ₹360 and FY28 FCFF is ₹400. WACC 10%. Using the mid-period convention, compute the PV of (a) the FY27 remaining flow and (b) the FY28 flow.

Solutions

1. NPV at 11%: −800 + 250/1.11 + 300/1.2321 + 350/1.3676 + 400/1.5181 = −800 + 225.23 + 243.49 + 255.92 + 263.49 = **₹188.1**. IRR ≈ **20.7%** (at 20.7% NPV = 0).

2. (a) P = 12 ÷ (0.13 − 0.05) = **₹150**. (b) With P = 180: g = Ke − D_1/P = 13% − 12/180 = 13% − 6.67% = **6.33%**.

3. NOPAT = 600 × 0.75 = 450; ΔNWC = 48. FCFF = 450 + 150 − 260 − 48 = **₹292**.

4. FCFE = 380 + 150 − 260 − 48 + 100 − 40 = **₹282**. (Check: FCFF 292 − after-tax interest + net borrowing 60. After-tax interest implied = 450 − 380 = 70, so 292 − 70 + 60 = 282 ✔.)

5. β_u = 1.15 ÷ (1 + 0.75 × 0.40) = 1.15 ÷ 1.30 = **0.8846**. Re-levered: 0.8846 × (1 + 0.75 × 0.30) = 0.8846 × 1.225 = **1.0837**. Ke = 7% + 1.0837 × 6% = **13.50%**.

6. WACC = (1,400 × 14% + 600 × 10% × 0.75) ÷ 2,000 = (196 + 45) ÷ 2,000 = **12.05%**.

7. PVs: 90.91 + 99.17 + 105.18 + 109.28 + 111.77 = **516.31**. TV = 180 × 1.04 ÷ (0.10 − 0.04) = 3,120; PV = 3,120 ÷ 1.61051 = **1,937.27**. EV = **2,453.59**. Equity = 2,453.59 − 500 + 80 = **2,033.59**; per share = **₹101.68**.

8. (a) Reinvestment rate = 5% ÷ 12.5% = 40%; FCFF = 250 × 0.60 = 150; TV = 150 ÷ 0.055 = **₹2,727**. (b) ROIC = WACC → TV = NOPAT ÷ WACC = 250 ÷ 0.105 = **₹2,381** (the value of growth at ROIC 12.5% is ₹346).

9. NOPAT = 420 × 0.75 = **315**; ROIC = 315 ÷ 2,000 = **15.75%**; EVA = 315 − 0.105 × 2,000 = 315 − 210 = **₹105**.

10. PV of explicit flows at t − 0.5: **541.52**; PV of TV at 4.5 years: 3,120 ÷ $1.10^{4.5}$ = **2,031.83**; EV = **2,573.35**, i.e., **+4.9%** (≈ $(1.10)^{0.5}$ − 1 = 4.88%).

11. Unlevered EV = 150 ÷ (0.12 − 0.05) = **2,142.9**. PV tax shield = 0.25 × 800 = **200**. APV = **₹2,342.9**.

12. P/B = (0.16 − 0.06) ÷ (0.13 − 0.06) = 0.10 ÷ 0.07 = **1.43×**. If ROE = Ke = 13%, P/B = (0.13 − 0.06) ÷ (0.13 − 0.06) = **1.00×** — growth adds no value.

13. Post-money = 400 ÷ 1.5^4 = 400 ÷ 5.0625 = **₹79.0 crore**; pre-money = 79.0 − 15 = **₹64.0 crore**; ownership needed at exit = 15 ÷ 79.0 = **19.0%**. With 20% dilution before exit: 19.0% ÷ 0.80 = **23.7%** today.

14. 800 × 0.65 + 100 × 0.35 = 520 + 35 = **₹555**.

15. Dollar WACC = 1.12 ÷ (1.072/1.040) − 1 = 1.12 ÷ 1.03077 − 1 = **8.66%**.

16. Revenue: 1,150; 1,288; 1,416.8. NOPAT: 120.8; 144.9; 170.0. Net reinvestment (ΔRev ÷ 2.5): 60.0; 55.2; 51.5. FCFF: **60.8; 89.7; 118.5**. PV at 10%: 55.3 + 74.1 + 89.0 = **218.4**. Terminal: Revenue_4 = 1,416.8 × 1.04 = 1,473.5; NOPAT_4 = 1,473.5 × 0.16 × 0.75 = 176.8; reinvestment rate = 4% ÷ 12% = 33.3%; FCFF_4 = 117.9; TV_3 = 117.9 ÷ 0.06 = 1,964.6; PV = 1,964.6 ÷ 1.331 = **1,476.1**. EV = **1,694.4**. Equity = 1,694.4 − 300 + 50 = **1,444.4**; per share = **₹57.78**.

17. (a) Remaining half-year of FY27 FCFF = 180; mid-point is 0.25 years away: $180 \div 1.10^{0.25}$ = **₹175.76**. (b) FY28 cash flows have a mid-point at 0.5 + 0.5 = 1.0 year: 400 ÷ 1.10 = **₹363.64**.

Appendix C — Glossary

Adjusted present value (APV). Value of the unlevered firm plus the present value of financing side effects (tax shields, less distress costs).

Asset beta. Beta of a firm's operating assets, free of the effect of financial leverage. Also called unlevered beta.

Beta (β). A measure of systematic risk: the covariance of a security's returns with market returns divided by the variance of market returns.

Bottom-up beta. A beta built by un-levering and averaging betas of comparable firms, then re-levering to the target capital structure.

Bridge (equity bridge). The adjustments from enterprise value to equity value: subtract debt and other claims; add cash and non-operating assets.

CAPM. Capital Asset Pricing Model: $K_e = R_f + \beta \times ERP$.

Capitalisation rate (cap rate). Net operating income divided by property value; the reciprocal of a value multiple.

Capital charge. Invested capital multiplied by WACC; the dollar cost of the capital employed.

Capex. Capital expenditure on fixed assets.

Cash conversion cycle. Receivable days + inventory days − payable days.

Circularity. A situation in which value depends on WACC, which depends on value (through market-value weights); resolved by iteration.

Competitive-advantage period. The time over which a firm can earn returns above its cost of capital.

Control premium. The additional value a buyer pays to obtain control of a company over the value of a minority stake.

Cost of debt (Kd). The current rate at which the firm can borrow long-term; pre-tax unless stated.

Cost of equity (Ke). The return required by shareholders for bearing the risk of the equity.

Country risk premium (CRP). An addition to the equity risk premium reflecting the additional risk of investing in a particular country.

DCF. Discounted cash flow: valuation by discounting expected future cash flows at a risk-adjusted rate.

DDM. Dividend discount model.

Debt beta. Beta of a firm's debt; typically close to zero for investment-grade debt.

Debt-like items. Obligations that behave like debt and are deducted in the equity bridge (pension deficits, lease liabilities, deferred consideration, etc.).

Discount factor. $1 \div (1 + r)^n$.

DLOC. Discount for lack of control.

DLOM. Discount for lack of marketability.

EBIT. Earnings before interest and taxes (operating profit).

EBITDA. EBIT plus depreciation and amortisation.

Enterprise value (EV). The value of operating assets: equity value plus debt and other claims, less cash and non-operating assets.

Equity risk premium (ERP). The extra return investors require from equity over the risk-free rate.

EVA. Economic value added: NOPAT − WACC × invested capital.

Excess-return model. An equity valuation as book value plus the present value of returns above the cost of equity.

Fair market value. The price at which property would change hands between a willing buyer and a willing seller, neither under compulsion and both reasonably informed.

Fair value (Ind AS 113). The price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants.

FCFE. Free cash flow to equity: cash available to shareholders after reinvestment and debt service.

FCFF. Free cash flow to the firm: after-tax operating cash flow after reinvestment, available to all capital providers.

First Chicago method. A valuation that probability-weights success, sideways and failure scenarios.

Going-concern value. Value of a business that will continue to operate.

Gordon growth model. Value of a growing perpetuity: $C_1 \div (r - g)$.

Growth identity. g = reinvestment rate × ROIC.

Hamada equation. Relationship between levered and unlevered beta: $\beta_l = \beta_u[1 + (1 - t)D/E]$.

Hurdle rate. The minimum acceptable return on a project or investment.

Interest tax shield. Tax saved because interest is deductible: t × interest.

IRR. Internal rate of return: the discount rate at which NPV is zero.

Invested capital. Operating capital employed: net fixed assets, working capital, intangibles and other operating assets (or, equivalently, equity plus debt less cash and non-operating investments).

LBO. Leveraged buy-out: acquisition financed largely with debt.

Liquidation value. Net proceeds from selling a business's assets and settling its liabilities.

Marginal tax rate. The tax rate applicable to the next rupee of income.

Mid-year convention. Discounting cash flows as if received half-way through each year.

Monte Carlo simulation. Repeated random sampling of uncertain inputs to produce a distribution of valuation outcomes.

MVA. Market value added: market value of the firm minus invested capital.

Net working capital (operating). Operating current assets less operating current liabilities (excluding cash and debt).

NOPAT. Net operating profit after tax: EBIT × (1 − t).

NPV. Net present value: sum of discounted cash flows, net of the investment.

Non-operating assets. Assets not used in the core business (surplus cash, investments, idle land).

Pre-money / post-money value. Value of a company before / after a new investment; post = pre + investment.

Premise of value. The assumption on which value is estimated: going concern or liquidation.

Real option. The right, not the obligation, to take a business action at a future time.

Relative valuation. Valuation by reference to the pricing of comparable assets, using multiples.

Reinvestment rate. Net investment divided by NOPAT.

Reverse DCF. A DCF solved for the assumptions implied by the market price.

Risk-free rate (Rf). Return on an asset with no default or reinvestment risk, in the currency of the cash flows.

ROIC. Return on invested capital: NOPAT ÷ invested capital.

ROE. Return on equity: net income ÷ book equity.

Sales-to-capital ratio. Incremental revenue per unit of incremental invested capital; the inverse is capital intensity.

Scenario analysis. Valuation under a small set of coherent alternative futures.

Sensitivity analysis. Measuring the change in value when one or more inputs change.

Standard of value. The type of value being estimated (fair market value, fair value, investment value, etc.).

Stub period. A partial first period when the valuation date is not a fiscal year-end.

Sum-of-the-parts (SOTP). Valuation of a diversified firm as the sum of its parts' values.

Terminal value (TV). The value at the end of the explicit forecast period of all cash flows beyond it.

Unlevered cost of equity (Ku). The required return on equity if the firm had no debt.

Value driver. An operating variable (growth, margin, reinvestment, ROIC) that determines value.

Value in use. Present value of cash flows expected from continuing use of an asset (Ind AS 36).

WACC. Weighted average cost of capital: $E/V \times Ke + D/V \times Kd \times (1 - t)$.

Appendix D — Discount Factor Tables

Table D1 gives the present-value factor $1/(1+r)^n$; Table D2 gives the annuity factor $[1 - (1+r)^{-n}]/r$. Rows are years; columns are discount rates.

Table D1: Present value of ₹1 received at the end of year n

Year	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%
1	0.9434	0.9346	0.9259	0.9174	0.9091	0.9009	0.8929	0.8850	0.8772	0.8696
2	0.8900	0.8734	0.8573	0.8417	0.8264	0.8116	0.7972	0.7831	0.7695	0.7561
3	0.8396	0.8163	0.7938	0.7722	0.7513	0.7312	0.7118	0.6931	0.6750	0.6575
4	0.7921	0.7629	0.7350	0.7084	0.6830	0.6587	0.6355	0.6133	0.5921	0.5718
5	0.7473	0.7130	0.6806	0.6499	0.6209	0.5935	0.5674	0.5428	0.5194	0.4972
6	0.7050	0.6663	0.6302	0.5963	0.5645	0.5346	0.5066	0.4803	0.4556	0.4323
7	0.6651	0.6227	0.5835	0.5470	0.5132	0.4817	0.4523	0.4251	0.3996	0.3759
8	0.6274	0.5820	0.5403	0.5019	0.4665	0.4339	0.4039	0.3762	0.3506	0.3269
9	0.5919	0.5439	0.5002	0.4604	0.4241	0.3909	0.3606	0.3329	0.3075	0.2843
10	0.5584	0.5083	0.4632	0.4224	0.3855	0.3522	0.3220	0.2946	0.2697	0.2472
11	0.5268	0.4751	0.4289	0.3875	0.3505	0.3173	0.2875	0.2607	0.2366	0.2149
12	0.4970	0.4440	0.3971	0.3555	0.3186	0.2858	0.2567	0.2307	0.2076	0.1869
13	0.4688	0.4150	0.3677	0.3262	0.2897	0.2575	0.2292	0.2042	0.1821	0.1625
14	0.4423	0.3878	0.3405	0.2992	0.2633	0.2320	0.2046	0.1807	0.1597	0.1413
15	0.4173	0.3624	0.3152	0.2745	0.2394	0.2090	0.1827	0.1599	0.1401	0.1229
16	0.3936	0.3387	0.2919	0.2519	0.2176	0.1883	0.1631	0.1415	0.1229	0.1069
17	0.3714	0.3166	0.2703	0.2311	0.1978	0.1696	0.1456	0.1252	0.1078	0.0929
18	0.3503	0.2959	0.2502	0.2120	0.1799	0.1528	0.1300	0.1108	0.0946	0.0808
19	0.3305	0.2765	0.2317	0.1945	0.1635	0.1377	0.1161	0.0981	0.0829	0.0703
20	0.3118	0.2584	0.2145	0.1784	0.1486	0.1240	0.1037	0.0868	0.0728	0.0611

Table D2: Present value of an annuity of ₹1 per year for n years

Year	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%
1	0.9434	0.9346	0.9259	0.9174	0.9091	0.9009	0.8929	0.8850	0.8772	0.8696
2	1.8334	1.8080	1.7833	1.7591	1.7355	1.7125	1.6901	1.6681	1.6467	1.6257
3	2.6730	2.6243	2.5771	2.5313	2.4869	2.4437	2.4018	2.3612	2.3216	2.2832
4	3.4651	3.3872	3.3121	3.2397	3.1699	3.1024	3.0373	2.9745	2.9137	2.8550
5	4.2124	4.1002	3.9927	3.8897	3.7908	3.6959	3.6048	3.5172	3.4331	3.3522

Year	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%
6	4.9173	4.7665	4.6229	4.4859	4.3553	4.2305	4.1114	3.9975	3.8887	3.7845
7	5.5824	5.3893	5.2064	5.0330	4.8684	4.7122	4.5638	4.4226	4.2883	4.1604
8	6.2098	5.9713	5.7466	5.5348	5.3349	5.1461	4.9676	4.7988	4.6389	4.4873
9	6.8017	6.5152	6.2469	5.9952	5.7590	5.5370	5.3282	5.1317	4.9464	4.7716
10	7.3601	7.0236	6.7101	6.4177	6.1446	5.8892	5.6502	5.4262	5.2161	5.0188
11	7.8869	7.4987	7.1390	6.8052	6.4951	6.2065	5.9377	5.6869	5.4527	5.2337
12	8.3838	7.9427	7.5361	7.1607	6.8137	6.4924	6.1944	5.9176	5.6603	5.4206
13	8.8527	8.3577	7.9038	7.4869	7.1034	6.7499	6.4235	6.1218	5.8424	5.5831
14	9.2950	8.7455	8.2442	7.7862	7.3667	6.9819	6.6282	6.3025	6.0021	5.7245
15	9.7122	9.1079	8.5595	8.0607	7.6061	7.1909	6.8109	6.4624	6.1422	5.8474
16	10.1059	9.4466	8.8514	8.3126	7.8237	7.3792	6.9740	6.6039	6.2651	5.9542
17	10.4773	9.7632	9.1216	8.5436	8.0216	7.5488	7.1196	6.7291	6.3729	6.0472
18	10.8276	10.0591	9.3719	8.7556	8.2014	7.7016	7.2497	6.8399	6.4674	6.1280
19	11.1581	10.3356	9.6036	8.9501	8.3649	7.8393	7.3658	6.9380	6.5504	6.1982
20	11.4699	10.5940	9.8181	9.1285	8.5136	7.9633	7.4694	7.0248	6.6231	6.2593

Table D3: Terminal-value multiples 1/(WACC − g)

WACC \ g	2%	3%	4%	5%	6%
8%	16.7×	20.0×	25.0×	33.3×	50.0×
9%	14.3×	16.7×	20.0×	25.0×	33.3×
10%	12.5×	14.3×	16.7×	20.0×	25.0×
11%	11.1×	12.5×	14.3×	16.7×	20.0×
12%	10.0×	11.1×	12.5×	14.3×	16.7×
13%	9.1×	10.0×	11.1×	12.5×	14.3×
14%	8.3×	9.1×	10.0×	11.1×	12.5×

Multiply the first post-horizon free cash flow by the factor to obtain the terminal value at the horizon date.

Appendix E — Building the Model: A Spreadsheet Blueprint

This appendix describes a clean layout for a single-company FCFF DCF in a spreadsheet, with the formulas that implement each step.

E.1 Sheet layout

Sheet	Contents
Inputs	All assumptions: valuation date, tax rate, growth by year, margin by year, sales-to-capital, terminal g and ROIC, WACC inputs, bridge items, shares, scenario selector
Historical	Three to five years of reformulated financial statements and ratios
Forecast	Revenue, EBIT, NOPAT, reinvestment, invested capital, ROIC
FCFF	Free cash flow build and cross-check by a second method
WACC	Peer betas, un-levering, re-levering, Ke, Kd, weights, WACC
DCF	Discount factors, PVs, terminal value, EV, bridge, per-share value
Sensitivity	WACC × g and ROIC × g tables, scenario table
Checks	All reconciliation tests, with a master TRUE/FALSE flag

Time runs left to right with the same column for each year on every sheet.

E.2 Core formulas (for a forecast starting in column D)

Assume row labels in column B and the first forecast year in column D; previous-year column C holds the base year.

Line	Formula (cell D)
Revenue	`=C_rev*(1+D_growth)`
EBIT	`=D_rev*D_margin`
Tax on EBIT	`=D_ebit*Inputs!tax`
NOPAT	`=D_ebit-D_tax`
Net reinvestment	`=(D_rev-C_rev)/Inputs!s2c`
FCFF	`=D_nopat-D_reinv`
Invested capital (closing)	`=C_ic+D_reinv`
ROIC (on opening capital)	`=D_nopat/C_ic`
Discount factor (end-year)	`=1/(1+WACC)^D_t`
Discount factor (mid-year)	`=1/(1+WACC)^(D_t-0.5)`
PV of FCFF	`=D_fcff*D_df`

Terminal value

Line	Formula
Terminal revenue	`=H_rev*(1+g)`
Terminal NOPAT	`=term_rev*term_margin*(1-tax)`
Reinvestment rate	`=g/term_roic`
Terminal FCFF	`=term_nopat*(1-reinv_rate)`
Terminal value	`=term_fcff/(WACC-g)`
PV of TV	`=TV*H_df`

Valuation

Line	Formula
Enterprise value	`=SUM(pv_range)+PV_TV`
Equity value	`=EV-debt+cash+investments-other_claims`
Per share	`=equity/diluted_shares`

E.3 The WACC sheet

Line	Formula
Unlevered beta (each peer)	`=beta/(1+(1-tax_peer)*DE_peer)`
Average / median	`=AVERAGE(range)` / `=MEDIAN(range)`
Re-levered beta	`=bu*(1+(1-tax)*target_DE)`
Cost of equity	`=Rf+beta_L*ERP`
After-tax Kd	`=Kd*(1-tax)`
Weights	`=D/V` with `D/V = target_DE/(1+target_DE)`
WACC	`=(1-wd)*Ke+wd*Kd_at`

E.4 Sensitivity tables

Use a two-input *data table* (What-If Analysis) with the row input cell set to terminal growth and the column input cell set to WACC. Because the WACC cell must be an *input* (not a formula) for the data table to flex it, include an "override" input: `=IF(override="",calculated_WACC,override)`. In newer spreadsheet versions the same effect can be achieved with a LAMBDA or by building the grid with explicit formulas.

E.5 Check cells

Check	Test
Invested capital identity	Operating-side = financing-side at base year
FCFF two-way	FCFF (EBIT-based) − FCFF (net-income-based) = 0

Check	Test
Terminal	g < Rf; WACC − g ≥ 0.03; reinvestment rate ≥ 0; ROIC_term ≥ WACC (flag if not)
Bridge	Every balance-sheet financing item accounted for
Sensitivity centre	Centre of data table = model output
Scenario selector	Active scenario matches the one reported
Circularity	Iteration flag, with convergence tolerance < 0.01%

A master flag cell sums all checks and turns the sheet's header red if any fails.

E.6 Common formula errors

- **Absolute vs relative references:** locking `$` on the wrong cell when copying across years.
- **Range off-by-one** in NPV/SUM ranges (omitting the last forecast year).
- **Mixing units** (₹ crore vs ₹ lakh; thousands vs millions).
- **Hard-coded numbers** inside formulas (e.g., `*0.75` instead of a tax input).
- **Inconsistent sign conventions** (outflows negative in one block, positive in another).
- **Percentages entered as whole numbers.**
- **Circular references** left unresolved or disabled.

E.7 Documentation

Each input should show its **source** (annual report note, broker report, central-bank website, management certificate), **date**, and **rationale**. A short "assumptions memo" at the front of the workbook, written in plain language, helps reviewers.

Appendix F — Answers to Numerical Review Questions

Selected numerical review questions from the chapters, with answers.

Chapter	Question	Answer
2	Convert a nominal 12% to real with 6% inflation	1.12 ÷ 1.06 − 1 = **5.66%**
3	FCFE: NI 120, D&A 40, capex 90, ΔNWC 15, new debt 30, repaid 10	120 + 40 − 90 − 15 + 30 − 10 = **75**
4	ROIC: EBIT 300, t 25%, IC 1,800	300 × 0.75 ÷ 1,800 = **12.5%**
5	FCFF: EBIT 500, t 30%, dep 120, capex 200, ΔNWC 45	**225** (worked in the text)
6	FCFE: NI 200, dep 60, capex 140, ΔNWC 20, borrowing 50, repayment 30	200 + 60 − 140 − 20 + 50 − 30 = **120**
6	Sustainable payout with ROE 18% and g 6%	1 − 6/18 = **66.7%**
9	Ke with Rf 7%, ERP 6%, β 1.2	7 + 1.2 × 6 = **14.2%**
10	Unlevered β: levered 1.2, D/E 50%, t 25%	1.2 ÷ 1.375 = **0.873**; re-levered at D/E 20%: 0.873 × 1.15 = **1.004**
11	WACC: E 700, D 300, Ke 14%, Kd 10%, t 30%	(98 + 21) ÷ 1,000 = **11.9%**
12	TV: $NOPAT_6$ 200, g 4%, ROIC 12%, WACC 10%	200 × (1 − 0.3333) ÷ 0.06 = **2,222**
17	PV of tax shield: debt 400, Kd 10%, t 30%	0.30 × 400 = **120**
18	P/B: ROE 18%, Ke 14%, g 6%	(0.18 − 0.06) ÷ (0.14 − 0.06) = **1.50×**
19	Property value: NOI 8, cap 7.5% / 8.5%	**106.7** / **94.1** (−11.8%)
20	Pre-money: exit 400 in 4 years, IRR 50%, investment 15	400 ÷ 5.0625 = 79.0 post; **64.0** pre
21	Blended equity: 800, 100, P = 35%	**555**
22	Dollar WACC from 12% rupee, Rf(INR) 7.2%, Rf(USD) 4.0%	**8.66%**
23	Pharma pipeline: Phase II cost 40 (p 50%), Phase III cost 120 (p 50%), launch value 1,200	Phase III start: 0.5 × 1,200 − 120 = 480; Phase II start: 0.5 × 480 − 40 = **200**

Selected conceptual answers

Chapter 1, Q4 — Five steps of DCF. (1) Understand the business and industry; (2) forecast free cash flows; (3) estimate the discount rate; (4) estimate the terminal value; (5) discount, sum, bridge to equity, and test with sensitivities and cross-checks.

Chapter 3, Q2 — FCFE at WACC overstates value. The WACC (a blend of the cost of equity and the cheaper after-tax cost of debt) is lower than the cost of equity. Discounting equity cash flows at a lower rate than the

equity holders require raises the present value. The debt in the capital structure has already been serviced in FCFE, so using a rate that also includes the cost of debt double-counts the benefit of debt.

Chapter 12, Q2 — ROIC = WACC. When new investment earns exactly the cost of capital, growth neither creates nor destroys value. TV = NOPAT × (1 − g/WACC) ÷ (WACC − g) = NOPAT ÷ WACC, in which g cancels.

Chapter 13, Q5 — DLOC and control premium. DLOC = 1 − 1/(1 + CP). A 30% DLOC implies 1/(1 + CP) = 0.70, so CP = 1/0.70 − 1 = **42.9%**.

Chapter 16, Q1 — Stable P/E. $P_0 = D_1 \div (Ke - g) = EPS_1 \times \text{payout} \div (Ke - g)$; dividing by EPS_0: P/E = payout × (1 + g) ÷ (Ke − g).

Chapter 22, Q2 — Rupee flows at a dollar rate. The dollar rate omits the expected depreciation of the rupee against the dollar, which is roughly the inflation (or interest-rate) differential. The resulting discount rate is too low and the value too high.

Appendix G — Further Practice Problems with Solutions

A second problem set, slightly more demanding than Appendix B. Amounts in ₹ crore unless stated.

Problems

G1. Growth identity. A firm earns ROIC of 16% on new investment and wants to grow NOPAT at 8% a year. What reinvestment rate is required? If NOPAT is ₹400, what is FCFF?

G2. Equity bridge. Enterprise value ₹5,000. Borrowings 1,200; lease liabilities 300; surplus cash 450; non-controlling interests (market-based) 400; a 30% associate with an equity value to the group of 250; an unfunded pension deficit of ₹80 pre-tax (tax 25%). Shares 50 crore. Compute equity value and value per share.

G3. Terminal-value share. PV of explicit cash flows is ₹800; terminal value at year 5 is ₹4,200; WACC 10%. Compute EV and the terminal value's share of EV.

G4. Beta and cash. The unlevered beta of a company *including* its cash is 0.80. Cash is 20% of firm value (cash beta ≈ 0). What is the beta of the operating business?

G5. Stable P/E. A stable firm has payout ratio 60%, growth 6% and cost of equity 12.5%. Compute the justified (forward-adjusted) P/E: P_0/EPS_0.

G6. EVA and MVA. NOPAT ₹270; invested capital ₹1,800; WACC 11%; market value of the firm ₹2,500. Compute EVA and MVA.

G7. Implied growth from an exit multiple. A buyer assumes an exit value of ₹4,500 at year 5 (9× EBITDA of 500). Year-6 FCFF would be ₹260 and WACC is 11%. What perpetual growth rate does the exit value imply?

G8. Options in the equity value. Equity value *before* option exercise is ₹1,000; 10 crore basic shares; 2 crore options with exercise price ₹40. Compute the value per share (options in the money).

G9. Lease consistency. Before Ind AS 116: EBITDA after rent ₹400; EV (excluding leases) ₹4,000. After Ind AS 116: rent of ₹60 is excluded from EBITDA and a lease liability of ₹450 is recognised. Compute EV/EBITDA on both bases and comment.

G10. Control and marketability. A minority, marketable stake is worth ₹100. Control premium 30%; DLOM for an unlisted minority stake 20%. Compute (a) the control value, (b) the DLOC implied relative to control value, (c) the value of an unlisted minority stake.

G11. Declining tax shield. A firm repays ₹1,000 of debt in four equal annual instalments (₹250 a year), interest at 10% on the opening balance, tax 25%, tax shield discounted at 10%. Compute the PV of the tax shield.

G12. Four-year mid-year DCF. Revenue ₹2,000 growing 10%, 10%, 8%, 8%; EBIT margin 15% throughout; tax 25%; marginal sales-to-capital 2.0; WACC 11%; terminal growth 5% and terminal ROIC 15% (margin 15%). Debt ₹800; cash ₹100; 40 crore shares. Mid-year discounting (TV discounted at 3.5 years). Compute the value per share.

G13. Scenario weighting. Bear, base and bull values are ₹60, ₹100 and ₹150 per share with probabilities 25%, 50% and 25%. Compute the expected value. What is the standard deviation?

G14. WACC. Ke 14%; pre-tax Kd 9%; tax 25%; D/V 30%. Compute WACC. If the debt ratio rises to 40% and Ke rises to 15.5% with Kd of 9.75%, compute the new WACC and comment.

Solutions

G1. Reinvestment rate = g/ROIC = 8% ÷ 16% = **50%**. FCFF = 400 × (1 − 0.50) = **₹200**.

G2. Equity = 5,000 − 1,200 − 300 + 450 − 400 + 250 − 60 (pension 80 × 0.75) = **₹3,740**. Per share = 3,740 ÷ 50 = **₹74.80**.

G3. PV of TV = 4,200 ÷ 1.10^5 = 2,607.9; EV = 800 + 2,607.9 = **₹3,407.9**; TV share = 2,607.9 ÷ 3,407.9 = **76.5%**.

G4. $\beta_{operating}$ = 0.80 ÷ (1 − 0.20) = **1.00**. (Cash dilutes the observed beta; the operating assets are riskier than the diluted figure suggests.)

G5. P/E = payout × (1 + g) ÷ (Ke − g) = 0.60 × 1.06 ÷ 0.065 = **9.78×**.

G6. EVA = 270 − 0.11 × 1,800 = 270 − 198 = **₹72**. MVA = 2,500 − 1,800 = **₹700**.

G7. 4,500 = 260 ÷ (0.11 − g) → 0.11 − g = 0.05778 → g = **5.22%**. This is a plausible but not undemanding rate; compare it with the risk-free rate and the economy's long-run growth.

G8. Exercise proceeds = 2 × 40 = 80. Equity after exercise = 1,000 + 80 = 1,080 over 12 crore shares: **₹90 per share**. (Check by treasury-stock method: 2 × (1 − 40/90) = 1.11 net new shares; 1,000 ÷ 11.11 = 90 ✔.)

G9. Before: EV/EBITDA = 4,000 ÷ 400 = **10.0×**. After: EBITDA = 460; EV including leases = 4,000 + 450 = 4,450; EV/EBITDA = 4,450 ÷ 460 = **9.67×**. The multiple drops slightly because the lease-adjusted EV rises by less (11.25%) than EBITDA (15%). *Never* divide an EV that excludes leases by an EBITDA that is before rent (it would be 4,000 ÷ 460 = 8.7×) — that mixes bases.

G10. (a) Control value = 100 × 1.30 = **₹130**. (b) DLOC = 1 − 1/1.30 = **23.1%**. (c) Unlisted minority stake = 100 × (1 − 0.20) = **₹80**.

G11. Interest: 100, 75, 50, 25; tax shields: 25.0, 18.75, 12.5, 6.25; PV at 10%: 22.73 + 15.50 + 9.39 + 4.27 = **₹51.9**.

G12. Revenue: 2,200; 2,420; 2,613.6; 2,822.7. NOPAT (15% × 0.75 of revenue): 247.5; 272.2; 294.0; 317.6. Net reinvestment (ΔRevenue ÷ 2.0): 100.0; 110.0; 96.8; 104.5. FCFF: **147.5; 162.2; 197.2; 213.0**. PV at 0.5, 1.5, 2.5, 3.5 years = **578.5**. Terminal: Revenue_5 = 2,822.7 × 1.05 = 2,963.8; NOPAT_5 = 333.4; reinvestment rate = 5% ÷ 15% = 33.3%; FCFF_5 = 222.3; TV = 222.3 ÷ 0.06 = 3,704.8; PV = 3,704.8 ÷ $1.11^{3.5}$ = **2,571.2**. EV = **3,149.7**. Equity = 3,149.7 − 800 + 100 = **2,449.7**; per share = **₹61.24**.

G13. Expected value = 0.25 × 60 + 0.50 × 100 + 0.25 × 150 = **₹102.5**. Variance = 0.25 × (60 − 102.5)² + 0.50 × (100 − 102.5)² + 0.25 × (150 − 102.5)² = 0.25 × 1,806.25 + 0.5 × 6.25 + 0.25 × 2,256.25 = 451.56 + 3.13 + 564.06 = 1,018.75; s.d. = **₹31.9**.

G14. First WACC = 0.70 × 14% + 0.30 × 9% × 0.75 = 9.80% + 2.025% = **11.83%**. Second: 0.60 × 15.5% + 0.40 × 9.75% × 0.75 = 9.30% + 2.925% = **12.23%**. WACC *rises* because Ke and Kd both rise with leverage by more than the weight shift lowers the blend; adding "cheap" debt does not automatically reduce WACC once the cost of equity is re-levered (Chapter 11).

Appendix H — Short Answers to Chapter Review Questions

Brief model answers to the review questions at the end of each chapter. Numerical answers repeat those in Appendix F where relevant. Use them to check understanding, not as a substitute for working the questions.

Chapter 1 — What Is Valuation?

1. *Cost* is what was paid (historical); *price* is what the asset trades for now (observed); *value* is the estimate of worth from expected cash flows and risk. Example: a listed company's book value (cost-based), its market capitalisation (price) and a DCF estimate (value).
2. M&A (negotiation ranges, synergies), financial reporting (fair value rules, market-participant assumptions) and tax/legal (prescribed standards). The purpose determines the standard of value and therefore the method and adjustments.
3. A multiple embeds assumptions about growth, risk and returns; applying a peer multiple assumes the subject matches the peer on those dimensions.
4. Understand the business; forecast free cash flows; estimate the discount rate; estimate terminal value; discount, bridge to equity and test.
5. Different views of growth, margins, reinvestment, risk and the durability of advantage; judgement is unavoidable, so a good valuation exposes it.

Chapter 2 — Mathematics of Discounting

1. Sum PV = $C_1/(1+r) \times [1 + q + q^2 + \ldots]$ with $q = (1+g)/(1+r) < 1$ gives $C_1/(1+r) \div (1 - q) = C_1/(r - g)$.
2. NPV() treats the first value as period 1; if the range includes today's investment it is discounted one period too much. Add the time-0 amount outside the function (or use XNPV).
3. A Gordon TV with mid-year flows is discounted at N − 0.5 years; an exit-multiple TV (sale at year end) at N years.
4. 1.12 ÷ 1.06 − 1 = 5.66%.

5. IRR ignores scale, assumes reinvestment at the IRR, and can be multiple or undefined when cash flows change sign repeatedly; NPV is the sounder criterion.

Chapter 3 — Architecture of a DCF

1. FCFF = EBIT(1 − t) + D&A − capex − ΔNWC: operating cash after tax and reinvestment, available to all capital providers.
2. WACC is below Ke; discounting equity cash flows at a lower rate overstates value and double-counts the benefit of debt.
3. Banks/insurers (debt is operating raw material), firms with stable leverage, or where debt flows are central; DDM for stable high-payout firms.
4. (Any four) cash flow ↔ discount rate; cash flow ↔ claim; currency/inflation; growth ↔ reinvestment; risk ↔ rate; tax treatment; cash/non-operating assets; timing convention.
5. 120 + 40 − 90 − 15 + 30 − 10 = **75**.

Chapter 4 — Reading Financial Statements

1. Interest income belongs to the financial/non-operating layer; its source (the cash) is valued in the bridge, so including it in FCFF double-counts.
2. 300 × 0.75 ÷ 1,800 = **12.5%**.
3. EBITDA rises (rent removed); lease liability is debt; FCFF is before lease payments, so EV includes the lease claim and the bridge deducts it; WACC should include lease debt.
4. When the item is genuinely non-recurring and the cost is not recurrent; test: if it occurred in three of the last five years, treat as recurring.
5. The business is harvesting its asset base — sustainable only temporarily; reported FCFF is flattered and a future catch-up in capex is likely.

Chapter 5 — Free Cash Flow to the Firm

1. It is captured in the after-tax cost of debt in WACC; including it in FCFF too would double-count.
2. FCFF = EBIT(1 − t) + D&A − … = (EBITDA − D&A)(1 − t) + D&A − … = EBITDA(1 − t) + t·D&A − …

3. With growth, capex exceeds depreciation: roughly $(1 + g)^{L/2}$ times depreciation for an asset life L (≈ 1.28 for g = 5%, L = 10).
4. Treat as a real cost: leave it deducted from EBIT; do not add it back.
5. Yes: a firm with large interest and depreciation but positive operating cash after tax and reinvestment can have negative net income yet positive FCFF.

Chapter 6 — FCFE and DDM

1. 200 + 60 − 140 − 20 + 50 − 30 = **120**.
2. Equity risk rises with leverage; if debt changes in the forecast, Ke must be re-levered or the FCFE value is inconsistent.
3. Payout = 1 − g/ROE = 1 − 6/18 = **66.7%**.
4. When payout differs from FCFE (non-payers, high borrowers), dividends irregular, or buyback-heavy firms.
5. Deposits and borrowings are operating inputs; enterprise value and net debt are not meaningful.

Chapter 7 — Forecasting

1. g = reinvestment rate × ROIC; 7% ÷ 14% = **50%**.
2. The marginal ratio shows how efficiently new capital is expected to work; an optimistic ratio inflates FCFF.
3. Five years (maybe seven) for a stable consumer company; short because growth and margins are near steady state.
4. Operating leverage (check fixed/variable split), mix (check segment margins), pricing/input costs (check pass-through history), scale/efficiency (check peer margins).
5. A period in which growth, margin and ROIC move gradually to terminal levels; avoids unrealistic cliffs.

Chapter 8 — Normalisation

1. Gain on asset sale (remove; asset already valued), restructuring cost (add back if one-off), related-party rent (restate to market), associate profit (remove and value the stake in the bridge).
2. Use a full-cycle average margin or a mid-cycle price deck; base terminal margin on it.

3. It is a real economic cost of compensating employees; adding it back overstates FCFF.
4. Debt, leases, unfunded pensions, deferred consideration, probable contingent liabilities, preference capital.
5. Because adjustments change taxable profit; ignoring tax effects misstates NOPAT.

Chapter 9 — Cost of Equity

1. Investors can diversify away specific risk, so only market (systematic) risk earns a premium.
2. 7 + 1.2 × 6 = **14.2%**.
3. It conflates the risk of the mean cash flow with volatility around it; better to reflect specific risks in cash-flow scenarios.
4. Historical (large standard error, survivorship); implied (depends on market-level and cash-flow assumptions); survey (opinion-based).
5. Compare with Rf, Kd, peers, the market return; check Ke > Kd > Rf and plausibility by risk.

Chapter 10 — Beta

1. 1.2 ÷ (1 + 0.75 × 0.5) = **0.873**.
2. 0.873 × (1 + 0.75 × 0.2) = **1.004**.
3. Errors in individual regressions are largely independent and partly cancel; standard error shrinks with √n.
4. Cash has near-zero beta and dilutes observed beta: operating beta = unlevered beta ÷ (1 − cash/firm value).
5. Betas tend toward 1 as firms mature; many valuers move the terminal beta toward 1 (or the industry average).

Chapter 11 — WACC

1. (700 × 14% + 300 × 10% × 0.7) ÷ 1,000 = **11.9%**.
2. Debt adds fixed claims ahead of equity, increasing equity volatility.
3. Iterate: guess D/V, compute WACC and value, recompute D/V, repeat until stable.
4. It double-counts risk if cash flows are already prudent; use scenarios instead.

5. Lease liabilities become debt: include in capital structure, cost of debt and bridge.

Chapter 12 — Terminal Value

1. Growth needs investment; assuming growth without reinvestment implies an unsustainable infinite ROIC.
2. TV = NOPAT ÷ WACC; growth adds nothing because new investment earns exactly its cost.
3. 200 × (1 − 4/12) ÷ 0.06 = **2,222**.
4. Gordon TV at N − 0.5; exit multiple TV at N.
5. TV share of EV, implied multiples, implied ROIC/reinvestment, g ≤ Rf, terminal margins, capex ≥ depreciation, terminal tax and WACC.

Chapter 13 — Equity Bridge

1. An item belongs in the bridge if its value is not already in the FCFF being discounted.
2. NCI at book ignores the subsidiary's profitability; market-based value is the claim.
3. Existing options are bridge items (option value); future grants are in the ESOP expense in EBIT.
4. FCFF is before lease payments (they are financing), so the lease claim is debt.
5. DLOC = 30% ⇒ 1/(1 + CP) = 0.70 ⇒ CP = **42.9%**.

Chapter 14 — Case Study

1. The downside scenario is further from base than the upside, so the weighted mean is below base.
2. Terminal ROIC and the year-5 margin: they drive TV, which is 81.5% of EV.
3. At ROIC = WACC value barely responds to g: growth adds nothing without excess returns.
4. Revenue 2,280 × 12.5% = 285.0 EBIT; NOPAT 213.75; reinvestment 140; FCFF **73.75**.
5. Sample: "We estimate fair equity value of about ₹2,560 crore (₹51 per share), within a range of ₹46–₹58, on a controlling, going-concern basis; terminal value forms 82% of EV."

Chapter 15 — Sensitivity and Simulation

1. Scenarios keep the discount rate for systematic risk and put specific risks in the expected cash flows; an add-on distorts long-dated flows.
2. A ranked bar chart of value swings per input; it shows which assumptions matter.
3. Subjective inputs, ignored correlations, spurious precision, incoherent extremes.
4. The TV formula is convex in WACC and g, so downside in WACC reduces value less than upside increases it.
5. Boom (price up, volumes up, margins up), base, downturn/over-capacity (prices down, volumes down, margins down, higher leverage risk).

Chapter 16 — Cross-Checks

1. $P_0 = D_1/(Ke - g) = EPS_1 \times \text{payout}/(Ke - g)$; dividing by EPS_0 gives payout $\times (1 + g)/(Ke - g)$.
2. It solves for the assumptions implied by the market price; it shows what must be true to justify a price.
3. Growth leaders' multiples embed higher growth/ROIC; applying them overstates value.
4. Trading multiples are minority prices of listed peers; transaction multiples include control premia and synergies.
5. Weights are subjective; averaging hides disagreements; method suitability matters more.

Chapter 17 — APV and LBOs

1. Without debt there is no tax shield, so equity holders require more than the blended WACC.
2. $0.30 \times 400 =$ **120**.
3. When debt is rebalanced to a constant ratio of value, the shield's risk equals the firm's (Ku).
4. EBITDA growth, debt paydown, multiple expansion.
5. They differ when debt policy differs; large when leverage changes materially or distress costs are significant.

Chapter 18 — Financial Institutions

1. Debt is operating, interest is an operating cost, and enterprise value/net debt are ill-defined.
2. (0.18 − 0.06) ÷ (0.14 − 0.06) = **1.50×**.
3. If ROE < Ke, growth lowers P/B because new equity earns less than its cost.
4. FCFE = net income − equity retained to maintain capital ratios as risk-weighted assets grow.
5. NIM, credit cost, cost-to-income, leverage, fee income.

Chapter 19 — Cyclicals, Resources, Real Estate

1. It overstates sustainable earnings and ROIC; the terminal state must be mid-cycle.
2. Explicit cash flows to reserve exhaustion with a price deck, costs, royalties, taxes, capex and closure costs; no perpetuity.
3. 8 ÷ 0.075 = **106.7**; 8 ÷ 0.085 = **94.1** (−11.8%).
4. The concession ends; cash flows stop at expiry.
5. Tax leakage, overhead, lack of control over cash flows, capital-allocation record.

Chapter 20 — Start-ups

1. It embeds failure: the target is on the *successful* outcome, not the expected return.
2. 400 ÷ 1.5^4 = 79.0 post-money; pre-money = **64.0**.
3. Forecast revenue and margin to maturity, accumulate losses as tax assets, interpolate, survival-adjust.
4. LTV/CAC well above 3×, strong retention, improving contribution margin, scalable take-rate.
5. It avoids mixing failure risk into the discount rate and exposes the probabilities.

Chapter 21 — Distressed Companies

1. Cash flows already reflect distress probabilities; adding it to the rate counts it twice.
2. 800 × 0.65 + 100 × 0.35 = **555**.

3. Limited liability: equity holders can lose only what they invested but gain from recovery — a call option.
4. Typically 20–50% of book, lower for specialised plant.
5. The valuer determines fair value and liquidation value, used by creditors to evaluate plans.

Chapter 22 — India and Emerging Markets

1. 1.12 ÷ (1.072/1.04) − 1 = **8.66%**.
2. The dollar rate omits expected rupee depreciation, understating the rate and overstating value.
3. Pre-tax rate, five-year budget limit, excludes future restructuring/enhancement and financing flows.
4. In oppression/appraisal contexts, fair value for a controller, or when valuing the whole company for a strategic buyer.
5. Model leakage in cash-flow scenarios or deduct the PV of non-arm's-length payments; avoid blanket discount-rate adders.

Chapter 23 — Real Options, Intangibles, ESG

1. Because the option to wait/expand has value; NPV today ignores flexibility.
2. S = PV of mine cash flows; K = development cost; T = lease life; σ = volatility of mine value (price volatility); r = risk-free; dividend yield = cost of delay.
3. Phase III start: 0.5 × 1,200 − 120 = 480; Phase II start: 0.5 × 480 − 40 = **200**.
4. The value is proportional to the royalty rate and is a market-evidence-based judgement.
5. Model the price path × emissions less abatement, after tax, in the forecast cash flows.

Chapter 24 — Pitfalls and Practice

1. Balance check; two-way FCFF reconciliation; sources/uses tie; terminal checks (g ≤ Rf, WACC − g ≥ 3 pts); sensitivity-centre check.
2. State assumptions in advance, document evidence, use ranges, disclose conflicts, retain the right to decline.

3. Conclusion and range, standard/premise of value, date, purpose, key assumptions, limitations.
4. Compare to history, to peers and industry, test drivers, reconcile cash, scrutinise incentives.
5. Choose three from the twenty and describe the direction and size of the distortion.

Chapter 25 — M&A

1. Revenue synergies depend on customer behaviour and competitor response; cost synergies are within management's control.
2. 70 ÷ 560 = **0.125**.
3. EPS accretion can come from cheap financing or a high acquirer P/E, regardless of value created.
4. The highest bidder tends to have the most optimistic estimate and overpays.
5. As a contingent claim: probability-weighted PV of expected payments.

Chapter 26 — Private Companies

1. Market beta ÷ correlation with market; for an undiversified owner's investment value.
2. Compare to a protective put over the restriction period; the put value approximates the cost of illiquidity.
3. 500 × 20% × (1 − 0.15) = **₹85** (in units of the same currency).
4. Owner pay, personal expenses, related-party purchases, non-operating assets, under-investment, unrecorded liabilities.
5. A minority forced out should receive its pro-rata share of the whole company's value.

Chapter 27 — Aster Pharma

1. EBITDA after Ind AS 116 excludes lease payments, so FCFF is before them; the lease claim is financing.
2. Mid-year flows continue in perpetuity; the Gordon TV is discounted with the same mid-year factor as $FCFF_5$.
3. Deduct 5% × 200 = 10 → 190 (and the per-share value falls by ₹0.10).
4. ROIC rises from 15% to 20% on margin expansion; terminal fade to 16% is a conscious conservative assumption.

5. Separate DCF/multiples of the subsidiary, minority rights, governance, any put options.

Chapter 28 — Case Library

1. Low reinvestment means growth is cheap, so the ROIC/WACC spread matters less.
2. If the forecast ended at a trough, naive extrapolation would understate value; use mid-cycle margins.
3. Growth consumes all or more than all of FCFE during the high-growth years.
4. Failure is handled in the probabilities; 25% reflects systematic risk and illiquidity.
5. Strong governance, high-quality assets, liquid stakes and a record of distributions.

Chapter 29 — Integrated Model

1. Net income → equity; D&A and capex → fixed assets; revenue → NWC; debt → interest; closing cash → balance sheet; cash-flow items → balance-sheet changes.
2. Different financing policies: the WACC assumes constant D/V while the FCFE model's debt follows book reinvestment.
3. Margin expansion raises ROIC and NI faster than equity grows.
4. Interest on opening balances; iteration; macro.
5. All FCFE is distributed, so cash stays constant and equity grows only by retained earnings.

Chapter 30 — Tax

1. Because the interest tax shield is in WACC; and the unlevered tax is independent of financing.
2. 25 per year × 25% = 6.25 a year; PV at 10% = 6.25 × 3.1699 = **₹19.8**.
3. When growth in capex keeps the DTL perpetually deferred and cash taxes are used in FCFF.
4. Remove them; apply the marginal rate in the terminal period.
5. It may be lost or restricted by law, so the shield's value could fall or vanish.

Chapter 31 — Sector Primers

1. Banks' debt is operating; FMCG's capital structure is a financing decision, so FCFF works.
2. Model the cliff explicitly: sharp revenue/margin fall after expiry.
3. In retail, leases are material and affect EBITDA, EV and WACC; in IT services they are small.
4. Value ends with the concession; no perpetuity.
5. Open answer: identify margin, growth and ROIC/capital intensity for the sector.

Chapter 32 — Limits of DCF

1. True if undisciplined; discipline (linked drivers, TV checks, sensitivities, reverse DCF) makes it defensible.
2. Compare with the reference class's base rate of sustaining such growth; require evidence for being exceptional.
3. Liquidation (asset value), holding company (NAV), bank (excess return); others per Section 32.3.
4. Build the value before looking at the price; record first estimate and reasons for change.
5. Open answer; common candidates: consistency (currency, financing), terminal assumptions, ranges.

Chapter 33 — Growth and ROIC

1. Each rupee reinvested yields less than a rupee of value; returning cash would create more value.
2. $(1 - 0.04/0.18) \div (0.10 - 0.04) = 0.7778 \div 0.06 =$ **12.96×**.
3. DCF value minus earnings-power value (NOPAT ÷ WACC).
4. 12 years of excess returns then ROIC ≈ WACC; with a possible cliff.
5. Average ROIC is NOPAT ÷ total capital; incremental ROIC is ΔNOPAT ÷ net new investment.

Appendix I — Projects and Exercises for Self-Study

Reading builds understanding; building models builds skill. These projects take a reader from a first spreadsheet to a defensible valuation. Each has a goal, steps and a self-check.

Project 1 — Rebuild Meridian from scratch (2–3 hours)

Goal. Reproduce the Chapter 14 valuation in a spreadsheet.

Steps.

1. Set up an Inputs sheet with growth, margins, tax, sales-to-capital, WACC inputs and bridge items.
2. Build the revenue → EBIT → NOPAT → reinvestment → FCFF forecast.
3. Compute WACC from the inputs (Chapter 11).
4. Discount FCFF; compute the terminal value from NOPAT × (1 − g/ROIC) ÷ (WACC − g).
5. Bridge to equity and divide by shares.

Self-check. FCFF FY27 = 73.7; EV = ₹2,913.3 crore; value per share ₹51.27.

Extensions. Switch to mid-year discounting (expect about ₹54.45); add a WACC × g data table; add a scenario selector.

Project 2 — Bottom-up beta (1–2 hours)

Goal. Estimate a cost of equity for a private company.

Steps.

1. Pick five listed peers; obtain regression betas (weekly, 2–5 years against a broad index), market capitalisations and debt.
2. Un-lever each beta using its own D/E and tax rate.
3. Compute the average and median; examine outliers.
4. Re-lever at a target D/E; compute Ke with a current risk-free rate and ERP.
5. Write one paragraph justifying every input.

Self-check. Unlevered betas should cluster; if one peer differs sharply, investigate its business mix or leverage before dropping it.

Project 3 — Reverse DCF on a listed company (3 hours)

Goal. Determine what the market expects.

Steps.

1. Choose a listed company with data you can obtain (annual reports, consensus forecasts).
2. Build a simple five-year FCFF model with revenue growth, margin, tax and a sales-to-capital ratio.
3. Compute WACC; use a terminal ROIC consistent with your view of the moat.
4. Solve (goal seek) for each of: margin, growth uplift, WACC and terminal growth that reconciles to the market price.
5. Judge plausibility of each implied assumption.

Self-check. Your conclusion should be a sentence of the form: "At today's price the market assumes..., which we consider [plausible/aggressive/conservative] because...".

Project 4 — The integrated model (6–8 hours)

Goal. Build the three-statement model of Chapter 29 and reconcile FCFF and FCFE.

Steps.

1. Start with Meridian's opening balance sheet.
2. Build the schedules: fixed assets, NWC, debt, equity, tax.
3. Create the income statement, cash-flow statement and balance sheet; check the balance.
4. Compute FCFF and FCFE; reconcile them.
5. Value by FCFF/WACC and by FCFE/Ke; explain any difference.
6. Re-run the FCFE model with debt at 20% of market EV and confirm the match.

Self-check. Balance check zero in every year; FCFE = ₹61.2 crore in FY27 under the book-debt policy; equity value ≈ ₹2,564 crore under the market-debt policy.

Project 5 — A mini valuation report (8 hours)

Goal. Write a valuation report of at least ten pages.

Contents. Executive summary; engagement terms; business and industry; historical analysis and normalisation; methodology; forecast and rationale; WACC; terminal value; DCF exhibits; bridge; sensitivity and scenarios; cross-checks; conclusion and range; limitations.

Rules. Every number has a source; every key assumption has a rationale; the TV share is disclosed; the range is shown with the drivers.

Self-check. Hand it to a colleague and ask them to find the three weakest assumptions. If they find them in the first read, the report is transparent.

Project 6 — Back-test (variable)

Goal. Learn from the past.

Steps.

1. Choose a company and a valuation date 5–8 years ago.
2. Build a valuation using only information available then.
3. Compare forecasts with outcomes: growth, margin, ROIC, share price.
4. Identify which assumptions were wrong and why: macro surprise, competition, management, your own bias.
5. Write down three lessons.

Self-check. Did the actual value fall within your sensitivity range? If not, what would a wider range have been?

Project 7 — Stress test the terminal value (1 hour)

Goal. See how much of a valuation depends on the terminal assumptions.

Steps.

1. Take any DCF you have built.
2. Compute the TV share of EV.
3. Compute EV with (a) terminal ROIC = WACC; (b) terminal growth = half your base; (c) terminal margin = peer median.
4. Rank the three by effect on value.

Self-check. If (a) cuts value by more than 25%, your valuation rests heavily on assumed excess returns — state it prominently.

A suggested study plan

Week	Reading	Practice
1	Chapters 1–3	Appendix B problems 1–6
2	Chapters 4–6	Appendix B problems 7–10; Project 1 (first half)
3	Chapters 7–8	Project 1 (second half)
4	Chapters 9–11	Appendix B 11–15; Project 2
5	Chapters 12–13	Appendix G 1–7
6	Chapters 14–16	Appendix G 8–14; Project 3
7	Chapters 17–21	Chapter review questions
8	Chapters 22–24	Project 5 (first draft)
9	Chapters 25–28	Project 7
10	Chapters 29–33	Project 4; Project 6

Index of Terms

Page numbers refer to this volume. Entries appearing on many pages are shown with their principal pages.

A

Adjusted present value (APV) 5, 72, 109, 111, 119, 134, 187, 213
Arbitrage pricing theory (APT) 62
Asset beta 24, 66, 68, 73, 140, 213, 219, 226

B

Balance sheet, integrated 148, 179, 180, 182, 183, 184, 242, 245
Berkus method 125
Beta 4, 43, 66, 81, 129, 161, 197, 221
Beta, bottom-up 47, 65, 67, 69, 115, 126, 137, 160
Beta, regression 65, 66, 68, 69, 137, 244
Black–Scholes 87, 131, 133, 142, 143, 146, 164, 214
Blume adjustment 66, 69, 213
Bridge, equity 12, 84, 86, 89, 92, 123, 127, 166

C

Cap rate 120, 121, 214, 219
Capex 12, 32, 40, 81, 118, 161, 191, 215
CAPM 23, 59, 61, 62, 63, 64, 65, 213
Cash conversion cycle 29, 32, 90, 106, 219
Cement industry 10, 54, 57, 117, 119, 152, 174, 194
Circularity 70, 72, 75, 111, 149, 184, 185, 219
Control premium 8, 87, 88, 89, 103, 158, 162, 166
Cost of debt 5, 23, 35, 70, 74, 145, 161, 187
Cost of equity 5, 23, 59, 70, 115, 147, 184, 219
Country risk premium 60, 61, 64, 73, 119, 135, 136, 140
Cyclical businesses 5, 55, 61, 65, 104, 118, 129, 174

D

Debt-like items 38, 56, 57, 84, 88, 89, 148, 195
Deferred tax 36, 39, 55, 85, 161, 184, 186, 188
Depreciation tax shield 34
Discount factor 14, 16, 18, 19, 42, 72, 78, 91
Discount for lack of control (DLOC) 63, 85, 88, 89, 161, 162, 165, 177
Discount for lack of marketability (DLOM) 88, 89, 137, 162, 163, 165, 220, 231
Distressed companies 5, 67, 73, 110, 129, 132, 213, 219
Dividend discount model 21, 25, 42, 44, 106, 193, 201, 220

E

EBITDA 10, 33, 68, 92, 110, 147, 174, 192
Economic value added (EVA) 9, 30, 80, 137, 161, 201, 209, 217
Embedded value 115, 116, 193, 197
Enterprise value 5, 24, 56, 88, 125, 133, 213, 220
Equity risk premium 4, 42, 69, 92, 128, 157, 197, 215
ESG 5, 142, 144, 145, 146, 202, 240
ESOPs and options 9, 37, 39, 40, 86, 87, 89, 111
Exit multiple 17, 19, 23, 79, 80, 81, 82, 97

F

Fair market value 9, 90, 127, 137, 161, 165, 220, 222
Fair value (Ind AS 113) 8, 9, 127, 138, 140, 220
FCFE 4, 24, 44, 116, 178, 183, 195, 220
FCFF 4, 35, 55, 88, 123, 181, 198, 230
First Chicago method 125, 128, 175, 176, 220
Football field 100, 105, 106

G

Goodwill 23, 28, 48, 131, 138, 139, 158, 181
Gordon growth model 15, 19, 42, 79, 81, 118, 129, 213
Growth identity 49, 51, 212, 221, 230, 235

H

Hamada equation 67, 69, 213, 221
Holding company discount 120, 121, 177, 178

I

Ind AS 103 (PPA) 8, 138, 139, 144, 158
Ind AS 116 and leases 30, 33, 53, 71, 75, 84, 166, 167
Ind AS 36 (value in use) 8, 138, 141, 158, 222
Insolvency and Bankruptcy Code 132, 134, 138
Intangible assets 10, 24, 28, 33, 130, 139, 144, 158
Interest tax shield 22, 36, 70, 74, 112, 187, 213, 221
Invested capital 24, 30, 44, 50, 166, 202, 209, 215
IRR 15, 19, 105, 111, 121, 127, 177, 205

K

Key-person risk 61, 160, 161, 165

L

Leases 27, 33, 71, 89, 146, 167, 195, 207
Leveraged buy-out (LBO) 72, 105, 108, 110, 111, 112, 221, 238
Liquidation value 9, 10, 80, 131, 132, 134, 138, 221

M

Marginal tax rate 28, 35, 36, 55, 70, 71, 74, 186
Market value added (MVA) 212, 221, 230, 231
Mid-year convention 16, 19, 82, 166, 170, 174, 215, 231
Minority interest (NCI) 9, 28, 84, 88, 139, 162, 166, 171
Modigliani–Miller 73, 108
Monte Carlo simulation 96, 98, 101, 214, 221

N

Net working capital 12, 28, 34, 48, 56, 111, 135, 158
NOL (tax losses) 5, 39, 61, 110, 123, 126, 139, 155
NOPAT 20, 33, 77, 92, 170, 208, 216, 228
Normalisation 4, 54, 56, 60, 90, 117, 121, 149
NPV 8, 15, 19, 143, 156, 192, 212, 216

P

P/B ratio 10, 102, 113, 114, 115, 116, 175, 193
P/E ratio 10, 94, 103, 106, 131, 159, 193, 213
PEG ratio 102, 158
Pension deficit 12, 39, 84, 89, 166, 169, 171, 230
Perpetuity 15, 19, 76, 80, 82, 118, 121, 169
Pre-money and post-money 124, 125, 127, 128, 176, 214, 216, 217
Present value 9, 13, 17, 60, 91, 130, 187, 202

R

Real options 5, 142, 146, 222, 240
Registered valuer 56, 132, 134, 137, 138, 139, 140
Reinvestment rate 49, 51, 78, 92, 169, 212, 221, 226
Relative valuation 10, 11, 13, 102, 131, 157, 192, 222
Reverse DCF 88, 90, 92, 97, 102, 103, 106, 170
Risk-free rate 5, 60, 73, 104, 135, 143, 161, 168
ROE 42, 45, 113, 115, 175, 179, 183, 193
ROIC 5, 51, 93, 106, 168, 191, 209, 225

S

Sales-to-capital ratio 24, 49, 52, 96, 99, 124, 207, 222
Scenario analysis 5, 74, 99, 127, 143, 157, 190, 214
Sensitivity analysis 5, 59, 77, 91, 96, 120, 202, 225
Size premium 61, 62, 63, 161
Start-ups 5, 46, 61, 122, 124, 125, 127, 128
Stock-based compensation 37, 55, 57, 126, 192
Stub period 17, 18, 19, 216, 222
Sum-of-the-parts 32, 73, 105, 120, 121, 176, 178, 194
Sustainable growth 42
Synergies 5, 9, 111, 151, 155, 157, 159, 163

T

Terminal value 4, 18, 58, 91, 114, 151, 177, 222
Treasury-stock method 86, 232

U

Unit economics 126, 128, 196

V

Value driver 116, 177, 178, 191, 192, 193, 194, 195
Value in use 138, 222
VC method 124, 127, 128, 176, 201

W

WACC 5, 43, 79, 99, 121, 156, 204, 224
Working-capital days 29, 183, 219